Program overview

19-May-2025 10:08

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master AUBS			
MSc 1 variant Architecture	**variant Architecture**		
AD	**Architecture and Dwelling**		
MSc 1 Architecture & Dwelling	**MSc 1 Architecture and Dwelling**		
AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
AR1A066	Delft Lectures on Architectural History and Theory	5	
AR1A080	Building Engineering Studios	10	
AR1AD014	Fundamentals of Housing Design	10	
MSc 2 AD	**MSc 2**		
Compulsory Choice			
AR2A011	Architectural History Thesis	5	
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	
Architecture electives (10 EC)			
AR0106	Architectural Ethnography	5	
AR0107	Housing Studies: An Open Intersectional Archive	5	
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
AR0109	City of Innovations Project	5	
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
AR0113	Tools of the Architect	5	
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
AR0118	Experiments in Drawing Theory	5	
AR0119	Figures	5	
AR0121	Analytical Models	5	
AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
AR0126	Bridge Design	5	
AR0132	Zero-Energy Design	5	
AR0133	Technoledge Glass Structures	5	
AR0134	Technoledge Façade Design	5	
AR0136	Making	5	
AR0137	Technoledge Health and Comfort	5	
AR0138	Technoledge Design Informatics	5	
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
AR0145	Circular Product Design	5	
AR0169	Materialisation: The Future Envelope	5	
AR0175	Campus Utopias	5	
AR0202	Computational Intelligence for Integrated Design	5	
AR0203	Eco-friendly Material Choices	5	
AR0215	Form & Inspiration	5	
AR0570	Worlding Theories and Concepts	5	
AR0796	Ornamatics	5	
AR0805	An Archeology of Digital Design	5	
AR0815	Idiosyncratic Infrastructures II	5	
AR0825	Building Stories: The Heteronomy of Urban Design	5	
AR0897	Van Gezel tot Meester	20	
AR0915	Online Digital Portfolio	5	
AR0916	Beyond 3D Computer Visualisation	5	
AR0917	Timber Architecture Design	5	
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5	
AR0919	Exteriorless: The Design of the Ordinary	5	
AR2AA010	Architectural Research and Design Seminar	5	
AR2AT041	Architecture and Philosophy Lecture Seminar	5	
AR2HA011	Building Green: Past, Present, Future	5	
AR2UA010	The Living City	5	
(Intra)disciplinary elective (15 EC)			
AR0139	MEGA	15	
AR0142	EXTREME technology	15	
AR0149	ON SITE: Landscape architectonic explorations	15	
AR0167	Architecture and Urban Design	15	
AR0177	The Why Factory MSc2 Design Studio	15	
		15	

AR0194	Bucky Lab A	
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15

MSc 3 and 4 Architecture and Dwelling

Compulsory Choice MSc 3 and 4

Advanced Housing Design

AR3A010	Research Plan	5
AR3AD100	Advanced Housing Design	55

Global Housing

AR3A010	Research Plan	5
AR3AD105	Dwelling Graduation Studio: Global Housing	55

Designing for Care — Designing for Care in an Inclusive Environment

AR3A010	Research Plan	5
AR3AD110	Dwelling Graduation Studio: Designing for Care in an Inclusive Environment	55

FSA — Form, Structure and Aesthetics

MSc 1 FSA — MSc 1 Form, Structure and Aesthetics

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1FO010	Form, Structure and Aesthetics	10

MSc 2 FSA — MSc 2

Compulsory Choice

AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5

AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15

MSc 3 and 4 Form, Structure and Aesthetics

AR3A010	Research Plan	5
AR3TA001	Technologies and Aesthetics	55

ADC — Architectural Design Crossovers

MSc1 ADC — MSc 1 Architectural Design Crossovers

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1DC010	Architectural Design Crossovers Studio	10

MSc 2 ADC — MSc 2

Compulsory Choice

AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5

AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
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AR2BO010	Borders and Territories International Design Studio	15
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AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
MSc 3 and 4 ADC	**MSc 3 and 4 Architectural Design Crossovers**	
AR3A010	Research Plan	5
AR3DC100	Architectural Design Crossovers Graduation Studio	55
AE	**Architectural Engineering**	
MSc 1 AE	**MSc 1 Architectural Engineering**	
AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1AE011	EXTREME architecture	10
MSc2 AE	**MSc 2**	
Compulsory Choice		
AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Architecture electives (10 EC)		

AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
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AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
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AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
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AR0216	Towards an inclusive living environment	15
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AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
MSc 3 and 4 AE	**MSc 3 and 4 Architectural Engineering**	
AR3A010	Research Plan	5

AR3AE100	Architectural Engineering Graduation Studio	55

A&PB Architecture and Public Building

MSc 1 Architecture and Public Building

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1AP012	Public Building Design Studio	10

MSc 2 AP MSc 2

Compulsory Choice

AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
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AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
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AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
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AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15

MSc 3 and 4 Architecture and Public Building

AR3A010	Research Plan	5
AR3AP100	Public Building Graduation Studio	55

CP Complex projects

MSc 1 CP MSc 1 Complex projects

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1CP011	Complex Projects Design Studio	10

MSc 2 CP MSc 2

Compulsory Choice

AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
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AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
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AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape	15

	of the Future	
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
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AR2MET011	Designing with Others	15
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ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
MSc 3 and 4 CP	**MSc 3 and 4 Complex projects**	
AR3A010	Research Plan	5
Choose one Graduation Studio		
AR3CP100	Complex Projects Graduation Studio	55
AR3CP120	Lunar Architecture and Infrastructure Graduation Studio	55
MAI	**Methods of Analysis and Imagination**	
MSc 1 MAI	**MSc 1 Methods of Analysis and Imagination**	
AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1MET015	Situating Architecture	10
MSc 2 MET	**MSc 2**	
Compulsory Choice		
AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Architecture electives (10 EC)		
AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
MSc 3 and 4 MofAI	**MSc 3 and 4 Methods of Analysis and Imagination**	
AR3A010	Research Plan	5
AR3MET105	A Matter of Scale	55
HA	**Heritage and Architecture**	
MSc 1 HA	**MSc 1 Heritage and Architecture**	
AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1AH010	Heritage and Architecture Design Studio: Architectonic Design	10
MSc 2 AH (nw)	**MSc 2**	
Compulsory Choice		
AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Architecture electives (10 EC)		
AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5

AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
MSc 3 and 4 HA	**MSc 3 and 4 Heritage and Architecture**	
MSc3 Adapting 20th century Heritage	**Graduation Studio Adapting 20th century Heritage**	
AR3A010	Research Plan	5
AR3AH105	Graduation Studio Adapting 20th century Heritage	55
MSc3 Revitalising Heritage	**Graduation Studio Revitalising Heritage**	
AR3A010	Research Plan	5
AR3AH115	Graduation Studio Revitalising Heritage	55
Interiors Buildings Cities		
MSc 1 AI	**MSc 1 Interiors Buildings Cities**	
AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1AI013	Interiors Buildings Cities MSc1 Design Project	10
MSc 2 AI	**MSc 2**	
Compulsory Choice		
AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Architecture electives (10 EC)		
AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
MSc 3 and 4 AI	**MSc 3 and 4 Interiors Buildings Cities**	
AR3A010	Research Plan	5
AR3AI100	Interiors Buildings Cities Graduation Project	55
The Why Factory		
MSc 1 TWF	**MSc 1 The Why Factory**	
AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5

AR1A081	Building Engineering Studios	10
AR1TWF011	The Why Factory MSc1 Design Studio	10
MSc 2 TWF (nw)	**MSc 2**	
Compulsory Choice		
AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Architecture electives (10 EC)		
AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15

AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
BO	**Borders and Territories**	
MSc 1 BO	**MSc 1 Borders and Territories**	
AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1BO010	Borders and Territories Design Studio	10
MSc 2 BO	**MSc 2**	
Compulsory Choice		
AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Architecture electives (10 EC)		
AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
		15

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
MSc 3 and 4 BO	**MSc 3 and 4 Borders and Territories**	
AR3A010	Research Plan	5
AR3BO100	Borders and Territories Graduation Studio	55
Urban Architecture		
MSc 1 UA	**MSc 1 Urban Architecture**	
AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1UA010	Urban Architecture MSc 1 Design studio	10
MSc 2 UA	**MSc 2**	
Compulsory Choice		
AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Architecture electives (10 EC)		
AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15

AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
MSc 3 and 4 UA	**MSc 3 and 4 Urban Architecture**	
AR3A010	Research Plan	5
AR3UA100	Urban Architecture Graduation Studio	55
Building Ideologies		
MSc 2 BI	**MSc 2**	
Compulsory Choice		
AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Architecture electives (10 EC)		
AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
AR0815	Idiosyncratic Infrastructures II	5
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5

AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
Design, Data and Society		
MSc 1 Design, Data and Society		
AR1A061	Delft Lectures on Architectural Design and Research Methods	5
AR1A066	Delft Lectures on Architectural History and Theory	5
AR1A080	Building Engineering Studios	10
AR1DDS005	Design, Data and Society MSc1 Design Studio	10
MSc 2 AD	**MSc 2**	
Compulsory Choice		
AR2A011	Architectural History Thesis	5
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Architecture electives (10 EC)		
AR0106	Architectural Ethnography	5
AR0107	Housing Studies: An Open Intersectional Archive	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0109	City of Innovations Project	5
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
AR0113	Tools of the Architect	5
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
AR0118	Experiments in Drawing Theory	5
AR0119	Figures	5
AR0121	Analytical Models	5
AR0122	1:1 Interactive Architecture Prototypes Workshop	5
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5
AR0136	Making	5
AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0175	Campus Utopias	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0215	Form & Inspiration	5
AR0570	Worlding Theories and Concepts	5
AR0796	Ornamatics	5
AR0805	An Archeology of Digital Design	5
		5

AR0815	Idiosyncratic Infrastructures II	
AR0825	Building Stories: The Heteronomy of Urban Design	5
AR0897	Van Gezel tot Meester	20
AR0915	Online Digital Portfolio	5
AR0916	Beyond 3D Computer Visualisation	5
AR0917	Timber Architecture Design	5
AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
AR0919	Exteriorless: The Design of the Ordinary	5
AR2AA010	Architectural Research and Design Seminar	5
AR2AT041	Architecture and Philosophy Lecture Seminar	5
AR2HA011	Building Green: Past, Present, Future	5
AR2UA010	The Living City	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0149	ON SITE: Landscape architectonic explorations	15
AR0167	Architecture and Urban Design	15
AR0177	The Why Factory MSc2 Design Studio	15
AR0194	Bucky Lab A	15
AR0216	Towards an inclusive living environment	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
AR0897	Van Gezel tot Meester	20
AR2AA015	Architectural Design Studio	15
AR2AA016	Architectural Design Studio	15
AR2AA017	Architectural Design Studio	15
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
AR2AT021	Architectural Technicities Design Studio	15
AR2BO010	Borders and Territories International Design Studio	15
AR2CP011	MSc2 Complex Projects Design and Research Studio	15
AR2DC010	Architectural Design Crossovers Studio	15
AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
AR2FO010	The Delta Shelter	15
AR2FST010	Studio 'High-Rise Culture'	15
AR2HA020	Urban Archipelago	15
AR2MET011	Designing with Others	15
AR2MM001	Building Ideologies Freedom	15
AR2UA020	Urban Architecture MSc2 design studio	15
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1501	Entrepreneurship in the Built Environment	15
Explore Lab		
MSc 3 and 4 Expl Lab	**MSc 3 and 4 Explore Lab, compulsory - for A students**	
AR3A010	Research Plan	5
AR3EX115	Explore Lab Graduation	55
Cross Domain City of the Future		
MSc 3 and 4 CS	**MSc 3 and 4 Cross Domain City of the Future - for A students**	
AR3A010	Research Plan	5
AR3CS100	Graduation Studio Cross Domain City of the Future	55
Resilient Rotterdam Graduation Studio - Veldacademie		
MSc 3 and 4 Resilient Rotterdam Studio		
AR3A010	Research Plan	5
AR3RE100	Resilient Rotterdam Graduation Studio - Veldacademie	55
Architectural Wood		
MSc 3 and 4 Architectural Wood		
AR3A010	Research Plan	5
AR3AW005	Architectural Wood Design Studio	55
variant Building Technology		
MSc 1 Building Technology		
AR1B011	Bucky Lab Design	10
AR1B022	Climate Design	5
AR1B023	Sustainable Architectural Materials and Structures	5
AR1B024	Introduction to Computational Design	5
AR1B032	Research and Methodology	5
MSc 2 Building Technology		
Building Technology electives (choose 3 courses)		
AR0126	Bridge Design	5
AR0132	Zero-Energy Design	5
AR0133	Technoledge Glass Structures	5
AR0134	Technoledge Façade Design	5

AR0137	Technoledge Health and Comfort	5
AR0138	Technoledge Design Informatics	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0145	Circular Product Design	5
AR0169	Materialisation: The Future Envelope	5
AR0202	Computational Intelligence for Integrated Design	5
AR0203	Eco-friendly Material Choices	5
AR0917	Timber Architecture Design	5
GEO5012	Land Administration	5
(Intra)disciplinary elective (15 EC)		
AR0139	MEGA	15
AR0142	EXTREME technology	15
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
ARFW1501	Entrepreneurship in the Built Environment	15
MSc 3 Building Technology		
AR3B025	Building Technology Graduation Studio	15
Compulsory Choice (choose 1 project)		
AR3B012	CORE	15
AR3B015	User-centred Sustainability Studio	15
MSc 4 Building Technology		
AR4B025	Building Technology Graduation Studio	30
variant Management in the Built Environment		
MSc 1 Management in the Built Environment		
AR1MBE015	Research Methods 1	5
AR1MBE020	Design and Construction Management	10
AR1MBE025	Building Economics	5
AR1MBE030	Real Estate Management	10
MSc 2 Management in the Built Environment		
AR2MBE011	Building Law	5
AR2MBE030	Urban Development Management	10
(Intra)disciplinary elective (15 EC)		
ARFW1501	Entrepreneurship in the Built Environment	15
ARFW1502	Peri-Urban Transformations: Analysis, Development, Design	15
MSc 3 Management in the Built Environment		
AR3MBE010	Research Methods 2	5
AR3MBE100	MSc 3 Graduation Laboratory Management in the Built Environment	10
Free electives 15 EC (MBE or other)		
AR0117	Didactic coaching skills for architecture and the built environment	5
AR0179	Value Capturing	5
AR0185	Research Methods 3	5
AR0835	Social Sustainability in Human Habitats	5
MSc 4 Management in the Built Environment		
AR4R010	MSc 4 Graduation Laboratory Management in the Built Environment	30
Cross Domain City of the Future		
MSc 3 Cross Domain City of the Future	**MSc 3 Cross Domain City of the Future - for MBE students**	
Compulsory for MBE students		
Compulsory		
AR3CS021	Seminar Cross Domain City of the Future	5
AR3CS081	Graduation Cross Domain City of the Future	20
Free electives/Research Methods 2	**Free electives/Research Methods 2 (5 EC)**	
MSc 4 Cross Domain City of the Future	**MSc 4 Cross Domain City of the Future - for MBE students**	
For MBE Students		
AR4CS030	Cross Domain City of the Future	30
variant Landscape Architecture		
MSc 1 Landscape Architecture		
AR1LA011	Architecture and Landscape: Design Studio	10
AR1LA021	Architecture and Landscape: Theory, Method and Critical Thinking	5
AR1LA051	Dutch Landscape: Design Studio	10
AR1LA061	Dutch Landscape: Theory, Method and Critical Thinking	5
MSc 2 Landscape Architecture		
AR2LA011	Urban Landscape: Design Studio	10
AR2LA021	Urban Landscape: Theory, Method and Critical Thinking	5
(Intra)disciplinary elective (15 EC or a combination of 10 + 5 EC)		
AR0149	ON SITE: Landscape architectonic explorations	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
AR2FO010	The Delta Shelter	15
ARFW0501	Applied Spatial Analysis for Sustainable Urban Development	5
ARFW0502	Plantscapes	5

ARFW1001	Integrated Metabolic Design for Sustainable Neighborhoods, Cities and Regions	10
ARFW1002	Heritage Landscape Design	10
ARFW1502	Peri-Urban Transformations: Analysis, Development, Design	15
MSc 3 Landscape Architecture		
AR3LA011	Landscape Architecture Analysis and Visualisation	5
AR3LA020	Research Methodology in Landscape Architecture	5
AR3LA031	Graduation Studio Landscape Architecture: Flowscapes	20
MSc 4 Landscape Architecture		
AR4LA010	Graduation Studio Landscape Architecture: Flowscapes	30
variant Urbanism		
MSc 1 Urbanism		
AR1U090	RandD Studio: Analysis and Design of Urban Form	10
AR1U100	RandD Studio: Designing Urban Environments	10
AR1U121	History and Theory of Urbanism	5
AR1U131	Sustainable Urban Engineering of Territory	5
MSc 2 Urbanism		
AR2U086	RandD Studio: Spatial Strategies for the Global Metropolis	10
AR2U088	Research and Design Methodology for Urbanism	5
(Intra)disciplinary elective (15 EC or a combination of 10 + 5 EC)		
AR0139	MEGA	15
AR0167	Architecture and Urban Design	15
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
ARFW0501	Applied Spatial Analysis for Sustainable Urban Development	5
ARFW0502	Plantscapes	5
ARFW0503	Urbidata: Responsible Design for Fair Data Cities	5
ARFW1001	Integrated Metabolic Design for Sustainable Neighborhoods, Cities and Regions	10
ARFW1002	Heritage Landscape Design	10
ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
ARFW1502	Peri-Urban Transformations: Analysis, Development, Design	15
MSc 3 Urbanism		
AR3U105	Graduation Orientation	3
AR3U115	Graduation Lab Urbanism	15
Choose 1 course		
AR3U110	Graduation Exploration	12
IFM4040	Joint Interdisciplinary Project	15
MSc 4 Urbanism		
AR4U010	Graduation Lab Urbanism	30
Cross Domain City of the Future		
MSc 3 Cross Domain City of the Future	**MSc 3 Cross Domain City of the Future - for U students**	
AR3U105	Graduation Orientation	3
AR3U110	Graduation Exploration	12
AR3U115	Graduation Lab Urbanism	15
MSc 4 Cross Domain City of the Future	**MSc 4 Cross Domain City of the Future- for U students**	
AR4U010	Graduation Lab Urbanism	30

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Master AUBS

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 variant Architecture

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

AD

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 Architecture & Dwelling

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1AD014	**Fundamentals of Housing Design**	**10**
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. H.A.F. Mooij	
Instructor	Ir. H.A.F. Mooij	
Instructor	T.W. Kupers	
Instructor	Ir. P.S. van der Putt	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	Variable 4 or 8 h per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Architecture & Dwelling MSc1 studio focusses on the fundamentals of the Dutch housing design and construction practice. It deals with the themes of stacking & linking, repetition, access & circulation, dwelling typology, construction methods, climate control and sustainability. Students of the Fundamentals of Housing Design Studio design a residential complex in an urban environment in the Netherlands. The design is accompanied/preceded by research into the design assignment and the specific topics of the studio. Though topics may vary from one semester to the next, at the core of each studio lies the design of dwellings and of the dwelling environment, complemented by research and literature study. Design work is done individually, while some of the research may be performed in teams. The Fundamentals of Housing Design Studio (AR1AD014) runs in conjunction with AR1A080, run by the Building Engineering department. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects of a medium sized building on MSC 1 level; implement the fundamentals of housing design (as defined by the course instructors) correctly, coherently and consistently, articulating the architectural appearance of a residential building; critically reflect on his/her own design and design process.	
Education Method	Studio sessions: 96 hours (for a group of 15 students) Self-study: 184 hours	
Literature and Study Materials	Bernhard Leupen and Harald Mooij, Housing Design a Manual (Rotterdam: NAi Publishers, 2011) Frederike Schneider, Grundrissatlas Wohnungsbau (Basel: Birkhäuser, 2011) Roger Sherwood, Modern Housing Prototypes (Cambridge: Harvard University Press, 1978)	
Assessment	The design studio Fundamentals of Housing Design (Ar1Ad014) has four (4) gradable items. The Final Grade (FG) of the studio is the weighted average of these four Item Grades (IG). Each item is graded with a precision of 5 decimal points, the Final Grade is rounded to 1 decimal point. Each item is seen as a partial exam. The grade for each partial exam may not be lower than 6,0. If one (or more) of the item grade(s) is 5,5 or lower, the Final Grade of the studio will be NVD (niet voldaan/not completed). Partial credits are not awarded. Repair regulations - If an item is deemed insufficient but repairable, the grade for that item will be a 5,0 or 5,5. - If an item is deemed insufficient and non-repairable, the grade for that item will be 4,5 or lower. An insufficient grade is deemed repairable if, in a period of two (2) weeks, certain shortcomings can reasonably be overcome. The tutor together with the course managers decide whether the aforementioned is the case. A maximum of one (1) item may be repaired.	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios.	
Period of Education	Semester1, Q1 & Q2	
Concept Schedule	Tuesday afternoon, Friday all day	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 AD

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5

Course Coordinator	Dr.ir. H. Sohn
Instructor	Dr.ir. H. Sohn
Instructor	Dr.ir. A. Radman
Instructor	Dr.ir. R.A. Gorny
Instructor	Dr.ir. S. Kousoulas
Responsible for assignments	Dr.ir. H. Sohn
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5

Course Coordinator	N.J. Amorim Mota
Instructor	Ir. P.S. van der Putt
Instructor	N.J. Amorim Mota
Instructor	R. Varma
Responsible for assignments	N.J. Amorim Mota
Contact Hours / Week x/x/x/x	4
Education Period	3
Start Education	3
Exam Period	none
Course Language	English

Course Contents

This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment.

In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences.

Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.

Study Goals

By the end of this course students will be able to:
1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective;
2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods;
3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media;
4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis;
5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences.
6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.

Education Method

The course Architectural Ethnography comprises group assignments and individual work.
The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course.

The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.

Literature and Study Materials

The course will use the following publications as main textbook references:

Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts.

Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto

Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126.

Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE.

Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.

Assessment

The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects.
The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment.
The summative assessment will be based on the following deliverables:

a) Analytical Assignment:
Ethnographic Research: Research Report (Group Work)

b) Practical Exercises:
Poster: Synthesis of (Quantitative) Data Collection (Group Work)
Ethnographic Research: Narrative and Visual Synthesis (Group Work)

c) Writing Assignment:
Critical Reflection (Individual Work)

For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c).

The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.

Retake and repair regulations
In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year.
Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Pereira Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Pereira Roders	
Co-responsible for assignments	Y. Zhang	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.	
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.	
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Book chapters, journal articles and other lecture materials.	
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	Technoledge Design Informatics	5
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5

Course Coordinator	Dr. B. Lubelli
Instructor	Dr. B. Lubelli
Instructor	S. Naldini
Instructor	T.P. Bennebroek
Responsible for assignments	Dr. B. Lubelli
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.
Period of Education	Q3
Concept Schedule	Wednesday afternoon
Minimum number of participants	15
Maximum number of participants	60
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarions	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

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Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Expected prior knowledge

The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval.

MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080).

Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.

Course Contents

Architectural and Urban Design in Times of Transitions.
Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies.

In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism.

The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.

Study Goals

At the end of this course, the student:
 knows key literature and recent research on people, movement and public space
 understands main theories on people, movement and public space
 applies these theories in analysing real urban settings
 evaluates critically on these theories
 creates presentations analysing the subject on an academic level.

And, the student:
 understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city
 evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning
 creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation.

Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.

Education Method

The course consists of interactive studio work and lectures.

Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory.

Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches.

Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared!

Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class.

*) the discussion groups ideally consist of four/five members, who divide topics and peer each other.

Literature and Study Materials

Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.

Assessment

Studio work 80% - Lectures 20%

Assessment of studio work:
Analyses and design, presented in drawing form with written commentary and a model.

Assessment of lectures:

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15

Course Coordinator	Dr. L. Cipriani
Responsible for assignments	Dr. L. Cipriani
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.
Concept Schedule	Wednesday, Friday
Minimum number of participants	15
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	**Van Gezel tot Meester**	**20**
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15

Course Coordinator	A.E. Rout
Course Coordinator	Prof.dr. T.G. Vrachliotis
Course Coordinator	M. Mateljan
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	

Course Contents

The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof.
Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent.
How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined?

The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers.

The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.

Study Goals

Upon completion of the Design Studio, the students will be able to:

convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level;
perform critical comparative research that results in a clearly formulated design hypothesis;
demonstrate how urgent societal issues are addressed in the design project;
demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal;
represent space in its complex interpenetration of people, architectures, technologies, materials;
confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.

Education Method

The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive.

The Studio will be structured across:
lectures
readings, writings and public discussion
design tutorials / workshops
specific exercises challenging innovative thinking

Literature and Study Materials

Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.

Assessment

Assessment of the students work will be based on the following:
Oral presentation;
Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.;
Writing assignments and analyses.

Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student.

The final grade reflects:
the position that is formulated with regard to the brief and its context;
the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation;
aesthetic and technical/functional qualities; the elaboration throughout the respective scales;
the quality of the presentation, the products and the argument;
the consistency, coherence and development of the students work during his/her process.

Products: will be described in the syllabus which will be published at the beginning of the course.

Repair regulations
The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5).

Retake regulations
Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator R. Ordonhas Viseu Cardoso

Course Coordinator Dr. D.A. Sepulveda Carmona

Responsible for assignments Dr. D.A. Sepulveda Carmona

Contact Hours / Week x/x/x/x 12 hours per week starting from week 1 and ending in week 9.

Education Period 4

Start Education 4

Exam Period 4

Course Language English

Course Contents

The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions.

One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales.

The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers.

The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).

Study Goals

At the end of this course the student is able to:
Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development.
Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation.
Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives.
Understand and develop planning strategies and actions across scales and borders using a variety of methods.
Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales.
Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.

Education Method

Approach:
The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework.

The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study.

Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others.

This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties.
Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions.

Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making)

Format:
The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals.
The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.

Literature and Study Materials

Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.

Assessment

Oral presentation and report.

A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 Architecture and Dwelling

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice MSc 3 and 4

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Advanced Housing Design

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AD100	Advanced Housing Design	55
Course Coordinator	Ir. O. Klijn	
Course Coordinator	Ir. H.A.F. Mooij	
Instructor	Ir. H.A.F. Mooij	
Instructor	Ir. O. Klijn	
Instructor	R.S. Guis	
Responsible for assignments	Ir. O. Klijn	
Co-responsible for assignments	Ir. H.A.F. Mooij	
Contact Hours / Week x/x/x/x	Research 6 hours p/w during 16 weeks in Q1 and Q2(start Semester 1)or Q3 and Q4 (start Semester 2) Design 8 hours p/w during 24 weeks in Q2, Q3 and Q4 (Start semester1) or Q1 and Q2 (Start semester 2)	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	There is an acute shortage of affordable and accessible dwellings in the Netherlands and a collectively acknowledged need to build one million homes until 2035. The societal task of The Advanced Housing Design Studio however, is not just about quantity or the architectural design of housing, but about a holistic understanding of dwelling as a social practice and of the city as an ecology. The studio focuses on the interrelationship between inhabitants and the architectural solutions of the design project and how this relation shifts within different territorial and governmental settings in The Netherlands. The design assignment is a mid to large size residential complex or neighbourhood, combined with collective and public functions depending on the specifics of the chosen location, displaying a variety of dwelling types for contemporary living. The design must take a long-standing notion of sustainability into account and include all three registers of sustainability: social, economic and ecological. The studio adopts a multi-dimensional research approach based on comparative design analysis and literature study. From this starting point, students develop an individual take on research which can include research methods from the fields of urban and environmental studies, social, architectural and urban history and architectural and urban ethnography. In a first phase of the studio, student collectively develop an urban analysis of the chosen site and discursive frame that delineate what an urban ecosystem and an inclusive housing project can mean in the urban context of the global North and the Netherlands. In parallel, students develop their your own specific angle of research and design approach based on the analysis of two to four housing projects that are exemplary for their individual research and design theme. In the second phase of the course, the design hypothesis is further developed in an iterative process between research insights and design decisions, taking the multi -scalarity of dwelling and an ecology of living together into account.	
Study Goals	Upon completion of the course, the student is able to: - design a mid to large size residential complex or neighbourhood (which responds to functional, cultural, structural, regulatory and aesthetic demands); - perform academic research (asking proper research questions, acquiring and implementing adequate research techniques and using astute modes of reporting); - establish a reciprocal relationship between research and design, and use the tools of academic research for the design process; - reflect on the architectural profession, on academic research, on his/her own position in the field of architecture, on the education she/he has received and on the ethical dimensions of architecture and academia.	
Education Method	Studio tutoring sessions, workshops, lectures.	
Assessment	Student work is assessed at P1 (ok or not ok) and at P2 (go or no-go) Only the assessment at P2 is decisive. Research and design will be assessed separately. A P2 may result in a 'Go', a 'No-Go' and a 'Retake' according to the regulations specified in the Graduation Manual.	
Period of Education	Semester	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Global Housing

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AD105	Dwelling Graduation Studio: Global Housing	55
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	Research 6 hours p/w during 16 weeks in Q1 and Q2 Design 8 hours p/w during 24 weeks in Q2, Q3 and Q4	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Summary	This course addresses the global need for more adequate housing, focusing on alternative solutions to engage architects in design processes that take into account a critical approach to the Sustainable Development Goals. The studio aims to produce knowledge on architectural concepts, models and instruments to deal with territories facing the challenge of rapid urban growth. The studio stimulates research on design approaches that explore alternative logics of modernization for emerging urban territories. Participants will be stimulated to reconsider established modes of analysis and intervention to support the emergence of urban welfare spaces; places of everyday wellbeing, commonality, conviviality, comfort, security, health, social and environmental justice.	
Course Contents	The Global housing graduation studio explores alternative approaches to the design of affordable housing in urban conditions characterized by rapid urban growth. The studio deals with the increasing cross-cultural character of contemporary architectural practice and the disciplinary challenges associated with the perennial global housing crisis. Focusing on the current challenges in low- and middle-income countries, participants in the studio are stimulated to critically review established modes of socio-spatial analysis and housing design decision-making processes, to support the emergence of urban welfare spaces; places of everyday wellbeing, commonality, conviviality, comfort, security, health, social and environmental justice. In the first phase of this course, students develop a collective knowledge base, unpacking the complexities of the projects context, as well as the particularities of mass housing design. The fieldwork in the projects site plays a key role in this phase, allowing students to advance a design hypothesis supported by empirical evidence of vernacular social and spatial practices collected through site surveys and ethnographic research. In the second phase, the design hypothesis is further developed and elaborated to deliver a housing project that explores a multi-scalar articulation between the urban realm, the sphere of the building, and the dwelling unit, considering a critical interrelation between social, economic and environmental factors.	
Study Goals	Upon completion of the course the student should be able to: 1. Elaborate a research plan, including problem statement, research question, goals and methodology; 2. Analyse the social and spatial practices of human settlements using transdisciplinary research tools and methods; 3. Communicate research outcomes using meaningful visual outputs and academic standards; 4. Synthesise processes of transformation of the natural and built environment using cross-disciplinary research methods to relate social, economical, cultural and environmental factors; 5. Formulate a design hypothesis based on a critical assessment of the societal issues pertaining to the project's context, a stakeholder's analysis and a programme of requirements; 6. Design a project for a housing complex, exploring a critical approach to the sustainable development goals, including the definition of a wider urban approach, appropriate architectural engineering and construction systems, and adequate spatial arrangements that cater for healthy and empowering livelihoods. 7. Produce meaningful visual and physical outputs to communicate an architectural project to an audience of experts; 8. Discuss the design principles of a housing project with other stakeholders. Note: The outcome of the graduation project demonstrates the students ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding Architectural ethics and his/her future role as architect. The individual graduation report should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which intercultural issues were addressed in the graduation project.	
Education Method	The tutorial system is the main educational method used in this course. The tutorial sessions will be focused on integrating research and design in the studio work. Throughout the whole first semester the work for the course Global Housing Graduation Studio will comprise work developed in group and individually. The fundamental research will be developed as team work, compiling a shared knowledge base focused on the theme of the studio. The research plan, including problem statement and research question will be developed individually and support the design hypothesis. In the second semester each student will develop the project individually. The studio will have weekly tutorial sessions. Some of these sessions will include lectures and (peer-)review sessions. In the first semester, a fieldtrip excursion to the project's site will be organised, and include a collaborative workshop with local students, researchers, educators and experts.	
Literature and Study Materials	Bhan, Gautam. Notes on a Southern Urban Practice. Environment and Urbanization (31-2), 2019, 639-654. Bredenoord, Jan, Paul Van Lindert, and Peer Smets, eds. Affordable Housing in the Urban Global South: Seeking Sustainable Solutions. Abingdon, Oxon: Routledge, 2014. Caldeira, Teresa. Peripheral Urbanization: Autoconstruction, transversal logics, and politics in cities of the global south. Environment and Planning D: Society and Space 35-1, 2017, 3-20. Gameren, Dick van, Frederique van Andel, and Pierijn van der Putt, eds. Global Housing: Affordable Dwellings for Growing Cities. DASH, 12/13. Rotterdam: NAi 010 Publishers, 2015. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Marcuse, Peter and David Madden. In Defense of Housing: The Politics of the Crisis. London and New York, Verso, 2016. Mota, Nelson, and Dick van Gameren. Affordable Housing and Sustainable Development: A Tale of Two Systems. The Architectural Review, April 2016. Mota, Nelson and Yael Allweil, eds. The Architecture of Housing after the Neoliberal Turn, Footprint 24. Prinsebeek: JapSam Books, 2019. Sennett, Richard. Building and Dwelling. Ethics for the City. Penguin Books, 2018.	

	Urban, Florian. Tower and Slab: Histories of Global Mass Housing. Oxon and New York: Routledge, 2013. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018. More specific literature and Study Materials will be made known 1 week prior to the start of the course in Brightspace.
Assessment	In the course of the graduation process two obligatory progress reviews (P1 and P3) and three formal assessments (P2, P4 and P5) take place. The P1 and the P2 are part of the Master 3 program and P3, P4 and P5 take place within the Master 4. Evaluation 1 (P1) - Compulsory progress review Goal: Assess whether the students working method and progress guarantee he / she will be able to meet the requirements for the P2 in time. Evaluation 2 (P2) - Formal assessment Goal: Completion of Master 3; assessment students admission to Master 4; the base for passing the P2 should be that the belief is that the student can graduate in the se-mester following the P2 period with a satisfactory result. Evaluation 3 (P3) - Compulsory progress review Goal: Survey whether the students working method and progress guarantee he or she will be able to meet the requirements for the P4 in time. Evaluation 4 (P4) - Formal assessment Goal: Assessment whether content of academic fields and presentation meet the requirements to admit the student to the final public presentation (P5). Evaluation 5 (P5) - Public final presentation Goal: Public final presentation and assessment graduation project. Students will receive marks at the P5 for the following aspects; - Architecture design - Building Technology - Research - Presentation - Final Mark. The assessment of Design and Research during the different evaluation moments will be based on the EMMA rubric and focus on the following criteria: -Coherence: internal consistency, integration, essence, concept -Significance: ethical, socio-cultural and/or scientific relevance, value, meaning -Elaboration: extensiveness, degree of detail of all aspects -Correctness: accuracy, efficacy, and evidence-based -Innovativeness: personal interpretation, creativity, new, unexpected, unique situation -Knowledge and know-how: effective study and use, processing of precedents and principles -Exploration: openness, discovering and investigation, analysis and testing -Reflection: careful consideration, evaluation, effects, comparing and positioning -Presentation: clarity, intelligibility, reflection and being engaged by it as a listener or a reader.
Remarks	The participants in this graduation studio will be encouraged to join a field trip to the site of the assignment during Q2. The cost of the field trip is approximately 1.000,00 - 1.500,00 (return flight and accommodation for 2 weeks). Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip. During the field trip excursion, participants in this studio will join seminars with local housing experts and workshops with local students and faculty.
Period of Education	Autumn Semester (P1, P2) and Spring Semester (P3, P4, P5)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Designing for Care

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AD110	Dwelling Graduation Studio: Designing for Care in an Inclusive Environment	55

Course Coordinator	Dr.ir. B.M. Jurgenhake
Instructor	Dr.ir. B.M. Jurgenhake
Instructor	J.H.A. Macco
Responsible for assignments	Dr.ir. B.M. Jurgenhake
Contact Hours / Week x/x/x/x	Research 6 hours p/w during 16 weeks in Q1 and Q2 Design 8 hours p/w during 24 weeks in Q2, Q3 and Q4
Education Period	1 2 3 4
Start Education	1
Exam Period	none
Course Language	English
Course Contents	We are more aware than ever of the role of our everyday (built) environment in our health and quality of life. Where we live affects our physical, social, and emotional well-being. Views of the outdoors, fresh air, daylight, control over our environment, privacy, contact, access to services, playgrounds, and sports facilities all impact our health. The amount of time we spend at home is increasing, with home offices being encouraged, and the changing healthcare system urging us to stay at home as long as possible, even if we are ill. In addition, the challenges of an ageing society require design solutions within our living environment. Healthcare will increasingly take place in our communities rather than in specialized facilities. This MSC3 studio explores how to design for health and care. How can architecture support a healthy society? How can we address current societal, health, and care-related issues? Instead of just treating disease, new models focus on disease prevention, health promotion, and improving quality of life. This is where we will focus, and where your creativity is needed. We start by understanding the people and identifying the problems. The integration of ethnographic research methods provides a unique opportunity to meet the target groups and understand their specific needs. For instance, you may stay at a nursing home, daycare facility, or home for people with disorders for a few days. You will participate in daily life, observe the situation, and talk to the users of the building. The outcomes of the research phase (fieldwork and literature studies) will be discussed, analyzed and translated to new situations. The research will guide you to ideas for design proposals, specifically focusing on health and prevention, or new care homes. You will draw connections between different ideas and justify your design decisions. This leads you to creative and innovative designs. Alternatively, you can focus on innovative building typologies that support health and life quality. We invite students who would like to question current healthcare approaches and rethink existing typologies and solutions in search of a healthy and sustainable living environment. The design location(s) will be in The Netherlands and may be within one chosen city. If your tutor agrees, you may choose a location outside The Netherlands.
Study Goals	Upon completion of the course, the student should be able to: 1. Understand and discuss health and care-related issues prevalent in our society. 2. Formulate and present a problem statement effectively. 3. Develop a research hypothesis pertinent to the identified problem. 4. Conduct human-centered research (fieldwork), gather data, analyze findings, and evaluate outcomes. 5. Establish connections between various research findings (from fieldwork and literature) and translate them into tangible design proposals. 6. Articulate and defend different design proposals while justifying the decisions made. 7. Formulate and implement architectural strategies aimed at creating a healthy and inclusive living environment, accommodating the diverse needs of all societal groups. 8. Identify suitable building techniques and construction systems to be utilized in the design proposal. 9. Create meaningful visual and physical representations to effectively communicate the project to an audience of experts. 10. Demonstrate moral sensibility, observational skills, analytical prowess, creativity, judgment, decision-making, and argumentation skills concerning architectural ethics and their future role as an architect through the graduation report. The report should not only elaborate on the societal and disciplinary relevance of the Graduation Project but also address design ethics and the manner in which interdisciplinary and intercultural issues were tackled in the graduation project.
Education Method	An important aspect of the graduation studio is ethnographic (human-centered) research, which involves understanding, preparation, and practice. This approach places the user at the center of the research process. Each year, we strive to closely collaborate with the target group itself. Establishing contact with the target group is crucial for gaining insights to inform the design process. Only after this initial phase does the design proposal begin. In addition to these participatory efforts, we collaborate with housing associations or other relevant stakeholders. Interdisciplinary research, which combines anthropological and architectural approaches through observation, participation, and/or interviews, serves as the foundation of our research activities, complemented by literature. Some research may be conducted in teams. Research tutoring takes place in our studio and informs the design process. During the first half of the semester, studio work is dedicated to the research phase, conducted partly in small groups with individual contributions, alongside the development of the initial individual design concept. In the latter part of the semester, following the P1 presentation, the focus will transition to design. Research findings will be translated into design guidelines at this stage. Various concepts will be tested, including through model-making and discussions with tutors. The conclusion of MSc 3 is marked by the P2 presentation, which encompasses the completed research report, including research conclusions, a master plan at the neighborhood level, and a preliminary design concept for the building.
Assessment	Examination is done with oral presentations and written paper, research report and a graduation report. The assessment of the work developed by the students will be determined by the following criteria: 1. Personal Development: Ability to self-reflect on the personal motivations to work on the assignment, capacity to elaborate a realistic planning, and demonstration of the awareness to evaluate his/her own strengths and weaknesses. 2. Research and Analysis: Ability to identify and interpret relevant information out of the research and of source material used to support the development of the design assignment. 3. Architectural Project: Ability to experiment, test and project concepts, processes, forms and materials demonstrating technical competence. 4. Materialization and Technology: Ability to articulate knowledge on the architecture discipline with other building technology, materials and construction processes. 5. Communication: Ability to create meaningful oral, written and visual communication, prepared using appropriate conventions and media. 6. Process: Demonstration of a critical attitude to the design assignment and ability to work collaboratively and independently within the main standards of the architecture profession. 7. Interdisciplinary Research: Ability to work with research methods, common in the field of anthropological and social research, like observation and interviews and combine them with the more visual methods of architectural research.

	Results: Study Plan with Problem Statement, Goal and Method description; Research Report, made individually or in small groups; Site assignment analysis report (group- or individual work, depending on the choice of the site); Design Proposal, including design program; All elements are integrated into an individual Graduation Report.
Exam Hours	The examination of this course will be done in several moments: P1 examination will be done after the first quarter. It is an oral examination. P2 examination will be done at the end of the second quarter within the obligated weeks (see graduation calendar) P3 examination will be done in the third quarter. It is an oral examination. P4 examination will be done in the fourth quarter and the Msc4 semester within the obligated weeks (see graduation calendar). P5 examination will be done at the end of the Msc4 semester within the obligated weeks (see graduation calendar). Retake and repair moments fall within the same academic semester, they must be workable and study able, and not cause study delays. A retake is an entirely new examination or assignment. A repair is a revision or addition to the existing assignment (with the same topic).
Period of Education	Fall semester 2024/25
Concept Schedule	A concept schedule will be distributed one week before the course starts.
Minimum number of participants	12-15
Maximum number of participants	24-26
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

FSA

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 FSA

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1FO010	**Form, Structure and Aesthetics**	**10**
Course Coordinator	G. Coumans	
Course Coordinator	Ir. R.R.J. van de Pas	
Instructor	S.M. Osinga	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc	
Course Contents	The central, essential theme in the studio is to come to spatial expression through Form, Structure and Material. The design will be mainly defined by the following: -The main guiding theme leads to a visibly readable expression, linked to the central themes of the studio. -Early experiments on a detailed scale on the basis of material use, assembly methods and technical solutions. The design brief is based on a cultural function and located in a transitional area in The Netherlands. Working together with Building Engineering Studios, the Form Studies studio can focus on the spatial expression, while sustainable, technical solutions are being integrated. The assignment very much embraces the integrated, sustainable design. The clear structure of the studio, shifting in scale and theme at predetermined dates, creates a creative laboratory from a students point of view. This introduces the students to a design method that is based on the use of physical models as research and design tool. The method steers to an active, experimental and flexible design attitude that enables expression and integration. In an early stage a limited amount of groupwork will take place, but the design project will be carried out individually.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design for a medium sized building on MSC 1 level. use and construct physical models, hand drawings and digital models to represent specific design and research themes. understand and practice the concept of the integrated architectural design. document and reflect on the design process and outcome.	
Education Method	Design Studio format with weekly assistances starting week 1. Workshops, excursions and lectures will be part of the studio.	
Assessment	Assessment based on process, documentation and final presentation. The final assessment is a presentation of visual representations of the project and a project dossier. The dossier will combine a documentation of the design process, necessary additions to the poster and a written description of the project. The final grade will be rounded to half or whole points. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios. The studio is not available for MSc 2 students.	
Period of Education	Semester 1, Q 1 & 2	
Concept Schedule	Week 1.1 Introduction Week 1.1-1.8 Research, Lectures, Workshops Week 1.8 Mid-term Presentation Week 1.8-2.8 Design Development, detailing Week 2.8 Final Presentation	
Leerstoel	Form Studies	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 FSA

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.	
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5

Course Coordinator	N.J. Amorim Mota
Instructor	Ir. P.S. van der Putt
Instructor	N.J. Amorim Mota
Instructor	R. Varma
Responsible for assignments	N.J. Amorim Mota
Contact Hours / Week x/x/x/x	4
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Pereira Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Pereira Roders	
Co-responsible for assignments	Y. Zhang	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.	
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.	
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Book chapters, journal articles and other lecture materials.	
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5

Course Coordinator	Prof.dr.ing. C.M. Hein
Course Coordinator	J.M.K.K. Hanna
Instructor	Dr. P. De Martino
Instructor	J.M.K.K. Hanna
Responsible for assignments	Prof.dr.ing. C.M. Hein
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	3rd Quarter
Concept Schedule	Friday
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

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Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.	
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.	
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:	

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15

Course Coordinator	Dr. L. Cipriani
Responsible for assignments	Dr. L. Cipriani
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.
Concept Schedule	Wednesday, Friday
Minimum number of participants	15
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15

Course Coordinator	A.E. Rout
Course Coordinator	Prof.dr. T.G. Vrachliotis
Course Coordinator	M. Mateljan
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15

Course Coordinator	Ir. P.S. van der Putt
Course Coordinator	Ir. S.M. Witteman
Instructor	Ir. P.A.M. Kuitenbrouwer
Instructor	T.W. Kupers
Instructor	Ir. S.M. Witteman
Responsible for assignments	Ir. P.S. van der Putt
Contact Hours / Week x/x/x/x	112 hours per quarter
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	completed MSc1
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 Form, Structure and Aesthetics

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3TA001	Technologies and Aesthetics	55
Course Coordinator	G. Coumans	
Course Coordinator	Prof.dr.ing. U. Knaack	
Instructor	Ir. C.N. van Leest	
Instructor	Ir. K.B. Mulder	
Instructor	G. Coumans	
Instructor	V.L. de Vries	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	...	
Expected prior knowledge	Completed MSc1 and MSc2 Architecture Design Studio and MSc1 Building Engineering Studio	
Course Contents	Climate changes and increasing demands on material use and re-use forces the architectural designer to constant technical innovations and re-thinking of the expression of architecture. The aim of the studio is to explore the relation between the technical design innovations and the specific architectural form language that comes along with and belongs to these developments. This will culminate in an architectural design of a small to medium size building where architectural design and techniques are fully integrated. The studio will offer each year three topics, based on technical innovations and/or climate related issues, from which the students can choose. The studio will focus on locations in Northern Europe where social or climate related problems are relevant and suitable for the studio topics. The design assignment will focus on the development of the architectural expression and integration of the topics of the studio. The chosen location will also create its own design leads; social, economical, geographical and ecological. The studio adopts a multi-dimensional research approach, initially based on comparative design analysis, technical analysis and literature study. From this starting point, students develop an individual take on research which can include research methods from the fields of Architecture, Arts and Building technology. The research will inform and initiate the design process, but will not limit the scope of the project. The interaction between the different research and design projects will be stimulated, which allows a multi-layered, innovative attitude.	
Study Goals	The student is able to: - present a coherent, significant, elaborated, correct and innovative design to a level fit for the architectural practice. - show an attained understanding in the design process with regard to building technology, human sciences, environmental aspects and sustainability. - take part in the debate about aesthetics, related to contemporary architectural design.	
Education Method	Lectures, workshops and groupwork will alternate the individual work and guidances during the research phase. The mentor team will be formed during the research phase, but will be set and unchanged when the design will be developed. Physical models will be seen and used as an important design tool. Students and their work will occupy the studio and the mentors will be visiting at fixed days.	
Assessment	Assessment will be based on the Emma Assessment tool for graduation studios. Assessment will take place at P1 (research progress), P2 (research outcomes and design proposal, go / no go, P3 (design progress), P4 (design outcome, go / no go) and P5 (Public Presentation and defense)	
Special Information	This graduation studio is a cooperation between the group of Formstudies from the department of Architecture and the Chair Design of Construction from the department of Architecural Engineering & Technology. The studio is only open for students of the Track of Architecture.	
Period of Education	The studio starts in Q1, final presentation / examination in Q4. The studio is offered only once a year.	
Concept Schedule	Q1: Preliminary research and Research plan; Q2: Research development / Design brief; Q3: Design developed as draft in all scales; Q4: Design developed in final drawings and models / Presentation.	
Leerstoel	Formstudies (Architecture) & Design of Construction (AE&T)	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

ADC

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc1 ADC

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1DC010	Architectural Design Crossovers Studio	10
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	Ir. S. Steenbruggen	
Instructor	Ir. C.M. Calis	
Instructor	Ir. A.M.R. van der Meij	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours/week (1.1-1.8 & 2.1-2.8)	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Architectural Design Crossovers MSc1 studio is structured as an introduction to the education groups objectives within the larger framework of a multi-disciplinary approach. The project sites and scopes within the studio intersect the disciplinary boundaries exploring the temporal and material aspects of the architectural project. We consider the built environment the accumulation of not only social, economic, cultural and political layers but also an intersection of environmental sources, material and immaterial flows. Shaped through intensive material and immaterial flows, these sites help the students define the "crossover" themes to be explored in the collective and individual design assignment. In this respect, the studio will focus on London, where "universal archive" will serve as a programmatic framework to explore new spatial configurations. The studio allows the students to investigate programmatic, typological, and material constraints under the new imperatives of im/material ecologies in the post-carbon design discourse: How does architecture respond to the new material paradigms? As designers, how should we renew material flows in architectural production, and what are the emergent building techniques, typologies, and programs? In this regard, the MS1 design studio allows the students to define the core aspects of the "universal archive" to be explored in a multi-disciplinary approach. During the design process, students will explore new architectural concepts and building types and they will build up a narrative supporting their architectural design proposal. Combined with the on-site survey data, the collective research serves the students with critical standpoints for their individual design work. An educational excursion (3-5 days) with an on-site workshop is part of the studio program. The expenses for the excursion are approximately ±250. The corresponding information is to be communicated prior to the start of the semester.	
Study Goals	Upon completion of the MSc1 studio the students are able to: - Analyze, evaluate and pursue a range of technical, programmatic, theoretical, historical and professional implications toward the final design proposal. - Integrate theoretical knowledge and practical skills into their design process. - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field. - Design and present a coherent, significant, elaborated, correct and innovative proposal for a medium sized building on MSc1-level.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Presentation and reviews	
Literature and Study Materials	Study materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio to be presented as part of the final review. The assessment criteria for the individual work are: - the position formulated by the student addressing the studio theme; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the elaboration level of the project throughout the respective scales addressed; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term review (Week 1.7/1.8) Workshop (Week 2.3/2.4) Final review (Week 2.8) The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Remarks	The Architecture Design Studios and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering tutors collaborate closely.	
Period of Education	Q1, Q2	
Concept Schedule	Wednesday morning & afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 ADC

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.	
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or	

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Pereira Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Pereira Roders	
Co-responsible for assignments	Y. Zhang	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.	
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.	
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Book chapters, journal articles and other lecture materials.	
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

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Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio 5
Course Coordinator	Ir. J.J.J.G. Hoogenboom
Course Coordinator	H.P. Kiksen
Instructor	H.P. Kiksen
Responsible for assignments	P. de Ruiter
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours
Computer Use	Laptop required
Literature and Study Materials	- The Brightspace section for this course
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Elective	Yes
Period of Education	Semester
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours
Leerstoel	Design Informatics
Minimum number of participants	12
Maximum number of participants	60
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.	
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.	
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:	

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday, Friday	
Minimum number of participants	15	
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	**Van Gezel tot Meester**	**20**
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more. The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.	
Study Goals	By the end of the MSc2 Studio, students should be able to: Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural) Visualise key design and data-related features using effective narrative and visual storytelling methods	
Education Method	The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include: Design tutorials (including individual design progress feedback and group discussions) Seminars (including lectures, workshops and reading discussions) Excursion (study trip including the site visit)	
Course Relations	Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues? The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas: CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives. ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort. RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges. Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.	
Literature and Study Materials	Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.	
Assessment	Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include: Posters (including architectural drawings and other visuals) Booklets and/or digital presentations Short animations (such as GIF animations) Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.	

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.	

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).	
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.	
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.	

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 ADC

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3DC100	Architectural Design Crossovers Graduation Studio	55
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	Dr.ir. R. Cavallo	
Instructor	Ir. S. Steenbruggen	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. A.M.R. van der Meij	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (1.1-1.8 & 2.1-2.5) 8 hours/week (2.6-2.10, 3.1-3.10, 4.1-4.10)	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 & MSc2	

Course Contents

The Architectural Design Crossovers graduation studio perceives the heterogeneity as the pervasive urban condition in the contemporary European cities, which cannot be addressed with the conventional disciplinary division in the design field. Therefore, the studio starts with issue-based lectures and seminars to reach beyond the problem oriented design solutions. In the studio, we consider the architectural project not only a synthesis of (im)material requirements of build-ing practices but also place it at the focus of an interdisciplinary collaboration with an emphasis on multi-modal design research.

In the studio and the seminars, the students develop an in-depth analysis of the context at different levels. Working on the different site conditions (networked/layered) the students are asked to challenge the disciplinary conventions, methodologies and approaches in a wider social, environmental or cultural context. Embracing the inherent complexities and conflicts, the studio embraces a careful curation of different modes of design representation. From spatial/morphological mapping practices to time-based analysis of urban transects, the students adopt appropriate design media pertaining to the scope and definition of their project. The subjects of study in the studio will start with the synthetic analysis of a complex phenomenon to investigate the site as a networked/layered entity.

Next to design and seminar research, the studio will be complemented by master classes involving experts from related (design) fields in which the themes, practices, experiences and positions closely related to the subject matter will be openly debated with the contribution of the students.

In Spring semester (MSc4) the studio elaborates on the design concepts and schemes developed during the Fall semester (MSc3) aiming at the final resolution of the graduation project that complies with the graduation and assessment criteria defined by the Board of Examiners.

More detailed information (i.e. literature, precedents, schedule) for the course will be provided in the course syllabus.
For questions or more information please contact the course coordinator.

Study Goals

By completing the course the students are able to:

- Define their design position addressing pressing issues specific to the project site
- Conduct in-depth research and analysis of the context, design brief and spatial program
- Analyse, assess and develop the technical and programmatic requirements relevant to the design position
- Use techniques and instruments for corresponding their design proposal
- Synthesise analytical findings into relevant spatial/architectural questions
- Demonstrate self awareness and insight about their design process and position with regard to historical, theoretical, cultural contexts

- Synthesise analytical design findings into a comprehensive research report
- Develop individually a design proposal integrating aesthetic, technical and functional requirements of the proposed program
- Present the process and the project integrally to the specialists within related fields and to a non-specialist audience
- Communicate and defend complex design ideas through verbal, visual and written media to both specialist and non-specialist audiences
- Integrate the relevant technical and material aspects into architectural design considering building techniques and technology
- Reflect on the the design and building practices, methods and technologies in a broader social, environmental and cultural context
- Demonstrate awareness of material and energy flows pertaining to the specific solutions in the proposal
- Present the technical aspects of the design process and the final result in a clear and systematic way in the form of drawings, words, texts and schemes
- Defend complex design position and solutions through verbal, visual and written media publicly

Education Method

In coordination with the Research Seminar (4 hours/week 1.1-2.5)) Research Plan lectures and tutorials (AR3A010) in weeks 1.1 -1.8

Issue-based pre-design research
Site survey and related fieldwork
Collective and individual research report
Presentations and reviews
Independent design & self-study

Until the mid-term review (P1), the students will work in groups that will be complemented with individual exercises as part of the seminar and tutorial classes.

Course Relations

Research Plan (AR3A010)

Literature and Study Materials

Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.

Reader

To be provided at the start of the semester

Assessment

- Design Presentation & Examination (See below P1-P5)
- Writing Assignment (Research Plan & Research Paper)

	(P1) Interim mid-term evaluation (week 1.9/1.10) (P2) Go-No go assessment (weeks 2.9/2.10) (P3) Progress review (week 3.8/3.10) (P4) Go-No go assessment (week 4.4/4.5) (P5) Public final presentation and diploma ceremony (week 4.9/4.10) P2 is a Go-No go examination that determines the students ability to to proceed to the final part of the studio. P4 is a Go-No go examination that determines the student's ability to proceed to the public graduation presentation and ceremony (P5). All the criteria for the evaluation are explained in the graduation manual and will determine the final mark obtained for the entire graduation studio.
Period of Education	Q1, Q2, Q3, Q4
Concept Schedule	Tuesday (seminar: 1.1-1.9 & 2.1-2.6) Friday morning & afternoon (studio: weekly except education free weeks)
Leerstoel	Architectural Design Crossovers
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

AE

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 AE

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1AE011	**EXTREME architecture**	**10**
Course Coordinator	A. Snijders	
Instructor	E.W.M. Hehenkamp	
Instructor	Ir.arch. G. Koskamp	
Instructor	Ir. R. Schroën	
Instructor	Ir. M. van Driel	
Practical Coordinator	Ir. R. Schroën	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in an extreme situation, with respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on the integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end, the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 1 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical and architectural solutions: the student is able to apply, reflect and transform principles concerning climate, construction, structure and architecture.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. If products are missing and the project is of sufficient quality, the teacher can offer the student a repair. For the repair, the student has to produce the missing products within two weeks after the original presentation. The products have to be uploaded to Brightspace, there will not be a new presentation. The result of the repair can be a V (=completed) or F (=fail).	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely.	
Period of Education	Semester 1, Q1 & 2	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc2 AE

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.	
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	

Course Contents

This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment.

In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences.

Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.

Study Goals

By the end of this course students will be able to:
1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective;
2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods;
3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media;
4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis;
5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences.
6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.

Education Method

The course Architectural Ethnography comprises group assignments and individual work.
The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course.

The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.

Literature and Study Materials

The course will use the following publications as main textbook references:

Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts.

Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto

Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126.

Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE.

Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.

Assessment

The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects.
The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment.
The summative assessment will be based on the following deliverables:

a) Analytical Assignment:
Ethnographic Research: Research Report (Group Work)

b) Practical Exercises:
Poster: Synthesis of (Quantitative) Data Collection (Group Work)
Ethnographic Research: Narrative and Visual Synthesis (Group Work)

c) Writing Assignment:
Critical Reflection (Individual Work)

For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c).

The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.

Retake and repair regulations
In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year.
Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Pereira Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Pereira Roders	
Co-responsible for assignments	Y. Zhang	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.	
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.	
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Book chapters, journal articles and other lecture materials.	
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5

Course Coordinator	Prof.dr.ing. C.M. Hein
Course Coordinator	J.M.K.K. Hanna
Instructor	Dr. P. De Martino
Instructor	J.M.K.K. Hanna
Responsible for assignments	Prof.dr.ing. C.M. Hein
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	3rd Quarter
Concept Schedule	Friday
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5

Course Coordinator	Dr. B. Lubelli
Instructor	Dr. B. Lubelli
Instructor	S. Naldini
Instructor	T.P. Bennebroek
Responsible for assignments	Dr. B. Lubelli
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.
Period of Education	Q3
Concept Schedule	Wednesday afternoon
Minimum number of participants	15
Maximum number of participants	60
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarions	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

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Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.	
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.	
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:	

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15

Course Coordinator	Dr. L. Cipriani
Responsible for assignments	Dr. L. Cipriani
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.
Concept Schedule	Wednesday, Friday
Minimum number of participants	15
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Prof.dr.ing. U. Pottgiesser	
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Prof.ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.	
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.	
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:	

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15

Course Coordinator	S. Corbo
Course Coordinator	Prof.ir. N.A. de Vries
Course Coordinator	Ir. A.M.F. van Dam
Instructor	Ir. A.M.F. van Dam
Instructor	S. Corbo
Responsible for assignments	Ir. A.M.F. van Dam
Contact Hours / Week x/x/x/x	112 hours per quarter
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more. The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.	
Study Goals	By the end of the MSc2 Studio, students should be able to: Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural) Visualise key design and data-related features using effective narrative and visual storytelling methods	
Education Method	The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include: Design tutorials (including individual design progress feedback and group discussions) Seminars (including lectures, workshops and reading discussions) Excursion (study trip including the site visit)	
Course Relations	Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues? The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas: CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives. ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort. RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges. Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.	
Literature and Study Materials	Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.	
Assessment	Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include: Posters (including architectural drawings and other visuals) Booklets and/or digital presentations Short animations (such as GIF animations) Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.	

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.	

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 AE

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AE100	Architectural Engineering Graduation Studio	55

Course Coordinator	Ir. M.J. Smit
Course Coordinator	A. Snijders
Instructor	Ir. S.H. Verkuijlen
Responsible for assignments	A. Snijders
Education Period	1 2 3 4
Start Education	1 3
Exam Period	none
Course Language	English
Course Contents	Climate change, resource depletion, urbanization, pandemics and economic instability face architects with new challenges; the need for carbon neutral buildings and nature inclusive neighborhoods, the lack of affordable housing, changing requirements for the existing building stock and infrastructure, etc. How can our living environment be improved whilst dealing with these challenges? What are suitable technical innovations and how can they be integrated within the architectural design? And how can transitional design and engineering strategies be organized in a socially responsible and equitable way? Specific focus areas of aE Studio are amongst others circularity, climate design, urban ecology, digital / robotic manufacturing, product design, material research, building physics, structural mechanics and computational modelling. Architectural Engineering encourages students to explore their role as architects in facing todays challenges. Under the guidance of a team of enthusiastic design tutors and together with various stakeholders, students search for innovative technical solutions for diverse problems in various contexts in both the Netherlands and abroad. The three main research by design domains promoted by Architectural Engineering are Make, Flow and Stock. Each domain requires a different approach and relates to very diverse research and design questions. As for example: o How to develop new (open)building systems and methods for reusable and renewable materials? o How to design demountable buildings and reusable building components for a circular economy? o How to integrate digital manufacturing techniques in the design process of buildings? o How to integrate energy production and smart management in the built environment? o How to (re)design the existing building stock in a sustainable way? o How to responsibly integrate infrastructure in the built environment? o How to guarantee nature inclusiveness within the architectural design? o How to integrate water, waste- and food flows in architectural design? o How to design healthy living conditions (f.e. dwellings) in an increasingly polluted environment?
Study Goals	AE graduating students achieve the following study goals: o Students are able to set up an accurate project definition (problem definition, objectives, research questions, methods, planning, etc.). o Students are able to incorporate a relevant technical research topic in their overall design question. o Students are able to incorporate specific technical knowledge in their architectural design. o Students develop skills in architectural design satisfying aesthetic as well as technical and functional requirements. o Students develop insights in and knowledge of the design process with regard to methods for research and design o Students are able to clearly and coherently present the project through the use of adequate means. o Students are able to reflect upon their research and design process and the related products. o Students are able to position themselves in the field of architecture and understand which role they can have within society as an architect. During the master 3 & 4 the complexity of the architectural design increases, leading to an optimal level required for architectural practice. The graduation report (combination of graduation products) demonstrates the students ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding architectural ethics and his/her future role as architect. The graduation products (research plan - AR3A010, research paper, reflection paper, design output, etc) should not only contain an elaboration regarding the graduation projects societal and disciplinary relevance, but also have to address design ethics and the way in which intercultural issues were addressed in the graduation project.
Education Method	aE Studio is characterized by the strong connection between technical research and architectural design. Within the MSc3 students will be guided by an architecture (first tutor) and a research tutor (third tutor), whom they will meet every week to discuss the progress of their projects. A diverse network of internal and external experts guide our students as research tutors. Within the MSc4 a building technology tutor will be added to the team. Within the Architectural Engineering Graduation Studio students define their individual graduation assignment, related to different selected contexts, societal challenges and technical topics. Based on various thematic assignments aE Studio organizes collective table meetings and presentations to discuss and reflect on each others projects.
Assessment	Within the MSc3 examination takes place through P1 (midterm) and P2 (final) presentations. Deliverables at the P2 are a graduation plan (for details see the Graduation Manual), research plan (part of AR3A010 course) and the aE Studio Research paper, which should at least have a grade 6 at the P2 for a Go towards the MSc4. After successfully finishing the MSc3 studio, the student will enroll in the MSc4 studio devoted to the elaboration of the design project. Within the MSc4 examination takes place through P3-, P4- and P5-presentations and a written Reflection Paper (see Graduation Manual).
Period of Education	MSc3 + MSc4
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

A&PB

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 Architecture and Public Building

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1AP012	Public Building Design Studio	10
Course Coordinator	S. Lee	
Instructor	S. Lee	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	The MSc. 1 Studio provides an introduction to the design of public buildings in the twenty-first century. It explores the design of hybrid buildings and sustainable public architecture and aims to address fundamental questions including: - What constitutes public architecture in the twenty-first century? - What role does public architecture play in global cities? - How can architecture promote publicness, inclusion and diversity? The Studio examines the significance of public architecture in the context of the broader social and cultural milieus. It encourages design processes and proposals that imagine innovative and diverse models of symbiosis among design intent, construction, users, and stakeholders. In addition, it encourages the exploration of hybridity to replace the limitations of monofunctional buildings and neighborhoods. Each year, the Studio is set in a different European city, exploring the unique site conditions as a foundation for the design of a medium-sized hybrid public building (approximately 3.000-5.000 square meters of floor area). The resulting design should act as a catalyst for new urban dynamics, featuring areas for leisure, contemporary media and cultures, education, and innovation. By designing a building that serves the diverse purposes and needs of the given community, the Studio advocates for vibrant and sustainable urban environments. The MSc. 1 Studio operates in close collaboration with the Building Engineering Studio (BES) to promote new sustainable designs, materials, and construction. The BES instructors take responsibility for the technical, structural, and performative aspects of the design exercise, implementing a collaborative educational agenda focused on sustainability, energy efficiency, designing for disassembly, and adaptability to climate change. The BES instructors prioritize specific materials, emphasize CO2-neutral and net-zero energy buildings, and reflect a circular economy. By working together, instructors from Architecture and BES aim to develop designs that are both innovative and environmentally responsible.	
Study Goals	Upon completion of the MSc 1 Studio, the students will be able to: present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects for a medium sized building on MSC 1 level; develop a design position that considers specific urban conditions and reflects on a coherent design strategy; relate the assigned functional program to a coherent architectural concept and to construction as well; grasp the complex and heterogeneous potential of the public realm through multiplicity, density, and hybridization; introduce properties that make architectural interventions less singular in function, but more productive, more transformative and resilient; ensure a design outcome where aesthetic and technical aspects are explicitly integrated and mutually related; design an innovative architecture that can improve adaptability to climate change and productivity of technical solutions, materials and building physics; represent space in its complex interpenetration of people, architectures, technologies, and materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Architectural Design Studio (AR1AP012) and the Building Engineering Studio (AR1A080) are fully integrated to ensure the comprehensive development of each proposal. By combining these two areas of study, students will be equipped to translate their design concepts into practical, buildable solutions. The Studio runs different activities, including: lectures public discussions field investigations / site-visits design-oriented assignments in drawing and modelling readings and writings	
Course Relations	AR1A080 Building Engineering Studio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, physical models, material samples, etc.; 3. Written statements and analyses. The final design proposal is individual. Architecture and Building Technology will provide respective grades. The assessment process consists of weekly in-class discussions of progress and the midterm and final reviews. The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Remarks	The Architectural Design Studio (AR1AP012) and the Building Engineering Studio (AR1A080) are integrated and form a coherent whole.
Period of Education	Fall Semester (Q1-Q2)
Concept Schedule	Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 AP

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.	
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or	

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5

Course Coordinator	Prof.dr. A.R. Pereira Roders
Instructor	Dr. I. Nevzgodin
Instructor	Ir. A.C. de Ridder
Responsible for assignments	Prof.dr. A.R. Pereira Roders
Co-responsible for assignments	Y. Zhang
Contact Hours / Week x/x/x/x	4 hours per week
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.
Literature and Study Materials	Book chapters, journal articles and other lecture materials.
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%
Period of Education	Q 3
Concept Schedule	Wednesday morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5

Course Coordinator	Prof.dr.ing. C.M. Hein
Course Coordinator	J.M.K.K. Hanna
Instructor	Dr. P. De Martino
Instructor	J.M.K.K. Hanna
Responsible for assignments	Prof.dr.ing. C.M. Hein
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	3rd Quarter
Concept Schedule	Friday
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar 5
Course Coordinator	Dr.ir. H. Sohn
Instructor	Dr.ir. A. Radman
Instructor	Dr.ir. S. Kousoulas
Responsible for assignments	Dr.ir. H. Sohn
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.
Elective	Yes
Period of Education	This course is taught only in Q3
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15

Course Coordinator	Dr.ir. M.G.A.D. Harteveld
Course Coordinator	Dr.ir. R. Cavallo
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld
Responsible for assignments	Dr.ir. R. Cavallo
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15

Course Coordinator: T. Maso

Instructor: T. Maso

Responsible for assignments: J. Arpa Fernandez

Contact Hours / Week x/x/x/x: 8 hours per week starting from week 4.1 and ending in week 4.10

Education Period: 4

Start Education: 4

Exam Period: none

Course Language: English

Expected prior knowledge: It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).

Summary

The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future.

Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions.

The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means.
A systematic, parametric exploration of parts of the design is an integral part of the research approach.

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO.

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

For more information about our previous studios, please visit:
https://thewhyfactory.com
https://thewhyfactory.com/education/

Course Contents

MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda.

During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)

Study Goals

- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.

- Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal.

There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios:

1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester.

2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc..

3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration.

4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors.

5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development.

6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.

Education Method

Number of studio hours: 80
Number of self study hours: 332

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program.

Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality.

Applications Program:In the applications program model cities both are tested in real cities. The different models become

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday, Friday	
Minimum number of participants	15	
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Prof.dr.ing. U. Pottgiesser	
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Prof.ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.	
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.	
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:	

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	**Van Gezel tot Meester**	**20**
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15

Course Coordinator	N.J. Amorim Mota
Instructor	Prof.ir. D.E. van Gameren
Instructor	N.J. Amorim Mota
Instructor	R. Varma
Responsible for assignments	N.J. Amorim Mota
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	

Course Contents

The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof.
Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent.
How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined?

The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers.

The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.

Study Goals

Upon completion of the Design Studio, the students will be able to:

convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level;
perform critical comparative research that results in a clearly formulated design hypothesis;
demonstrate how urgent societal issues are addressed in the design project;
demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal;
represent space in its complex interpenetration of people, architectures, technologies, materials;
confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.

Education Method

The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive.

The Studio will be structured across:
lectures
readings, writings and public discussion
design tutorials / workshops
specific exercises challenging innovative thinking

Literature and Study Materials

Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.

Assessment

Assessment of the students work will be based on the following:
Oral presentation;
Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.;
Writing assignments and analyses.

Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student.

The final grade reflects:
the position that is formulated with regard to the brief and its context;
the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation;
aesthetic and technical/functional qualities; the elaboration throughout the respective scales;
the quality of the presentation, the products and the argument;
the consistency, coherence and development of the students work during his/her process.

Products: will be described in the syllabus which will be published at the beginning of the course.

Repair regulations
The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5).

Retake regulations
Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).	
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.	
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.	

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 Architecture and Public Building

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AP100	Public Building Graduation Studio	55
Course Coordinator	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Lee	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. H.J. Bultstra	
Instructor	Prof.ir. N.A. de Vries	
Instructor	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	X/X/X/X	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Public Building Group investigates the future of public buildings, public spaces, and their role in the built environment. This requires developing new types of places, buildings and building elements, re-inventing past structures and questioning existing typologies through research and design as well as research by design. The Graduation Studio aims to produce future proof designs that are sustainable, and investigate the possibilities of design thinking in a world where the definition of what an architect is and does, ceaselessly shifts. Architectural designs will be made for culture, health, education, work and leisure influenced by contemporary contexts, sites and conditions. Each year, the studio focuses on a specific, and for the above representative, topic or combination of topics. Research-by-Design will occupy the core to the students individual development of the studio project throughout the MSc3 and MSc4 periods. The design of the project starts with conceptual frameworks, driven by ambitions against which the ultimate result has to be tested. The student will formulate these ambitions early in the process in an individual Design Manifesto. Between concept and design lies a creative process augmented by research, with design and research as intertwined and often indistinguishable acts. Design and research simultaneously mean creating design loops, variants, reiterations, intersections, impressions, and still progressing. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities. Public architecture should respond to and accommodate todays needs while anticipating the future. Within the course of research by design, multiplicity provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Building Technology will be treated as an inseparable part of this process.	
Study Goals	Upon completion of the course, students will be able to: - to present a coherent, significant, elaborated, correct and innovative design of a (hybrid) public building, complex or space - on main issues and on aspects to a level fit for the architectural practice; - to comprehend design processes in relation to architectural history and architectural theory, art, building technology, human sciences, environmental aspects and sustainability, digitalization, climate control and the materialization of buildings and interiors; - to become familiar with a variety of tools and media: advanced forms of architectural visualization and representation, AI, GIS mapping, augmented / virtual reality, digital fabrication.	
Education Method	As a response to ever-changing political and societal needs, public buildings evolve and require new spatial solutions. Research by Design allows to investigate and envision such solutions. Through a series of individual and collective assignments, students will learn to reflect on the needs of contemporary societies, to develop architectural positions and to translate their concepts into buildable spatial solutions. The Graduation Studio is organized across literature reviews, tutorials, site visits, and pre-design analysis, leading to schematic design. The Graduation Studio takes two semesters of full-time study. Semester 1 consists of Research by Design that merge Project Design, Theory and Delineation Research. They relate to analysis and collection of data, pre-design, site studies, and schematic design. As a part of Project Design, tutorials on Building Technology will be included. Building on the preceding semester, Semester 2 includes design development, materialization, and building technologies, emphasizing sustainability. Theory and Delineation Research A constitutive and integrated component of the Graduation Studio, Theory and Delineation Research is a territory of experimentation: it consists of themes, discourses, typologies, that encourage coherent design thinking and reflection. Theory and Delineation [TD] aims to facilitate the Project Design [PD] work in various ways and to enrich the outcome of the design process. TD examines a variety of literature in science, history, architecture, and related disciplines in combination with hands-on practice. TD also provides the backbone of the required MSc 3 course Research Plan (AR3A010), leading to the completion of the Graduation Plan required for the P2 at the end of the fall semester. The TD sequence provides a more precise and relevant focus on the rubric Research by Design. As part of the Studios work, each student should maintain the specificity of his/her own individual motives, interests, and approaches. TD has been therefore conceived as a journey that will gradually introduce the students to a variety of techniques and media, both analog and digital. Such media are not only intended as representational or explanatory devices. On the contrary, coherently with the nature of a Research by Design approach, emphasis will be placed on the generative character of such media their ability to act as vectors for new and unforeseen scenarios. TD addresses various topics from ideation to materialization in visionary and canonical projects. While in the first weeks of the program conventional media will be explored sketches, drawings, models , in the second block students will be introduced to more advanced media in order to develop their conceptual and spatial narratives: 3D modelling, diagramming, data visualization, scripting.	
Course Relations	AR3A010 Research Plan	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will take place during the Design Examinations, in the form of: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Writing assignments and analyses: Research-by-Design Journal.	

	Criteria are: a. Ability to identify a relevant topic vis-á-vis the design assignment; b. Ability to research the topic both from a present-day point of view as from a historical perspective; c. Ability to identify and interpret relevant information or source material and use it to support the development of the design assignment; d. Ability to experiment, test and project concepts, processes, forms and materials demonstrating technical competence; e. Ability to self-reflect on the personal motivations to work on the assignment, capacity to elaborate a realistic planning, and demonstration of the awareness to evaluate his/her own strengths and weaknesses; f. Ability to create meaningful oral, written and visual communication, prepared using appropriate conventions and media; g. Demonstration of a critical attitude to the design assignment and ability to work collaboratively and independently within the main standards of the architecture profession. For the final grade compound, we refer to the Graduation Manual, Master of Science Architecture, Urbanism and Building Sciences, Academic Year 2023-2024. The final grade is the average or may deviate from the average depending on the extent to which the whole does (or not) exceed the sum of its parts, or due other exceptional qualities of the work.
Special Information	The Public Building MSc Graduation Studio will require one excursion outside of the Netherlands.
Period of Education	Two semesters (Q1-Q2-Q3-Q4)
Concept Schedule	This course will be given on Thursday. Global setup in phases: Phase 1 (in semester 1) up to P1 The foundation for the individual project. In small groups the overall framework of the assignment is analysed through site-research and a site visit. Through individual assignments focused on a research-by-design approach, students will progressively build a thematic and conceptual framework that will drive their design explorations. Individually, students work towards (a) Spatial Diagram(s) that explain(s) the relations between program, site specific conditions, technical possibilities and so on, within the individual design framework. The work is conducted and guided in Project Design, Theory Research, Delineation Research, and Building Technology Research to be thoroughly compiled in the Research by Design Journal. Result (individual): P1 Presentation and Research by Design Journal Phase 2 (in semester 1) up to P2 The Spatial Diagram is developed to architectural concept design The Concept Design for the individual project is made and elaborated upon. The concept design shows the architectural potential and the ambition of the project within the boundaries of the individual design framework. It offers possibilities that can be assessed and shows the direction in which the design can be developed. The Concept Design is a result of, or fed by the research by design. Students produce their Graduation Plan with the design objectives for Msc4. Result (individual): P2 presentation of Concept Design, Research by Design Journal, Graduation Plan Phase 3 (in semester 2) up to P3 From Concept Design to Preliminary Design Phase 4 (in semester 2) up to P4 & P5 Final Design Awarded Diploma
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

CP

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 CP

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1CP011	Complex Projects Design Studio	10
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Ir. H.A. van Bennekom	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1 and ending in week 2.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	MSc 2	
Expected prior knowledge	BSc completed	
Course Contents	As an introduction to Complex Projects, the MSc1 design studio called 'Dutch Change' focuses on the Netherlands ground. By getting familiar with the Dutch environment, the MSc students will learn to understand the complex system behind buildings and establish a connection between building design and social changes. During the design process, students will explore new architectural concepts and building types and they will build up a narrative supporting the architectural design proposal. By identifying and making decisions on the balance between what changes and what remains in Dutch architecture, students will be challenged to explore both the tradition of Dutch planning and building culture. In the end, students will do an architectural design project in a Dutch city. The assignment starts from a broader understanding of the physical context, its past and future planning, environmental issues and climate adaptation on the one hand and an in-depth understanding of the building typological developments on the other hand. In fall 2024, the 'Dutch Change' studio focuses on the architecture of buildings for science and learning in the context of the Dutch Delta. Site visits to examplary educational buildings in Europe are included in the program.	
Study Goals	Upon completion of the MSc1 Complex Projects studio trajectory: - The student will be able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects from a medium-sized building on MSc 1 level. - The student will be able to be aware of the connection between history and urban context and use architectural instruments to respond to new challenges. - The students will be able to work both with a group and independently, to coordinate different opinion and formalize an efficient design process.	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology, and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relationship between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, the materialization of buildings, comfort, and climate control.	
Education Method	Tutorial once a week. Tutorials in group, supported by lectures, field trip, self study and group work. Students will be working both in groups and individually or in pairs. The individual assignment is a medium-sized building. Building Engineering Studio is integrated.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Books	Recommended Literature: Wagenaar, Cor (2011) Town Planning in the Netherlands since 1800: Responses to Enlightenment. Ideas and Geopolitical Realities, Rotterdam: 010 Publishers Barbieri, U., van Duin, L. (2003) Hundred Years Of Dutch Architecture, A: Trends, Highlights, SUN NAi Publishers. Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to the start of the studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The proportion of different assessments will be described in the syllabus. Marks will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for	

	evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Marking period within 10 working days
Remarks	The Architecture Design Studio (AR1CP011) and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely.
Period of Education	Quarter 1-2 (fall semester)
Concept Schedule	Tuesdays morning
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 CP

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.	
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or	

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5

Course Coordinator	Prof.dr. A.R. Pereira Roders
Instructor	Dr. I. Nevzgodin
Instructor	Ir. A.C. de Ridder
Responsible for assignments	Prof.dr. A.R. Pereira Roders
Co-responsible for assignments	Y. Zhang
Contact Hours / Week x/x/x/x	4 hours per week
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.
Literature and Study Materials	Book chapters, journal articles and other lecture materials.
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%
Period of Education	Q 3
Concept Schedule	Wednesday morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

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Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.	
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.	
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:	

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15

Course Coordinator	T. Maso
Instructor	T. Maso
Responsible for assignments	J. Arpa Fernandez
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday, Friday	
Minimum number of participants	15	
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15

Course Coordinator	N.J. Amorim Mota
Instructor	Prof.ir. D.E. van Gameren
Instructor	N.J. Amorim Mota
Instructor	R. Varma
Responsible for assignments	N.J. Amorim Mota
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.	

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	Urban Archipelago	15
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).	
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.	
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.	

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.	
Course Contents	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.	
Study Goals	After completion of the course the student is able to: Overarching study goals - Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment. - Assess the role of entrepreneurship in changes in the architectural world and the built environment. - Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge. - Critically evaluate and reflect on entrepreneurial skills and plans. - Design and deliver a compelling and succinct vision and business pitch. Additional study goals for Variant: Develop your architectural firm (open to A en BT students) - Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio. - Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network. - Develop a design vision to guide the future direction of their own design firm. - Reveal the design vision in an actual design. - Propose the next steps for establishing a self-owned design firm. Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students) - Assess building sector organisations, including their business models, revenue streams, and network. - Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities. - Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues. - Defend their business plan in a convincing matter to industry experts and utilise expert input to improve. - Architecture students will also develop a design as part of the impactful business model - Propose the next steps for implementing the business model in an existing or new venture.	
Education Method	The course's learning activities comprise: - lectures: theory - workshops - self-study: developing vision and translation into entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop vision and entrepreneurial plan	
Assessment	Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may	

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 CP

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Choose one Graduation Studio

AR3CP100	Complex Projects Graduation Studio	55

Course Coordinator	H. Smidihen
Course Coordinator	Prof.ir. C.H.C.F. Kaan
Instructor	Dr. O. Caso
Instructor	Prof.ir. C.H.C.F. Kaan
Instructor	M. Triggianese
Instructor	H. Smidihen
Responsible for assignments	Dr. O. Caso
Education Period	1 2 3 4
Start Education	1 3
Exam Period	none
Course Language	English
Course Contents	In the graduation studio, all elements of the building and design will be seamlessly integrated into a clear and coherent project that touches on all the design challenges, from analysis and problem statement, through building concept and urban implementation, all the way to materialisation and critical details. The assignment for the MSc3 CP Graduation Studio will be to develop a design brief: a clear design assignment based on the narrative generated by the research of globally relevant topics, and analyses of given urban contexts. During the MSc4, the design brief will be answered by an architectural proposal which will be developed into a clear and comprehensive architectural design to scale 1:5. The architectural design proposal needs to have a strong relation with the research and urban strategy made during MSc3.
Study Goals	The complex projects graduation programme consists of two connected semesters: MSc3 & MSc4. Successfully completed courses from previous semesters, (MSc1 & MSc2) are a precondition for the successful start of the graduation. In the first semesters, MSc3, the students focus on the research, design investigation, and the thorough understanding of all the elements that are rendering the given context. The final goal of the semester is to transform personal fascination and research conclusions into a clear design task. The design task should be formulated as a design brief, in which design assignment, program, site and ambition of the project are clearly stated. Upon completion of the first semester, students will be required to provide an answer to the previously defined design brief in a form of a design proposal. During the development of the design, students need to keep a strong link to the previous research, as well as to the group work and to the overall site vision. The graduation ends with the final presentation of the design proposal, linked together in a group strategy. Graduated students should develop the skills in architectural design to satisfy both aesthetic and functional requirements. Additionally, skills related to understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects will be developed. Together with the training on building technology, students will be able to incorporate an understanding of the design process with regards to structural design, materialisation of buildings, comfort and climate control. The final graduation thesis demonstrates the students ability to employ analysis, creativity, judgment, decision and argumentation skills regarding architectural ethics and the students future role as an architect. The individual graduation thesis should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which intercultural issues were addressed in the graduation project.
Education Method	The main methodology in Complex Projects is to develop each students organised and structured approach to the design process. Students are confronted with the very rigorously defined design schedule, and at the same time rather complex and demanding design assignment. By following the given structure, students will learn how to deal with complex design assignments, how to process and organise a large amount of data, how to meet deadlines and prioritise workflow. Work process, consistency & quality of weekly development will have influence on the final result. Students will be encouraged to record progress and development of the work and use it in the final narrative as well as in the argumentation of the design. Product based design development is a design method being used in Complex Projects. By having strictly defined deliverables, within the dense schedule, students are encouraged to constantly produce whilst developing a design in parallel. In this way, the design progress and the final products are much more integrated, but even more importantly deliverables are used as a mean to develop, explore and test the design. The goal is to continually refine the methods of representing, presenting and communicating the urban and architectural work. The student should explore methods of presentation, storytelling and design documentation. Group work, collaboration and collective deliverables are strongly promoted in Complex Projects. Through practising group work, students learn how to discuss, how to divide work and how to make compromises. Development of communication skills, constant usage of different communication mediums (sketches, models, presentations, reports...) is becoming more and more important in the architectural profession. Furthermore, constant communication, updates and sharing of information are becoming a medium to develop and progress design. One of the central group deliverables is a collective research book - a medium to develop a research part of the graduation. This book will be continuously updated throughout the year and will show the creative and analytic process of the group. Next to the research, students will be encouraged to investigate spatial aspects of the design questions through making large urban models. Those models will be used throughout the whole graduation to facilitate discussions, group presentations and design ideas.
Assessment	The assessment of the studio will be based on the individual performance of each student combined with involvement and result of the group work. The students performance will be determined by the quality of his or her work, their commitment, effort and improvement over the entire course of the semester. The final evaluation will be based on the performance throughout the whole year as well as the final result. During the first MSc3 semester of graduation, there will be in total four presentation moments. The first presentation (P0.5) is an informal group presentation, where initial findings regarding graduation assignments are shared, presented and discussed. The second assessment (P1.0) is a formal presentation where students have to present their individual graduation plan and their research topic. The third presentation (P1.5) is informal, which is focusing on group work and development of the links between individual work of students, connecting them into the group strategy. The first semester ends with P2.0, a formal presentation which is go-no go review where the students continuation to MSc4 is decided upon. The evaluation of the result will be based on the performance of the student throughout the semester, the collaborative group work and the final presentation. In the final presentation, the design assignment needs to be presented in the form of a design brief, with a clearly formulated design question and a determined program, size and location for the project. After completion of the first semester, students will be required to continue developing their project further through the Msc4, and focus on the structure, materialization, comfort and climate control of the building. During MSc4, there will also be four presentation moments. The first presentation (P2.5) has and informal character and focuses mostly on presentation of the urban strategy and concept of the design. The second presentation (P3) is when students are asked to present preliminary design of their proposals. In this moment functional and programmatic aspects of the design should be developed and frozen. P4 is the main presentation of MSc4, where students receive another go-no go. In P4 a complete design should be presented, including

	materialization, structure, climate and façade design. After P4, students have several weeks to polish the design, create visuals and models for final presentation, P5.
Period of Education	Complex Projects graduation consists of the two semesters. First semester, MSc3 is focusing on the research, analyses of the context, and the search for the thesis topic. Second semester, MSc4 is more dealing with the design, and providing design answers to created tasks in MSc3. Both semesters are divided in four phases, each of them lasting for five weeks. At the end of each phase the group pin-up, and individual presentation are organized. Complex Projects graduation starts in both the fall and spring semester.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR3CP120	Lunar Architecture and Infrastructure Graduation Studio	55

Course Coordinator	Dr. Dipl.-Ing. H.H. Bier
Instructor	Dr. Dipl.-Ing. H.H. Bier
Instructor	Ir. F. Adema
Instructor	A.J. Hidding
Instructor	L. Peternel
Instructor	J.M. Prendergast
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier
Contact Hours / Week x/x/x/x	8 per week
Education Period	1 2 3 4
Start Education	1
Exam Period	none
Course Language	English
Course Contents	Mirroring the interdisciplinary nature of space colonization projects where architects, roboticists, and space scientists work together to achieve common goals, this new inter-faculties graduation course, Lunar Architecture and Infrastructure (LA&I), is designed to foster collaboration and shared objectives among students from various faculties. It is open to all graduating TU Delft students interested in addressing challenges of space colonization. To survive in outer space, humans need protection from radiation. Such protection requires architecture and infrastructure to ensure habitation and accommodate various activities ranging from living and working (e.g., habitat, roads), to generating energy (e.g., equipment, storage), and facilitating communication (e.g., base stations), while relying heavily on in-situ resource utilization and a high degree of automation. In this regard, the automation scale and use of robots plays an important role in minimizing the risks for humans. Robots stand out in operating within harsh environments, such as those with extreme temperatures, radiation, and low gravity, which are typically inhospitable to humans. Their capacity to perform tasks that are repetitive, dangerous, or demand high precision makes them a perfect fit for activities like excavation, construction, maintenance, and other critical off-Earth tasks. Additionally, use of robots significantly mitigates the cost and risk associated with human missions by preparing the lunar surface and infrastructure even prior to the arrival of astronauts. They can be utilized to mine local materials, construct structures, and extract essential resources like fuel, water, and oxygen, thereby facilitating human habitation in extraterrestrial environments. Lunar architecture and infrastructure are designed by computational means and are constructed by robotic means involving Robot-Robot and Human-Robot Interaction (R/ HRI) supported Design-to-Robotic-Production-Assembly and -Operation (D2RPA&O). The construction of habitats underground on the Moon offers natural protection from radiation and lesser temperature fluctuations, changing the solicitations on the structure. The construction relies on a swarm of mobile robots equipped with various end-effectors that are deployed to map the terrain and mine for materials used to 3D print building components. These are then assembled using R/ HRI-supported methods. The assembled structure is equipped with a Life Support System (LLS), which relies on D2RO processes. Both habitat construction and inhabitation are powered by an energy system involving space-based solar power. The ultimate goal is to develop an autarkic R/ HRI-supported D2RPA&O system employing In-situ Resource Utilisation (ISRU) for building habitats and to transfer developed technology to terrestrial applications in due time.
Study Goals	The LA&I graduation project is implemented in two consecutive semesters with study goals aligned with the respective stages of design. The goal in the first semester is to extract from the analysis and problem statement phase a clear design task formulated as a design brief, in which design assignment, program, site and ambition of the project are identified. While general knowledge in space colonisation will be developed in lectures with invited speakers from a.o. ESA, knowledge and skills in HRI-supported D2RPA&O will be developed in workshops. The overall goal is to develop necessary knowledge and skills to advance the design in the second semester up to materialization level by addressing the requirements formulated in the brief. In this context, students develop: knowledge and skills in architectural design to satisfy both aesthetic and functional requirements within the larger architectural history, theory, technology, and human sciences context; knowledge and skills related to understanding the design process concerning the relationship between natural and constructed spaces and societal needs, including sustainability requirements; knowledge and skills in computational design, robotic construction, and building technology; ability to understand the design process inclusive of structural and materialization requirements for buildings with integrated climate control; competence to employ analysis, judgment, creativity, innovativeness, decision, and argumentation skills regarding architectural ethics and the future role as an architect; competence to identify the societal and disciplinary relevance of their project and address, where relevant, intercultural issues; ability to present adequately and coherently significant results that are correct, thorough, and complete with all relevant aspects elaborated in greater depth; ability to present with a degree of clarity, intelligibility, and reflection.
Education Method	The pedagogical methodology relies on a problem-based learning approach that involves defining problem(s), brainstorming, structuring, hypothesizing, and synthesizing. This implies that students identify what they already know, what they need to know, and how and where to access new information that may lead to the resolution of the design and engineering problem(s). With this approach, students learn how to deal with complex design assignments, process and organize large amounts of data, meet deadlines and prioritize milestones. Collaboration involving group and individual work with corresponding deliverables is strongly promoted. Through group and individual work, students learn how to discuss, divide work, and compromise while relying on personal autonomy and accountability to produce deliverables on time. Deliverables require advancing methods of representing, presenting, and communicating the architectural project using various media (sketches, models, simulations, prototypes, videos, presentations, reports, etc.). The workflow methodology relies on Design-to-Robotic-Production-Assembly and -Operation (D2RPA&O) methods that integrate computational design with robotic construction and operation of buildings. These involve the use of various Artificial Intelligence (AI) and Human/ Robot-Robot-Interaction (H/ RRI) supported computational and robotic processes that fundamentally change the architectural project, which increasingly emerges from human and non-human interaction. In this context, the H/ RRI-supported D2RPA&O process is implemented by ABE students in collaboration with students and tutors from ME, EEMCS, and AE. For instance, ABE students develop habitat designs, while ME and EEMCS students develop robots for constructing the habitat, and AE energy generation systems for constructing and operating the habitat.
Assessment	Evaluation is based on the group and individual performance of each student. The students performance will be determined by the quality of his/ her work, commitment, effort, and improvement over the entire course of each semester. In both semesters, the assessment is implemented at midterm and final review, with a pre-final or go/no go review scheduled a month before

	graduation. Examinations schedule and formal requirements are described in the Graduation Manual. Assessment is implemented by reviewing following deliverables: presentations showing analysis, concept, parametric models, and prototypes; 3-4D parametric models showing the design within the site at the phase of concept design, schematic design, design development and construction design; structure and materialization design for robotic production, assembly, and operation; from 3D model obtained sections, plans, and views at appropriate scales ranging from 1:1000 to 1:1; physical models and 1:1 prototypes developed from the 3D parametric model; photos/ videos documenting production, assembly process, and final result; digital documentation of all above including 300 words abstract describing project on the LA&I wiki (http://cpa.roboticbuilding.eu/) and TU Delft repository.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams. Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration. Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below). The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Graduation projects start in the fall semester and are implemented in two semesters, where each semester is divided into four phases, each of them lasting for five weeks. At the end of each phase, the group and individual presentations take place.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment). However, since this course is offered for the first time in 2024 there are no evaluations available.

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MAI

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 MAI

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1MET015	Situating Architecture	10
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. P.H.M. Jennen	
Instructor	J.A. Mejia Hernandez	
Instructor	A. Stanii	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	...	
Expected prior knowledge	Completed BSc	
Course Contents	Well situated buildings are evidence of architectural knowledge and skills. In a series of thematic modules you will analyze and imagine different (e.g. material, perceptual, economic, narrative) layers of a given context through concrete design tasks, leading to the individual production of a complete and well situated architectural project. Through intense design practice, you will train in the use of analytical and projective methods required to develop architectural designs that are sensibly inscribed within a particular context, understood as an entanglement (Lat. con + texere = interwoven) of natural and cultural variables. The course will be integrated with the AR1A080 Building Engineering Studio.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze a given context using selected instruments and methods of architectural analysis and imagination studied in the course. - Design a coherent, significant, elaborated, correct and innovative architectural project - on mainline and on aspects for a medium sized building on MSC 1 level.* - Develop a complete technical strategy that is integral to your project and is coherent with your analysis of the context. - Present your research findings and design proposal in clear and engaging way by using various representational tools. * Standard learning objective required for all MSc1 design studios.	
Education Method	A first part of the course develops a series of thematic modules, which present you with different tasks that require the analysis of a particular context and the design of an architectural project. Special emphasis is put on the instruments and methods used in both the analytical and the design processes. In the second part of the studio you will develop one of your designs into a complete architectural project for a medium sized building, including the constructive method required to build it. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to situate your project within its context.	
Literature and Study Materials	...	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow you to decide which of your designs to develop further. The three final outcomes (research booklet, project, and presentation) are assessed and graded individually, using the aspects and criteria determined in the EMMA feedback and assessment tool, adjusted for the MSc1 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	semester 1, Q1 & Q2	
Concept Schedule	...	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 MET

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5

Course Coordinator	Dr.ir. H. Sohn
Instructor	Dr.ir. H. Sohn
Instructor	Dr.ir. A. Radman
Instructor	Dr.ir. R.A. Gorny
Instructor	Dr.ir. S. Kousoulas
Responsible for assignments	Dr.ir. H. Sohn
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or	

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Pereira Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Pereira Roders	
Co-responsible for assignments	Y. Zhang	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.	
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.	
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Book chapters, journal articles and other lecture materials.	
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5

Course Coordinator	Dr. B. Lubelli
Instructor	Dr. B. Lubelli
Instructor	S. Naldini
Instructor	T.P. Bennebroek
Responsible for assignments	Dr. B. Lubelli
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.
Period of Education	Q3
Concept Schedule	Wednesday afternoon
Minimum number of participants	15
Maximum number of participants	60
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarions	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.	
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.	
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:	

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15

Course Coordinator	Dr.ing. M. Bilow
Responsible for assignments	Dr.ing. M. Bilow
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Special Information	The maximum marking period is 10 work days.
Period of Education	Quarter 4
Concept Schedule	Tuesday and Thursday - details may vary
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday, Friday	
Minimum number of participants	15	
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	**Van Gezel tot Meester**	**20**
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15

Course Coordinator S. Corbo

Course Coordinator Prof.ir. N.A. de Vries

Course Coordinator Ir. A.M.F. van Dam

Instructor Ir. A.M.F. van Dam

Instructor S. Corbo

Responsible for assignments Ir. A.M.F. van Dam

Contact Hours / Week x/x/x/x 112 hours per quarter

Education Period 4

Start Education 4

Exam Period 4

Course Language English

Expected prior knowledge It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).

Course Contents

In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal.

Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future.

Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.

Study Goals

Upon completion of the Design Studio, the students will be able to:

convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level;
investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions;
demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal;
focus on the qualitative aspects of multiplicity in society and design;
design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics;
confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal;
represent space in its complex interpenetration of people, architectures, technologies, materials.

Education Method

The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression.

Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive.

Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues.

The Studio will be structured across:
lectures
field trips
readings, writings and public discussion
experimental research
design tutorials
specific exercises challenging innovative thinking

In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.

Literature and Study Materials Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.

Assessment

Assessment of the students work will be based on the following:
1. Oral presentation;
2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.;
3. Written statements and analyses.

The design proposal is individual.
The assessment process implies:
in-class participation on a weekly base with public discussion of in-between results;
Midterm and Final Reviews;

The final grade reflects:
Critical analysis and the urban context (25%)
Design quality of the final proposal (55%)
Participation, collegiality, commitment (20%)

Repair regulations
The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.

The maximum marking period is 10 working days.

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15

Course Coordinator	A.E. Rout
Course Coordinator	Prof.dr. T.G. Vrachliotis
Course Coordinator	M. Mateljan
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.	

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 MofAI

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3MET105	A Matter of Scale	55
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Prof.dr.ir. K.M. Havik	
Instructor	Ir. W.W.L.M. Wilms Floet	
Instructor	Ir. P.H.M. Jennen	
Instructor	J.A. Mejia Hernandez	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	Contact Hours: 8 hours per week	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	- you can design a complete architectural project, including its technical aspects, independently. - you can communicate your architectural ideas via multiple representations, including (but not limited to) technical drawings and scale models.	
Course Contents	The quality of life in European cities is inseparable from their human scale. Through meticulous and insightful analysis and imaginative design practice you will develop a comprehensive architectural project for a European city, with particular attention to the ethical and aesthetical consequences of scaling architectural design accurately in relation to human proportions. In partnership with a local academic institution you will develop thorough analyses of (a) selected aspects of the context, and of (b) one or more buildings that contain valuable knowledge about design and construction in that context. Analytical findings will serve as basis for the formulation of a design proposal, which will be developed into a comprehensive architectural project, with special emphasis on the conditions of exploration and innovation defined by the EMMA feedback and assessment tool (see assessment, below). The conclusions of the entire process are presented as an evaluation of the instruments and methods used to achieve a human scale for your project, and its expected urban impact.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze specific layers from a chosen context using selected instruments and methods of architectural analysis and imagination. - Formulate a clear and concise design assignment for the development of a complete architectural project for that context, with special attention to its scale. - Design a comprehensive (formal, technical, functional, meaningful) architectural project as development of that proposal, expressed in a complete, consistent, and elegant set of architectural representations, prepared for an audience of experts and non-experts. - Evaluate the innovative urban impact of the resulting architectural design in ethical and aesthetical terms, with special focus on all design decisions concerned with architectural scale.	
Education Method	The studio is divided into five parts. As part of a small group, you will first develop in-depth analyses of a key aspect of the context, and of a building that contains valuable disciplinary knowledge to design and build in that context. To collect empirical evidence and carry out the necessary fieldwork you are required to participate in a studio excursion. The results of all analyses will be edited and printed into a booklet produced by the studio as a whole. In the second part of the studio you will start working individually (and continue doing so onwards), to formulate a clear and concise design proposal consisting of (a) the selection of a concrete site or situation for your project, (b) a justification for its chosen use and purpose, and (c) a first attempt to determine its overall form. In the third and fourth part of the studio all formal, functional, and technological aspects of your project are developed thoroughly, through different projective methods and multiple representations, and with special attention to scale. The fifth and final part of the course focuses on communicating the complete and final outcome of your work, including an evaluation of the instruments and methods used to design and scale your project, and its expected urban impact.	
Assessment	The analyses (P1) and designs (P3) produced in the first and third parts of the studio are subject to group and individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your design assignment (P2) and final designs (P4), produced in parts two and four of the studio, are assessed and graded with go/no-go decisions. Your final presentation (P5), at the end of the fifth and last part of the course, is assessed and graded with a final mark for the whole course. All reviews and assessments will be based on the aspects and criteria determined in the EMMA feedback and assessment tool. Related to the research carried out by the chair of Methods of Analysis and Imagination, particular attention is given to the criteria exploration and innovativeness, as defined in the tools glossary of terms, and in relation to the different aspects that are evaluated in your research and design.	
Period of Education	One academic year.	
Concept Schedule	Tuesday, Friday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

HA

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 HA

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1AH010	**Heritage and Architecture Design Studio: Architectonic Design**	**10**
Course Coordinator	Ir. W. Willers	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W. Willers	
Contact Hours / Week x/x/x/x	4-8 hours per week starting from week 1.1 and ending in week 2.8.	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The object of this Heritage & Architecture studio is the architectural design for the re-use of a building or building-ensemble to meet requirements of contemporary and future use. A transformation framework will be made by the interpretation of the analysis of the urban context, the building and the program requirements. Various aspects of designing in existing built structures are investigated by studying reference projects and literature. By working on different scale-levels a coherent design will be made. At atelier meetings different aspects like relation existing new, urban context, re-use and sustainability, functionality, spatial quality, technical aspects, material aspects will be discussed. The current debate of transformation and intervention with topics like authenticity, sustainability, layers of history, and so on is very present during this course on every single scale. Different presentations will help students to develop their presentation skills. There is a strong relation between the H&A Architectural design studio and the Building Engineering studio.	
Study Goals	Upon completion of the Master 1 design project the student is able to: - produce an urban, context-, architectural-, ecological and building technical analysis. - interpret a given value assessment, interpret the project brief and formulate a transformation framework - translate the transformation framework to an intervention strategy - explore and select a scenario combining contextual, architectural, ecological and technical qualities, cultural values and program requirements - create an elaborated design that shows coherence between the Analysis- Value Assessment- Transformation Framework- Designconcept - create an elaborated and innovative design that shows coherence, significance and correctness in the urban, ecological, spatial, functional and material aspects. - reflect on her/ his position in designing with Heritage - explain and present the design as a result of the study - explain the design clearly; graphically and orally - reflect on the design process and the design product - relate the design proposal to a wider discourse	
Education Method	Design coaching, 4- 8 hours per studio during education weeks, total 96 hours	
Assessment	Assessment wil be based on the Emma assessment tool for MSc1 studios. Detailed information will be published in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6. A student has the possibility to take a repair. Repair has to be completed within 2 weeks after completion of the studio. The grading is based on the EMMA Assessment (grading)	
Special Information	The visit to the building, the subject of the assignment, is mandatory.	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios. The visit to the building, the subject of the assignment, is mandatory.	
Period of Education	Semester 1, Q1 & 2	
Minimum number of participants	the minimum number of participants is 12	
Maximum number of participants	the maximum number of participants is 60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 AH (nw)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5

Course Coordinator	Dr.ir. H. Sohn
Instructor	Dr.ir. H. Sohn
Instructor	Dr.ir. A. Radman
Instructor	Dr.ir. R.A. Gorny
Instructor	Dr.ir. S. Kousoulas
Responsible for assignments	Dr.ir. H. Sohn
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or	

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5

Course Coordinator	Prof.dr. A.R. Pereira Roders
Instructor	Dr. I. Nevzgodin
Instructor	Ir. A.C. de Ridder
Responsible for assignments	Prof.dr. A.R. Pereira Roders
Co-responsible for assignments	Y. Zhang
Contact Hours / Week x/x/x/x	4 hours per week
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.
Literature and Study Materials	Book chapters, journal articles and other lecture materials.
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%
Period of Education	Q 3
Concept Schedule	Wednesday morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5

Course Coordinator	Prof.dr.ing. C.M. Hein
Course Coordinator	J.M.K.K. Hanna
Instructor	Dr. P. De Martino
Instructor	J.M.K.K. Hanna
Responsible for assignments	Prof.dr.ing. C.M. Hein
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	3rd Quarter
Concept Schedule	Friday
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15

Course Coordinator	Dr.ir. M.G.A.D. Harteveld
Course Coordinator	Dr.ir. R. Cavallo
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld
Responsible for assignments	Dr.ir. R. Cavallo
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	

Summary

The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future.

Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions.

The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means.
A systematic, parametric exploration of parts of the design is an integral part of the research approach.

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO.

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

For more information about our previous studios, please visit:
https://thewhyfactory.com
https://thewhyfactory.com/education/

Course Contents

MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda.

During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)

Study Goals

- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.

- Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal.

There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios:

1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester.

2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc..

3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration.

4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors.

5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development.

6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.

Education Method

Number of studio hours: 80
Number of self study hours: 332

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program.

Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality.

Applications Program:In the applications program model cities both are tested in real cities. The different models become

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15

Course Coordinator	Dr.ing. M. Bilow
Responsible for assignments	Dr.ing. M. Bilow
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Special Information	The maximum marking period is 10 work days.
Period of Education	Quarter 4
Concept Schedule	Tuesday and Thursday - details may vary
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15

Course Coordinator	Dr. L. Cipriani
Responsible for assignments	Dr. L. Cipriani
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.
Concept Schedule	Wednesday, Friday
Minimum number of participants	15
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	**Van Gezel tot Meester**	**20**
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	

Course Contents

This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change.

To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.

Study Goals

Upon completion of the design studio the student:
1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment.
2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition.
3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals.
4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale.
5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community.
6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project.
7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context.
8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.
9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.

Education Method

The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio.

In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts.

In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.

Assessment

Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase.
The summative feedback will be based on the assessment of the process of each individual student and an individual dossier.
The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases:

PHASE 1_Foundations
Assignment 1. Fieldwork / Contextual Research (group of 4-5 students).
Assignment 2. Problem Statement and Design Manifesto (Individual work).

PHASE 2_Project
Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students).
Assignment 4. Housing Project (Individual Work).
Assignment 5. Project Description and Reflection (Individual Work).

The following processes will be part of the course management and organisation:
- Different group composition for assignments 1 and 3
- Peer Evaluation will be used in Assignment 1.
- Self-Evaluation will be used in Assignment 3.
- The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal.

The final grade will be calculated with the following weight:
Group Work: 30%
Individual Work: 70%

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	

Course Contents

The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof.
Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent.
How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined?

The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers.

The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.

Study Goals

Upon completion of the Design Studio, the students will be able to:

convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level;
perform critical comparative research that results in a clearly formulated design hypothesis;
demonstrate how urgent societal issues are addressed in the design project;
demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal;
represent space in its complex interpenetration of people, architectures, technologies, materials;
confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.

Education Method

The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive.

The Studio will be structured across:
lectures
readings, writings and public discussion
design tutorials / workshops
specific exercises challenging innovative thinking

Literature and Study Materials

Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.

Assessment

Assessment of the students work will be based on the following:
Oral presentation;
Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.;
Writing assignments and analyses.

Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student.

The final grade reflects:
the position that is formulated with regard to the brief and its context;
the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation;
aesthetic and technical/functional qualities; the elaboration throughout the respective scales;
the quality of the presentation, the products and the argument;
the consistency, coherence and development of the students work during his/her process.

Products: will be described in the syllabus which will be published at the beginning of the course.

Repair regulations
The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5).

Retake regulations
Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.	
Course Contents	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.	
Study Goals	After completion of the course the student is able to: Overarching study goals - Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment. - Assess the role of entrepreneurship in changes in the architectural world and the built environment. - Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge. - Critically evaluate and reflect on entrepreneurial skills and plans. - Design and deliver a compelling and succinct vision and business pitch. Additional study goals for Variant: Develop your architectural firm (open to A en BT students) - Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio. - Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network. - Develop a design vision to guide the future direction of their own design firm. - Reveal the design vision in an actual design. - Propose the next steps for establishing a self-owned design firm. Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students) - Assess building sector organisations, including their business models, revenue streams, and network. - Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities. - Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues. - Defend their business plan in a convincing matter to industry experts and utilise expert input to improve. - Architecture students will also develop a design as part of the impactful business model - Propose the next steps for implementing the business model in an existing or new venture.	
Education Method	The course's learning activities comprise: - lectures: theory - workshops - self-study: developing vision and translation into entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop vision and entrepreneurial plan	
Assessment	Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may	

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 HA

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc3 Adapting 20th century Heritage

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AH105	Graduation Studio Adapting 20th century Heritage	55

Course Coordinator	Ir. L.G.K. Spoormans
Instructor	A. de Ridder
Instructor	Prof.ir. W. de Jonge
Instructor	W.J. Quist
Instructor	Ir. J. Roos
Instructor	Ir. L.G.K. Spoormans
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Instructor	Dr. M.T.A. van Thoor
Instructor	F. Marulo
Instructor	Prof.dr. A.R. Pereira Roders
Instructor	Prof.dr.ing. U. Pottgiesser
Responsible for assignments	Ir. L.G.K. Spoormans
Education Period	1 2 3 4
Start Education	1
Exam Period	none
Course Language	English

Course Contents

The AR3AH105 Graduation Studio focusses on Adapting 20C Heritage. What is the essence of this young heritage and what is its status? How can we adapt these buildings and ensembles to make them sustainable and future proof without losing their values?
The section of Heritage & Architecture holds three chairs and related research fields: Heritage & Design, Heritage & Technology and Heritage & Values. These are represented in the research and design approach of the H&A graduation studio, the curriculum and the tutors involved. The studio focusses on selected themes regarding built heritage, such as vacancy, preservation of monuments, new heritage, shared heritage or heritage communities. Research and design of the studio relates to H&A research projects and expertise of H&A staff.

Students of Heritage & Architecture have to position themselves as architects in the debate on the architectural and technical characteristics of built heritage, its tangible and intangible values and design strategies for conservation and adaptive re-use. The articulation and understanding of architectural and technical values is of great importance when design decisions have to be made regarding what to conserve, what to adapt.

In MSc3, students conduct research on the studio theme in a larger context and on the specific characteristics of selected cases. They develop a research question, methodology and final proposal for their individual research project. Methods can be literature study, comparative case study, mapping, archival research, field work and survey, interviews, research by design etc. Students observe, explore, identify and prioritise urban, architectural and technical values of the existing building and site in order to formulate starting points for a meaningful redesign. The research forms a theoretical base for the design project and results in an individual academically substantiated report or journal article. The individual work and the collective research by the studio contribute to the body of knowledge for the graduation studio.
Based on both the thematic research and the mapping of a specific building or area that serves as a case, students develop a design brief and a design method. They explore a range of scenario's combining heritage values and program requirements, and study references on concept, strategy, materialisation, detailing, program etc. By a creative process, logical argumentation and evidence-based choices, students create a concept design that shows coherence and correctness.

In MSc4, students individually develop an elaborated design based on the research, design brief and concept design of MSc3. The academic framework is applied and theoretical and practical references are studied. Strategies of conservation, intervention and transformation are explored and translated into an architectural and technical design. Students individually create a design that shows coherence and correctness, and a meaningful translation of an intervention strategy. Furthermore, they will position the project within a broader perspective of socio-cultural, historical, philosophical, economic and environmental contexts. In the graduation project the societal and disciplinary position and its relevance in relation to design ethics and intercultural issues are discussed. After reflective exploration and consideration, students present a detailed design project that addresses all important aspects of architectural design, technology and cultural value. During the course, students practice to present their design proposals both graphically and orally, and demonstrate logical argumentation and evidence-based choices.

Study Goals

MSc3
The student:
- is able to develop a full research proposal within the context of the studio
- is able to conduct, synthesise and present the research using substantiated methods
- is able to explain architectural and technical values, as well as their place in historical, societal, economic and environmental context
- can critically reflect on research results and translate these into opportunities, obligations and dilemmas, and can prioritize and incorporate the key values into meaningful preliminary design
- develop and present a design brief, method, scenarios and concept design that is relevant for contemporary academic and social society,
- individually draw conclusions and present these in an academically substantiated and comprehensive report or paper as well as through a verbal presentation,

MSc4
The student:
- is able to position the project and its heritage aspects in a broader perspective of socio-cultural, historical, philosophical, economic and environmental contexts,
- is able to apply professional knowledge and design tools related to architecture, building technology and values,
- is able to argue and reflect on the design product and process in relation to current architectural discourse,
- is able to demonstrate and employ moral sensibility, analysis, creativity, judgement, decision-making and argumentation skills regarding architectural ethics and his/her future role as architect,
- is able to communicate complex design ideas at an advanced level through verbal presentations, visual and written media.

Education Method	Lectures, workshops, atelier counselling
Literature and Study Materials	Literature and study materials will be available prior to the start of the course in Brightspace.
Assessment	Written report/ article, paper/ digital/ oral presentation

Period of Education	Academic year
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc3 Revitalising Heritage

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AH115	Graduation Studio Revitalising Heritage	55
Course Coordinator	Ir. L.G.K. Spoormans	
Instructor	A. de Ridder	
Instructor	Prof.ir. W. de Jonge	
Instructor	W.J. Quist	
Instructor	Ir. J. Roos	
Instructor	Ir. L.G.K. Spoormans	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Instructor	Dr. M.T.A. van Thoor	
Instructor	F. Marulo	
Instructor	Prof.dr. A.R. Pereira Roders	
Instructor	Prof.dr.ing. U. Pottgiesser	
Responsible for assignments	Ir. L.G.K. Spoormans	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3AH115 Graduation Studio focusses on Revitalising Heritage. How to bring new life to abandoned or dilapidated buildings or areas? What combination of architectural interventions and functional program leads to preservation by revitalisation? What is the capacity for change? The section of Heritage & Architecture holds three chairs and related research fields: Heritage & Design, Heritage & Technology and Heritage & Values. These are represented in the research and design approach of the H&A graduation studio, the curriculum and the tutors involved. The studio focusses on selected themes regarding built heritage, such as vacancy, preservation of monuments, new heritage, shared heritage or heritage communities. Research and design of the studio relates to H&A research projects and expertise of H&A staff. Students of Heritage & Architecture have to position themselves as architects in the debate on the architectural and technical characteristics of built heritage, its tangible and intangible values and design strategies for conservation and adaptive re-use. The articulation and understanding of architectural and technical values is of great importance when design decisions have to be made regarding what to conserve, what to adapt. In MSc3, students conduct research on the studio theme in a larger context and on the specific characteristics of selected cases. They develop a research question, methodology and final proposal for their individual research project. Methods can be literature study, comparative case study, mapping, archival research, field work and survey, interviews, research by design etc. Students observe, explore, identify and prioritise urban, architectural and technical values of the existing building and site in order to formulate starting points for a meaningful redesign. The research forms a theoretical base for the design project and results in an individual academically substantiated report or journal article. The individual work and the collective research by the studio contribute to the body of knowledge for the graduation studio. Based on both the thematic research and the mapping of a specific building or area that serves as a case, students develop a design brief and a design method. They explore a range of scenario's combining heritage values and program requirements, and study references on concept, strategy, materialisation, detailing, program etc. By a creative process, logical argumentation and evidence-based choices, students create a concept design that shows coherence and correctness. In MSc4, students individually develop an elaborated design based on the research, design brief and concept design of MSc3. The academic framework is applied and theoretical and practical references are studied. Strategies of conservation, intervention and transformation are explored and translated into an architectural and technical design. Students individually create a design that shows coherence and correctness, and a meaningful translation of an intervention strategy. Furthermore, they will position the project within a broader perspective of socio-cultural, historical, philosophical, economic and environmental contexts. In the graduation project the societal and disciplinary position and its relevance in relation to design ethics and intercultural issues are discussed. After reflective exploration and consideration, students present a detailed design project that addresses all important aspects of architectural design, technology and cultural value. During the course, students practice to present their design proposals both graphically and orally, and demonstrate logical argumentation and evidence-based choices.	
Study Goals	MSc3 The student: - is able to develop a full research proposal within the context of the studio - is able to conduct, synthesise and present the research using substantiated methods - is able to explain architectural and technical values, as well as their place in historical, societal, economic and environmental context - can critically reflect on research results and translate these into opportunities, obligations and dilemmas, and can prioritize and incorporate the key values into meaningful preliminary design - develop and present a design brief, method, scenarios and concept design that is relevant for contemporary academic and social society, - individually draw conclusions and present these in an academically substantiated and comprehensive report or paper as well as through a verbal presentation, MSc4 The student: - is able to position the project and its heritage aspects in a broader perspective of socio-cultural, historical, philosophical, economic and environmental contexts, - is able to apply professional knowledge and design tools related to architecture, building technology and values, - is able to argue and reflect on the design product and process in relation to current architectural discourse, - is able to demonstrate and employ moral sensibility, analysis, creativity, judgement, decision-making and argumentation skills regarding architectural ethics and his/her future role as architect, - is able to communicate complex design ideas at an advanced level through verbal presentations, visual and written media.	
Education Method	Lectures, workshops, atelier counselling	
Literature and Study Materials	Literature and study materials will be available prior to the start of the course in Brightspace.	
Assessment	Written report/ article, paper/ digital/ oral presentation	

Period of Education	Academic year

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Interiors Buildings Cities

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 AI

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10

Course Coordinator	M. Parravicini
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10
Education Period	1 2
Start Education	1
Exam Period	none
Course Language	English
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.
Period of Education	1st Semester

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1AI013	Interiors Buildings Cities MSc1 Design Project	10

Course Coordinator	Ir. S. Pietsch
Instructor	Ir. L.M.M. de Wit
Instructor	Ir. S. Pietsch
Responsible for assignments	Ir. S. Pietsch
Contact Hours / Week x/x/x/x	Variable 4 till 8 hours per week
Education Period	1 2
Start Education	1
Exam Period	none
Course Language	English

Course Contents

Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses how people within or affected by the scenes made by architecture can be situated in relation to place and conditions, time and the historical moment, and social and material culture.

Thinking beyond individual students and courses, Interiors Buildings Cities considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. Interiors Buildings Cities engages in common questions concerning the city and its territories, public space, the public interior, questions of interiority, and relations within the social and physical fabric of the city. It does so with the well-being of the entire environment shared by people and other living beings as a central concern.

House
The House of the MSc1 project is neither the representative monument of collective public life, nor is it the private house of an individual. Instead, it suggests and considers those buildings that stand between; intimate figures, which both inform and are informed by the physical and social fabric of their immediate locality.

Such buildings are carefully wrought and spatially rich, both generous and adaptable, yet they also embody a degree of typicality and a measure of the material culture of their place. Their interiors host the rooms and spaces that structure and draw a community into consciousness of and in dialogue with itself.

Through a process of iterative drawing and large-scale modelling, supported by lectures and workshops, students will resolve the design and technical integration of such a building and its principal interior spaces, considering it as a figure set within the immediate context a carefully documented urban setting.

The MSc1 Design Project is paired with the Building Engineering Studio (AR1A080) coordinated by the Department of Architectural Engineering & Technology.

Study Goals

Upon completion of the Master 1 studio trajectory the student:

- is able to develop a coherent, meaningful, elaborated, correct and innovative design - in general and on specific aspects - for a medium sized building at MSC 1 level
- has developed skills and knowledge to embed an architectural intervention appropriately in its social, cultural and physical context, from the urban scale to the interior.
- is able to apply skills and knowledge related to structural design, materialisation of buildings and interiors, comfort and climate design in an appropriate and purposeful manner

A specific description of the studio and its aims will be published in the Studio Manual, to be distributed at the beginning of the course.

Education Method

The studio consists of individual project work that builds on a foundation of intensive group work at the beginning of the semester. In addition, it features project-specific studies and thematic exercises, as well as lectures that relate to the topic of the studio.

A characteristic working method of the Interiors Buildings Cities course is the production of models at different scales appropriate to the subject matter, in which ideas about the design project, from form to space and other relationships, are tested and materialised.

Please note:
In the event of unannounced and unexcused absences from the course, the tutors may decide not to continue tutoring the student.

Literature and Study Materials

Reading material relating to the context of the design assignment, as well as a number of thematic texts, will be provided in the Studio Manual at the beginning of the semester.

Assessment

The design studio concerns the development of an architectural project on all scale levels, from its urban setting to its materiality and elaboration of its details. The project will be assessed during an intermediate, pre-final and final presentation on the following aspects:

Coherence and Significance
Within the context of the project and the design task, there is a meaningful theme, which gives direction to and is recognisable in the concrete design. The various relevant aspects and scale levels (domains) form a coherent whole.

Correctness and Elaboration
The design is based on relevant knowledge, which has been properly applied. The relevant aspects and scale levels are elaborated.

Creativity / Sensibility
The design shows a creative spark, a personal interpretation or an innovative contribution.

Presentation
The design, the themes and the research are documented, visualised and argued in an insightful way, using appropriate means.

Study
Within the context of the project, the design task and the theme. Relevant aspects have been analysed and relevant questions formulated and researched. Experiments have been conducted: various ideas and variants were explored and evaluated at the relevant scale levels.

	The product to be assessed is the final design proposal, which is the result of an iterative design process and is represented by model photos, drawings, a written description and the project journal (process document). The above listed assessment criteria all count equally. The assessment is carried out by at least two tutors (4-eyes principle). A final grade of 6.0 or higher is needed to successfully complete this course. Each partial grade should be graded with at least with a 5.0. In case a partial grade is lower than 6.0, students get the opportunity to repair the assignment within 2 weeks after the initial grade was announced. A successful repair will result in a grade no higher than 6.0. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.
Special Information	The maximum marking period is 10 working days.
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios.
Period of Education	Autumn semester, first and second quarter
Leerstoel	Interiors Buildings Cities
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 AI

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.	
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or	

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5

Course Coordinator	Prof.dr. A.R. Pereira Roders
Instructor	Dr. I. Nevzgodin
Instructor	Ir. A.C. de Ridder
Responsible for assignments	Prof.dr. A.R. Pereira Roders
Co-responsible for assignments	Y. Zhang
Contact Hours / Week x/x/x/x	4 hours per week
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.
Literature and Study Materials	Book chapters, journal articles and other lecture materials.
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%
Period of Education	Q 3
Concept Schedule	Wednesday morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0109	City of Innovations Project	5

Course Coordinator	M. Triggianese
Instructor	M. Triggianese
Instructor	Y. Söylev
Instructor	I. Zaid
Responsible for assignments	M. Triggianese
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Required for	MSc3
Expected prior knowledge	BSc and MSc1 completed
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarions	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

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Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Expected prior knowledge

The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval.

MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080).

Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.

Course Contents

Architectural and Urban Design in Times of Transitions.
Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies.

In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism.

The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.

Study Goals

At the end of this course, the student:
 knows key literature and recent research on people, movement and public space
 understands main theories on people, movement and public space
 applies these theories in analysing real urban settings
 evaluates critically on these theories
 creates presentations analysing the subject on an academic level.

And, the student:
 understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city
 evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning
 creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation.

Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.

Education Method

The course consists of interactive studio work and lectures.

Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory.

Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches.

Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared!

Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class.

*) the discussion groups ideally consist of four/five members, who divide topics and peer each other.

Literature and Study Materials

Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.

Assessment

Studio work 80% - Lectures 20%

Assessment of studio work:
Analyses and design, presented in drawing form with written commentary and a model.

Assessment of lectures:

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15

Course Coordinator	Dr.ing. M. Bilow
Responsible for assignments	Dr.ing. M. Bilow
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Special Information	The maximum marking period is 10 work days.
Period of Education	Quarter 4
Concept Schedule	Tuesday and Thursday - details may vary
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday, Friday	
Minimum number of participants	15	
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter 15
Course Coordinator	G. Coumans
Course Coordinator	M.G. Vink
Course Coordinator	D. Piccinini
Course Coordinator	F.T. Toni
Instructor	G. Coumans
Instructor	M.G. Vink
Responsible for assignments	M.G. Vink
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.
Literature and Study Materials	Syllabus
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	coordinator
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.
Period of Education	Q4, 10 weeks, starting in week 4.1
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.
Used Materials	Model materials - for students account

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.	

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 AI

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AI100	Interiors Buildings Cities Graduation Project	55
Course Coordinator	M.W. Pimlott	
Instructor	M.W. Pimlott	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Instructor	Prof. D.J. Rosbottom	
Instructor	S. De Vocht	
Instructor	Dr. A.R. Thomas	
Responsible for assignments	M.W. Pimlott	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	

Course Contents

Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which people within or affected by the scenes made by architecture can be situated in relation to place and conditions, to time and the historical moment, and to social and material culture.

Thinking beyond individual students and courses, Interiors Buildings Cities considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. Interiors Buildings Cities engages in common questions concerning the city and its territories, public space, the public interior, questions of interiority, and relations within the social and physical fabric of the city. It does so with the well-being of the entire environment shared by people and other living beings as a central concern.

Palace
The graduation studio theme, Palace, does not concern literal palaces, but those complex public buildings that have used the palace as a formative motif, and play a significant role in the fabric of the city. These accommodate the publics image of itself and the sense of collective and individual agencies within society. The theme responds to the history and position of such representative public buildings in the urbanised environment, and acknowledges that the figure, idea and role of the Palace have been constantly reinterpreted over time. It questions what these are in the present, and the ways through which the Palace might engage with both the contemporary city and its citizens in the present and future.

To aid the study of the nature of the project and its conditions, collective research is undertaken into precedents of related exemplary projects over relevant historical periods, and in workshops and seminars aimed towards understanding their relevance to the project and participants architectural knowledge. Further studies are made directly in relation to the conditions surrounding the project itself, from the urban environment to attendant social, cultural and political circumstances that have bearing upon them. Beyond working in existing physical and cultural contexts, we frequently work with existing buildings, their histories, meanings and materials.

In the design of a large-scale project, largely the focus of our studies, we are concerned not only in the making of a building for the representations of the city and its publics, but how, and of what, it is made: the acts of building, the use and re-use of material, the sustainability of strategies of building, of the origin of materials, their extraction, processing, transport and associated labour, and the levels of responsibility of the architect, are all central considerations. We do this in association with the Department of Architectural Engineering and Technology from the outset, and regularly thereafter, so as to generate deeper questions and understanding of how to build intelligently and responsibly now, in the hope that this might be a model for future practices.

Study Goals

Upon completion of a trajectory in the Masters programme from MSc1 through MSc4, the student:
- Has developed the skills in architectural design satisfying aesthetic, relational, functional and technical requirements. During the trajectory, the complexity of the architectural design increases, leading to a level fit for architectural practice and a training period for professional accreditation as architect.
- During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to research and reading in urban and architectural history, relevant theory, cultural practices, human sciences and technology.
- Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects and sustainable practices.
- During the Master 1, 2, 3 & 4 process, skills are acquired to incorporate insights into and knowledge of the process attained with regard to methods of investigation and architectural design.
- Together with the training with regard to aspects of building technology, during the Master 1, 2, 3 & 4 process, skills are acquired to incorporate an understanding of the design process in response to truly sustainable principles, affecting approaches to building design, structural design, materialisation of buildings and their interiors, human comfort, natural diversity, and climate design.

The Graduation Report, in the form of a Project Book and Project Journal, demonstrates the students ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding Architectural ethics and his/her future role as architect. The individual Book and Journal not only contains an elaboration regarding the Graduation Projects societal and disciplinary relevance, but addresses design ethics and the way in which intercultural issues have been addressed within the graduation project itself.

Education Method

The object of the course is the formulation of a design project for an urban or civic institution on a representative site. This is achieved over a series of stages, ranging from discrete thematic studies to analyses of relevant models, integrated with project proposals with an increasingly significant level of definition. This includes:
1. Workshops in analysis of local urban contexts, characteristics particular to the public interior, programme, precedents and use, conducted by the design mentors.
2. Research seminars on literature, theory and history specific to the theme of the programme within the design studio, related to both the Research Plan, and research questions integrated into the development of the project, followed throughout the year through a project journal, conducted by the research mentors.
3. Research and design supervision in studio, in groups and for individuals in relation to precedents and models, in which comparative historical analysis plays a central part, and studies of local conditions, contexts and the specific conditions of both site and building, conducted by the design mentors.
4. Preliminary design development, guided in workshops in MSc3 and individual tutorials towards detailed design development in MSc4, with studio presentations and critique, conducted by the design mentors.

A characteristic working method of the Chair are specific design briefs on aspects of the greater project, often resolved through

	the making of physical models of varying scales appropriate to matters under consideration, in which ideas about the design project, from form to space and other relationships are tested and materialised.
Literature and Study Materials	A core list of reading or other material is determined for each design programme by the design and research mentors, ranging from literature and essays on cultural matters to urban and architectural theory, and critical analysis of architecture. This is published in the Studio Manual issued to participants at the beginning of the first semester. Additional reading material pertaining to issues of design and research are set forward in the Studio Manual, specific to the programme of the studio.
Assessment	The project will be assessed during intermediate and final presentations on: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the organisation, composition, presence and 'appearance' and the potential for functional technical qualities; the potential for elaboration throughout the various scales; the potential for integration of the disciplines, from aesthetic to technological; the quality of the presentation, the argument and the product; the consistency, coherence and development of the students work during the process of design, as demonstrated through prepared project journal, project portfolio and project book. Intelligent conventional architectural drawing and accessible models are particularly important.
Enrolment / Application	Students can enrol for this course during the pre-enrolment period for entry in first or third quarters. The studio beginning in the third quarter is based on individually generated design and research proposals, and so follows an individually-oriented research and design programme, with the same products as the course beginning in the first quarter. Enrolment for the studio beginning in the first quarter follows the convention of registering interest in the pre-enrolment period. For the studio beginning in the third quarter, students are invited to make design and research proposals before the pre-enrolment period. Those proposals which are suitable for development within the studio are enrolled through the appropriate department.
Special Information	The marking periods are directly related to the examination periods: the Go/ No Go examination (P2), and the final examination (P5), as set by the university and the department of architecture.
Period of Education	Two semesters of twenty weeks, beginning either in the first or third quarters, with a Go/No Go examination (P2) at the end of the first two quarters.
Leerstoel	Interiors Buildings Cities
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

The Why Factory

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 TWF

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A081	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Contact Hours / Week x/x/x/x	An average of 4-6 hours a week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. The main topic of the Building Engineering Studio (BES) is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. The aim of materialization - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole.	
Education Method	Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. The presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual. In general, the BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Special Information	For questions please contact Ferry Adema (F.Adema@tudelft.nl). The Building Engineering Studios are taught during the 2nd quarter. Only students who choose the MSc 1 studio of The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter.	
Period of Education	2nd Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1TWF011	The Why Factory MSc1 Design Studio	10

Course Coordinator T. Maso

Course Coordinator Prof.ir. W.G.M. Maas

Responsible for assignments J. Arpa Fernandez

Contact Hours / Week x/x/x/x 8 hours per week starting from week 1.1 and ending in week 1.10

Education Period 1

Start Education 1

Exam Period none

Course Language English

Summary

The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future.

Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions.

The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means.
A systematic, parametric exploration of parts of the design is an integral part of the research approach.

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

For more information about our previous studios, please visit:
https://thewhyfactory.com
https://thewhyfactory.com/education/

Course Contents

MSc1 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda.

During the Why Factory MSc1 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)

MSc1 Design Studio at the Why Factory starts in week 1.1 and ends in week 1.10. After the design studio, students follow their work in the Building Technology Seminar.

Study Goals

The student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects from a medium sized building on MSC 1 level.

Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC1 level.

Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal.

There are six qualitative aspects expected from students at the end of their MSc1 Design Studio:

1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester.

2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc..

3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration.

4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors.

5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development.

6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.

Education Method

Number of studio hours: 72
Number of self study hours: 208

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program.

Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality.

	Applications Program:In the applications program model cities both are tested in real cities. The different models become counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Remarks	Students who choose the MSc 1 studio of The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. This design studio is not integrated with Building Engineering Studios.
Period of Education	From week 1.1 thru week 1.10 in the Fall semester
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 TWF (nw)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.	
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or	

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Pereira Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Pereira Roders	
Co-responsible for assignments	Y. Zhang	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.	
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.	
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Book chapters, journal articles and other lecture materials.	
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design 5
Course Coordinator	A.E. Rout
Course Coordinator	Prof.dr. T.G. Vrachliotis
Course Coordinator	M. Mateljan
Responsible for assignments	Prof.dr. T.G. Vrachliotis
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.
Period of Education	Quarter 3 (spring semester)
Concept Schedule	Friday morning
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.	
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.	
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:	

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	

Summary

The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future.

Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions.

The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means.
A systematic, parametric exploration of parts of the design is an integral part of the research approach.

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO.

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

For more information about our previous studios, please visit:
https://thewhyfactory.com
https://thewhyfactory.com/education/

Course Contents

MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda.

During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)

Study Goals

- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.

- Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal.

There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios:

1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester.

2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc..

3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration.

4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors.

5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development.

6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.

Education Method

Number of studio hours: 80
Number of self study hours: 332

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program.

Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality.

Applications Program:In the applications program model cities both are tested in real cities. The different models become

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday, Friday	
Minimum number of participants	15	
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	**Van Gezel tot Meester**	**20**
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	

Course Contents

In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal.

Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future.

Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.

Study Goals

Upon completion of the Design Studio, the students will be able to:

convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level;
investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions;
demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal;
focus on the qualitative aspects of multiplicity in society and design;
design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics;
confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal;
represent space in its complex interpenetration of people, architectures, technologies, materials.

Education Method

The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression.

Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive.

Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues.

The Studio will be structured across:
lectures
field trips
readings, writings and public discussion
experimental research
design tutorials
specific exercises challenging innovative thinking

In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.

Literature and Study Materials

Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.

Assessment

Assessment of the students work will be based on the following:
1. Oral presentation;
2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.;
3. Written statements and analyses.

The design proposal is individual.
The assessment process implies:
in-class participation on a weekly base with public discussion of in-between results;
Midterm and Final Reviews;

The final grade reflects:
Critical analysis and the urban context (25%)
Design quality of the final proposal (55%)
Participation, collegiality, commitment (20%)

Repair regulations
The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.

The maximum marking period is 10 working days.

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more. The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.	
Study Goals	By the end of the MSc2 Studio, students should be able to: Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural) Visualise key design and data-related features using effective narrative and visual storytelling methods	
Education Method	The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include: Design tutorials (including individual design progress feedback and group discussions) Seminars (including lectures, workshops and reading discussions) Excursion (study trip including the site visit)	
Course Relations	Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues? The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas: CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives. ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort. RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges. Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.	
Literature and Study Materials	Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.	
Assessment	Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include: Posters (including architectural drawings and other visuals) Booklets and/or digital presentations Short animations (such as GIF animations) Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.	

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.	

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

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Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 BO

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1BO010	Borders and Territories Design Studio	10
Course Coordinator	N. Sanaan Bensi	
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week starting from week 1.1 and ending in week 2.10.	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	BORDER AGENCIES: The Design of an Institute [Introduction] The educational program of Borders & Territories (B&T) focuses on the territory as a problematic and complex field of operation, with its borders as the spatial tools for differentiation. The B&T MSc 1 studio is understood as a laboratory for the investigation of territorial, spatial, ecological, technical and architectural agencies in which participants will develop a Border-Zone Institute. [Focus] As a result of the Unification of Europe, places where once the periphery was located, have become central. Before the formation of the EU, most of the European border regions had been treated as the backyards of their respective countries. The NL/BE border zone is an interesting example of this, where the accumulation of historical military sites, refugee camps, extraction, and industrial complexes overlap with the natural reserves, such as Natura 2000 (as an EU-initiated program). [procedure] The studio privileges the section as a drawn speculation revealing complexity. It establishes the particular relationship between the human body, the ground and architectural form. The section tends to the human scale on one side, and on the other side of the spectrum to the dimension of the territory at large. [Program] The students will design a series of Border-Zone Institutes (the Border Agencies) to initiate a proper spatial and material study of the border areas and create knowledge and awareness of the differentiation in utilizations, appropriation, and perceptions of the territory itself. Starting from a generic given program for an institute, the students will then adjust the program based on the particularities of the investigated site condition. [Site] By studying the fragments of the western NL/BE border zone- extending between the Noord Zee and Scheldt River, this B&T studio will investigate the transformation of these territories under initiated EU programs, such as Interreg Europe which strives to reduce disparities in the levels of development, growth and quality of life in and across Europes regions. Each student will produce an innovative design for a medium sized building. B&Ts MSc_1 Design Studio is integrated with BES.	
Study Goals	The student is able to: LO1: establish an independent critical reflection and position regarding the design process and architectural representation. LO2: integrate the research and analysis into the design process and concepts. LO3: communicate key ideas and design skills. LO4: express theoretical knowledge addressing the design procedure. LO5: develop a coherent, significant, elaborated, correct and innovative design for a medium-sized building.	
Education Method	Group work (site analysis). Pin-up collective presentations. Individual Consultation. Independent design & self-study. Workshops and seminars.	
Assessment	Participation in collective presentation. Individual presentations & evaluations: midterm and final presentation. (Refer to the calendar for specific weeks). Assessment Scheme Final design (70 %) Midterm presentation (10 %) Participation (initiative, in-class discussion) (10 %) Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely.	
Period of Education	Semester 1, Q1 & 2	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 BO

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	

Expected prior knowledge

The student:
- Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor.
- Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course.
- Demonstrates a general historical understanding of the architecture profession and the role of the architect in society.
- Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design.

Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links:

https://learn.saylor.org/course/view.php?id=42

https://learn.saylor.org/course/view.php?id=43

(These links lead to the English courses offered for free to all by the online Saylor Academy.)

Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.

Course Contents

Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated.

The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant.

Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations.

Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).

Study Goals

Learning objectives
After completion of the course the student:
Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context.
Is able to plan and develop a scientific research project.
Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images.
Is able to evaluate, interpret and make proper reference to available sources.
Is able to build on existing knowledge and develop new knowledge.

Education Method

Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.

Literature and Study Materials

Course material--readings and documents on research and writing--is available on the course Brightspace page.

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5

Course Coordinator	Dr.ir. H. Sohn
Instructor	Dr.ir. H. Sohn
Instructor	Dr.ir. A. Radman
Instructor	Dr.ir. R.A. Gorny
Instructor	Dr.ir. S. Kousoulas
Responsible for assignments	Dr.ir. H. Sohn
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	

Course Contents

This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment.

In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences.

Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.

Study Goals

By the end of this course students will be able to:
1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective;
2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods;
3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media;
4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis;
5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences.
6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.

Education Method

The course Architectural Ethnography comprises group assignments and individual work.
The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course.

The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.

Literature and Study Materials

The course will use the following publications as main textbook references:

Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts.

Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto

Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126.

Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE.

Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.

Assessment

The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects.
The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment.
The summative assessment will be based on the following deliverables:

a) Analytical Assignment:
Ethnographic Research: Research Report (Group Work)

b) Practical Exercises:
Poster: Synthesis of (Quantitative) Data Collection (Group Work)
Ethnographic Research: Narrative and Visual Synthesis (Group Work)

c) Writing Assignment:
Critical Reflection (Individual Work)

For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c).

The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.

Retake and repair regulations
In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year.
Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5

Course Coordinator Prof.dr. A.R. Pereira Roders

Instructor Dr. I. Nevzgodin

Instructor Ir. A.C. de Ridder

Responsible for assignments Prof.dr. A.R. Pereira Roders

Co-responsible for assignments Y. Zhang

Contact Hours / Week x/x/x/x 4 hours per week

Education Period 3

Start Education 3

Exam Period 3

Course Language English

Summary

Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.

Course Contents

Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study.

The five domains are:
[C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign.
[R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign.
[A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign.
[S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign.
[H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.

Course Contents Continuation

Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage).
Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects.
Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors.
Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc.

By the mid-presentation, students should have collected all the data needed to perform the analysis.
By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students.
By the end of the course, the students will compile the research in a written report.

Study Goals

Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign.

In terms of knowledge, the students will:
Assess a selected domain, comparing before and after architectural redesign
Reach consensus on a co-created assessment, making use of a pre-defined framework

In terms of skills, the students will:
Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations.
Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language

In terms of attitude, the students will:
Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage.
Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions.
Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.

Education Method

Lectures (including Seminars): 12 hours
Tutorials/Presentations: 16 hours
Group field work: 4 hours
Independent study (including additional field work): 108 hours (77%)

Course Relations

The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.

Literature and Study Materials

Book chapters, journal articles and other lecture materials.

Assessment

Dataset 40%
Pechakucha 20%
Final Report 30%
Attitude 10%

Period of Education Q 3

Concept Schedule Wednesday morning

Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarions	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

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Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Expected prior knowledge

The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval.

MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080).

Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.

Course Contents

Architectural and Urban Design in Times of Transitions.
Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies.

In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism.

The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.

Study Goals

At the end of this course, the student:
 knows key literature and recent research on people, movement and public space
 understands main theories on people, movement and public space
 applies these theories in analysing real urban settings
 evaluates critically on these theories
 creates presentations analysing the subject on an academic level.

And, the student:
 understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city
 evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning
 creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation.

Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.

Education Method

The course consists of interactive studio work and lectures.

Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory.

Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches.

Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared!

Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class.

*) the discussion groups ideally consist of four/five members, who divide topics and peer each other.

Literature and Study Materials

Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.

Assessment

Studio work 80% - Lectures 20%

Assessment of studio work:
Analyses and design, presented in drawing form with written commentary and a model.

Assessment of lectures:

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday, Friday	
Minimum number of participants	15	
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15

Course Coordinator	A.E. Rout
Course Coordinator	Prof.dr. T.G. Vrachliotis
Course Coordinator	M. Mateljan
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.	

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.	
Course Contents	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.	
Study Goals	After completion of the course the student is able to: Overarching study goals - Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment. - Assess the role of entrepreneurship in changes in the architectural world and the built environment. - Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge. - Critically evaluate and reflect on entrepreneurial skills and plans. - Design and deliver a compelling and succinct vision and business pitch. Additional study goals for Variant: Develop your architectural firm (open to A en BT students) - Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio. - Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network. - Develop a design vision to guide the future direction of their own design firm. - Reveal the design vision in an actual design. - Propose the next steps for establishing a self-owned design firm. Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students) - Assess building sector organisations, including their business models, revenue streams, and network. - Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities. - Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues. - Defend their business plan in a convincing matter to industry experts and utilise expert input to improve. - Architecture students will also develop a design as part of the impactful business model - Propose the next steps for implementing the business model in an existing or new venture.	
Education Method	The course's learning activities comprise: - lectures: theory - workshops - self-study: developing vision and translation into entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop vision and entrepreneurial plan	
Assessment	Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may	

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 BO

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3BO100	Borders and Territories Graduation Studio	55
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	O.R.G. Rommens	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 2.1 and ending in week 2.10. 8 hours per week starting from week 3.1 and ending in week 4.10.	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The graduation studio of B&T investigates contemporary border conditions within the larger urban and territorial scale. The studio focuses on sites where new spatial conditions have emerged that are teeming with new meanings and unexpected potential but are hardly analysed within contemporary architectural discourse. Nowadays, it seems no longer possible to investigate space without considering global political developments, nor to ignore ecological anxieties, pressured coexistences or even economically driven migratory movements. B&Ts graduate studio intends to map these emerging spatial complexities, focusing on the fundamental changes regarding radical spatial differentiation, biodiversity, atmospheric and soil conditions, increased technological spatial control and economic asymmetries. The spatial patchworks of superposed political and economic processes and juxtaposed social and spatial practices will constitute the (almost) solid basis for alternative architectural interventions proposing radical counterstrategies to the current and no longer sustainable status quo. B&T considers the current developments in the border-territories related to the New Silk Road particularly intriguing testing ground(s) for speculative approaches, as the spatial transformations in these areas between the Far East, Africa and Europe are resulting in a fascinating array of spatial changes, where different regimes of spatial planning, (inter-) continental infrastructural projects, political (dis-) agreements, border tensions, global capitalism, extra-state utopias and the like, are re-formatting the contemporary territorial and urban landscapes. B&Ts 2024/25 graduate studio will explore the current state of the New Silk Road by globally cross-sectioning these territories. Central to the investigation and intrinsic to the design elaboration is the production of specific mappings addressing: Border and Migration, Infrastructure and Congestion, Soil and Fault Lines, Water and Aquifers, and Contaminated Ecologies. B&Ts 2024/25 graduate studio will investigate three sites: Zeebrugge (BE), Tbilisi (GE) and Chongqing (CH).	
Study Goals	The student is able to develop effective tools and techniques for implementing a design position. The student is able to analyze, evaluate and pursue a range of technical, programmatic, theoretical, historical and professional implications toward the final design proposal. The student is able to integrate and express theoretical knowledge and practical skills into design process. The student is able to develop a definitive project on her/his own that qualifies functional, spatial and aesthetic qualities through relevant research and preparation. The student is able to integrate building technology in adequate ways into the architectural design. The student is able to synthesize requirements of building technology and the architectural design. The graduation report demonstrates the students ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding Architectural ethics and his/her future role as architect. The individual graduation report should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which intercultural issues were addressed in the graduation project.	
Education Method	(first semester) Pre-design research. Site visit and field investigation. Formal exercises in drawings and models. Material exercises. Presentation and critique. The studio work will include and be supplemented by seminar session, informal/intermediate reviews, and a two-weeks workshop. During the first half of the semester, until the midterm review (P1), the students will work in groups. (second semester) Individual Consultation. Independent design & self-study.	
Assessment	(first semester) *P1 (Week 1.10) PDF/PPT presentation. Project synopsis, problem statement, concepts and analytical mappings. Requirements specified by the faculty rules & regulations for graduation. *P2 (Week 2.9) PDF/PPT presentation. Urban analysis/mappings (incl. methods and results) and design project proposal. The definition of the design project consists of site, site analysis, program, program analysis, site model (1:500) and a sketch design. Requirements specified by the faculty rules & regulations for graduation. The P1 & P2 weeks may be subject to change. Consult the graduation regulations for the submission requirements. (second semester) Presentations & evaluations: P3, P4 and P5. Refer to the graduation calendar for specific weeks. Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and	

	Examination Regulations Masters).
Period of Education	Refer to the graduation calendar.
Concept Schedule	First semester: Tuesday, and Thursday Second semester: Thursday
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Urban Architecture

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 UA

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1UA010	**Urban Architecture MSc 1 Design studio**	**10**
Course Coordinator	Ir. M.G.J. van Gelderen	
Instructor	Ir. M.G.J. van Gelderen	
Responsible for assignments	Ir. M.G.J. van Gelderen	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Msc1: What to keep? At the chair of Urban Architecture, we study medium-size urban sites that are inaccessible for the regular tools of urbanism, and where expectations of urban design can only be fulfilled by architecture. Adding a building thus means (re)designing an environment. The Msc1 studio opposes tabula rasa and addresses questions prior to the new design, concerning the existing structures on the site: whether to demolish (and what exactly) and how to re-use. The motives for preservation are not necessarily architectural value, but also rest-value of written-off material for social, non-profit uses and, especially, ecological value of used material in itself and in economizing the use of new material. A particular urban site must be restructured with the design of a new building, whilst maintaining selected parts of an existing building. Students are challenged to re-interpret as-found constellations, to re-integrate them into a new design concept and to enjoy contingencies. We want to make the best possible use of resources observed near the site, both socially, physically and technically. Other resources are buildings nearby and far-off, realized or only imagined, as an archive of human efforts to accommodate life. We favour programs that communicate their social role to the public realm sensitively and intervene in daily life, in urban areas where the right to the city, as Henri Lefebvre has put it, is in need of strengthening. The course is in close collaboration with Building Technology.	
Study Goals	- The student is able to present a coherent, significant, elaborated, correct and innovative design in general and on specific aspects from a medium-sizes building on MSC 1 level. - The student is able to creatively design meaningful relationships between the existing and the new. - The student develops awareness of the responsible uses of building materials	
Education Method	Excursion, both to the site and to relevant architectural projects Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 working days.	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios.	
Period of Education	Semester 1, Q1 & 2	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 UA

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5

Course Coordinator	Dr.ir. H. Sohn
Instructor	Dr.ir. H. Sohn
Instructor	Dr.ir. A. Radman
Instructor	Dr.ir. R.A. Gorny
Instructor	Dr.ir. S. Kousoulas
Responsible for assignments	Dr.ir. H. Sohn
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or	

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5

Course Coordinator	Prof.dr. A.R. Pereira Roders
Instructor	Dr. I. Nevzgodin
Instructor	Ir. A.C. de Ridder
Responsible for assignments	Prof.dr. A.R. Pereira Roders
Co-responsible for assignments	Y. Zhang
Contact Hours / Week x/x/x/x	4 hours per week
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.
Literature and Study Materials	Book chapters, journal articles and other lecture materials.
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%
Period of Education	Q 3
Concept Schedule	Wednesday morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5

Course Coordinator	Prof.dr.ing. C.M. Hein
Course Coordinator	J.M.K.K. Hanna
Instructor	Dr. P. De Martino
Instructor	J.M.K.K. Hanna
Responsible for assignments	Prof.dr.ing. C.M. Hein
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	3rd Quarter
Concept Schedule	Friday
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarions	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15

Course Coordinator	Dr.ir. M.G.A.D. Harteveld
Course Coordinator	Dr.ir. R. Cavallo
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld
Responsible for assignments	Dr.ir. R. Cavallo
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15

Course Coordinator	T. Maso
Instructor	T. Maso
Responsible for assignments	J. Arpa Fernandez
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15

Course Coordinator	Dr. L. Cipriani
Responsible for assignments	Dr. L. Cipriani
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.
Concept Schedule	Wednesday, Friday
Minimum number of participants	15
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15

Course Coordinator	A.E. Rout
Course Coordinator	Prof.dr. T.G. Vrachliotis
Course Coordinator	M. Mateljan
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.	

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.	
Course Contents	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.	
Study Goals	After completion of the course the student is able to: Overarching study goals - Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment. - Assess the role of entrepreneurship in changes in the architectural world and the built environment. - Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge. - Critically evaluate and reflect on entrepreneurial skills and plans. - Design and deliver a compelling and succinct vision and business pitch. Additional study goals for Variant: Develop your architectural firm (open to A en BT students) - Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio. - Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network. - Develop a design vision to guide the future direction of their own design firm. - Reveal the design vision in an actual design. - Propose the next steps for establishing a self-owned design firm. Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students) - Assess building sector organisations, including their business models, revenue streams, and network. - Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities. - Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues. - Defend their business plan in a convincing matter to industry experts and utilise expert input to improve. - Architecture students will also develop a design as part of the impactful business model - Propose the next steps for implementing the business model in an existing or new venture.	
Education Method	The course's learning activities comprise: - lectures: theory - workshops - self-study: developing vision and translation into entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop vision and entrepreneurial plan	
Assessment	Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may	

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 UA

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3UA100	Urban Architecture Graduation Studio	55
Course Coordinator	Prof.ir. P.E.L.J.C. Vermeulen	
Course Coordinator	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 & 2	
Course Contents	Msc3: Where to build? At the chair of Urban Architecture, we study medium-size urban sites that are inaccessible for the regular tools of urbanism, and where expectations of urban design can only be fulfilled by architecture. Adding a building thus means (re)designing an environment. This graduation studio aims at critically assessing a given situation, in order to either complete or re-direct it. The position of the building(s) relative to existing structures, its role in delineating the private and the public realm, awareness of scale and material relationships are issues that are addressed in the studio. We want to make the best possible use of resources observed near the site, both socially, physically and technically. Other resources are buildings nearby and far-off, realized or only imagined, as an archive of human efforts to accommodate life. Therefore, whether to demolish and how to re-use are considered vital questions. We favour programs that communicate their social role to the public realm sensitively and intervene in daily life, in urban areas where the right to the city, as Henri Lefebvre has put it, is in need of strengthening. The themes and design task in the msc3 studio are focussed on an elaborate contextual research, integrating the obtained knowledge of the research course (AR3UA110) the development of an urban strategy, an appropriate massing, typologies and plans, and a conscious addressing of the urban realm. The project develops over the course of two semesters. The first semester is dedicated to a collective and individual fieldwork research as well as an urban plan. The second semester focuses on the further development of an architectural project and its spatial, material and technical components. Research and design are treated as integral components that inform and mutually reinforce each other. Emphasis will be laid upon experimental research methods and notation/documentation.	
Study Goals	Upon completion of the Master 3 & 4 studio the student can: - analyse and interpret the physical and social fabric within a given urban context in relation to architectural themes by means of different experimental research methods; - formulate strategies to strengthen urban assets on a specific site, develop an appropriate programme and implement these in a design proposal for a master plan; - integrate architectural concepts derived from the research into the design of a complex architectural proposal that deals with different aspects of the research outcomes in a coherent way; - develop an architectural design that is satisfying aesthetically and answers to technical / functional requirements as well as integrates a coherent structural, material and climatic design; - acquire and incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. - acquire and incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. - acquire and incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing.	
Education Method	Excursion, both to the site and to relevant architectural projects. Group work and individual work in the studio Independent design & self-study.	
Assessment	Research & design examination. This examination is an active assessment, during and at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. All relevant studies, text(s) and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals and the general P2 and P5 criteria. The assessment criteria will be provided by the tutors at the beginning of the semester, depending on the study and tasks. A text will form part of the content to be assessed.	
Special Information	The maximum marking period is 10 working days.	
Period of Education	Semester 1&2	
Concept Schedule	Friday morning and afternoon	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Building Ideologies

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 BI

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.	
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5

Course Coordinator	N.J. Amorim Mota
Instructor	Ir. P.S. van der Putt
Instructor	N.J. Amorim Mota
Instructor	R. Varma
Responsible for assignments	N.J. Amorim Mota
Contact Hours / Week x/x/x/x	4
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment. In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences. Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition. The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design. Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; Understand the concept of intersectionality and its relation with the architecture of housing; Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens; Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Housing Studies consists of three interrelated parts: Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments. Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions. Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.	
Literature and Study Materials	Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.	
Assessment	The evaluation methods in the course Housing Studies comprise a combination of two assessments: - Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10	
Concept Schedule	Friday Morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5

Course Coordinator	Prof.dr. A.R. Pereira Roders
Instructor	Dr. I. Nevzgodin
Instructor	Ir. A.C. de Ridder
Responsible for assignments	Prof.dr. A.R. Pereira Roders
Co-responsible for assignments	Y. Zhang
Contact Hours / Week x/x/x/x	4 hours per week
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.
Literature and Study Materials	Book chapters, journal articles and other lecture materials.
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%
Period of Education	Q 3
Concept Schedule	Wednesday morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5

Course Coordinator	Prof.dr.ing. C.M. Hein
Course Coordinator	J.M.K.K. Hanna
Instructor	Dr. P. De Martino
Instructor	J.M.K.K. Hanna
Responsible for assignments	Prof.dr.ing. C.M. Hein
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	3rd Quarter
Concept Schedule	Friday
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5

Course Coordinator	Dr. B. Lubelli
Instructor	Dr. B. Lubelli
Instructor	S. Naldini
Instructor	T.P. Bennebroek
Responsible for assignments	Dr. B. Lubelli
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.
Period of Education	Q3
Concept Schedule	Wednesday afternoon
Minimum number of participants	15
Maximum number of participants	60
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarions	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

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Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5

Course Coordinator	S. Corbo
Responsible for assignments	S. Corbo
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Required for	The course is also open to non-architecture majors.
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Half semester (Q3)
Concept Schedule	Wednesday afternoon
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15

Course Coordinator	Dr.ir. M.G.A.D. Harteveld
Course Coordinator	Dr.ir. R. Cavallo
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld
Responsible for assignments	Dr.ir. R. Cavallo
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)
Education Period	4
Start Education	4
Exam Period	4
Course Language	English

Expected prior knowledge

The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval.

MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080).

Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.

Course Contents

Architectural and Urban Design in Times of Transitions.
Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies.

In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism.

The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.

Study Goals

At the end of this course, the student:
knows key literature and recent research on people, movement and public space
understands main theories on people, movement and public space
applies these theories in analysing real urban settings
evaluates critically on these theories
creates presentations analysing the subject on an academic level.

And, the student:
understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city
evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning
creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation.

Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.

Education Method

The course consists of interactive studio work and lectures.

Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory.

Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches.

Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared!

Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class.

*) the discussion groups ideally consist of four/five members, who divide topics and peer each other.

Literature and Study Materials

Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.

Assessment

Studio work 80% - Lectures 20%

Assessment of studio work:
Analyses and design, presented in drawing form with written commentary and a model.

Assessment of lectures:

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15

Course Coordinator	T. Maso
Instructor	T. Maso
Responsible for assignments	J. Arpa Fernandez
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).

Summary

The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future.

Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions.

The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means.
A systematic, parametric exploration of parts of the design is an integral part of the research approach.

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO.

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

For more information about our previous studios, please visit:
https://thewhyfactory.com
https://thewhyfactory.com/education/

Course Contents

MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda.

During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)

Study Goals

- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.

- Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal.

There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios:

1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester.

2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc..

3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration.

4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors.

5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development.

6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.

Education Method

Number of studio hours: 80
Number of self study hours: 332

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program.

Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality.

Applications Program:In the applications program model cities both are tested in real cities. The different models become

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15

Course Coordinator	Dr. L. Cipriani
Responsible for assignments	Dr. L. Cipriani
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.
Concept Schedule	Wednesday, Friday
Minimum number of participants	15
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15

Course Coordinator	N.J. Amorim Mota
Instructor	Prof.ir. D.E. van Gameren
Instructor	N.J. Amorim Mota
Instructor	R. Varma
Responsible for assignments	N.J. Amorim Mota
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15

Course Coordinator	A.E. Rout
Course Coordinator	Prof.dr. T.G. Vrachliotis
Course Coordinator	M. Mateljan
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15

Course Coordinator	Ir. P.S. van der Putt
Course Coordinator	Ir. S.M. Witteman
Instructor	Ir. P.A.M. Kuitenbrouwer
Instructor	T.W. Kupers
Instructor	Ir. S.M. Witteman
Responsible for assignments	Ir. P.S. van der Putt
Contact Hours / Week x/x/x/x	112 hours per quarter
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	completed MSc1

Course Contents

The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof.
Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent.
How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined?

The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers.

The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.

Study Goals

Upon completion of the Design Studio, the students will be able to:

convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level;
perform critical comparative research that results in a clearly formulated design hypothesis;
demonstrate how urgent societal issues are addressed in the design project;
demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal;
represent space in its complex interpenetration of people, architectures, technologies, materials;
confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.

Education Method

The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive.

The Studio will be structured across:
lectures
readings, writings and public discussion
design tutorials / workshops
specific exercises challenging innovative thinking

Literature and Study Materials

Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.

Assessment

Assessment of the students work will be based on the following:
Oral presentation;
Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.;
Writing assignments and analyses.

Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student.

The final grade reflects:
the position that is formulated with regard to the brief and its context;
the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation;
aesthetic and technical/functional qualities; the elaboration throughout the respective scales;
the quality of the presentation, the products and the argument;
the consistency, coherence and development of the students work during his/her process.

Products: will be described in the syllabus which will be published at the beginning of the course.

Repair regulations
The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5).

Retake regulations
Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15

Course Coordinator	Prof.dr.ir. J.W.F. Wamelink
Instructor	Ir. H.J.M. Vande Putte
Instructor	Ir. A.C. Bergsma
Instructor	Dr.ing. V.E. Scholten
Instructor	Dr. M. Bos-de Vos
Instructor	Dr. D.M. Hall
Instructor	Dr.ir. M.U.J. Peeters
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink
Education Period	4
Start Education	4
Exam Period	4
Course Language	English

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Design, Data and Society

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 Design, Data and Society

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr. A. Sioli	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Co-responsible for assignments	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods ("Positions") highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. During the course you will get a wider overview of and insights into distinct architectural approaches, their knowledge systems, analytic and designerly tools, and values. Thereby you gain a more comparative and systematic understanding of their differences and possible complementarities, so as to recognize the wider values of scientific research as linked to design practice. By applying these insights in a case study, you will establish an intellectual sensitivity to this relationship between research and design. This stimulates you to intellectually position these approaches amongst each other, or within wider discourses; it also invites you to critically reflect on your own position within the larger knowledge systems of architectural practice.	
Study Goals	After following the course, you should be able to: LO1: identify different methodological positions (i.e. knowledge systems, epistemological frameworks, and concepts) within architectural research and design; LO2: Analyze an architectural case study in its reflective as well as practice aspects by means of self-selected theories, methods, and tools; LO3: Present a critical reflecting upon positions in the field	
Education Method	The course consists of several lectures and seminars that stand in relation to each other. In the lectures, 5 different societal themes are presented, and related to architectural research methods and design positions. In seminar groups, students explore these research and design positions further by means of a case study analysis. Each seminar consists of 4 sessions, with three tutorials and a presentation and reflection to their tutors and peers in an academic setting. The analysis is written in a group of 4-6 students, and includes a collectively written introduction and analytical framework, individually-written chapters, and a collective conclusion and reflection.	
Literature and Study Materials	The compulsory literature for the course is the book T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded solely on the basis of the final assignment (comprising the collective written analysis and presentation). Graded (in principle) as a group work, the collective assignment will constitute the principal grade assignable to the Lecture Series. The group assignment will be evaluated based on the groups ability to position multiple possible ways in which to study the project. It is evaluated according to four grading components. The first aspect concerns the general ability to understand positions regarding architectural research and design, as well as related methods (30 %); the second component concerns the academic skill of conducting of research (30 %); the third component the ability of constructing, formulating and presenting a well-structured and critical argument (40 %).	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; - analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	Writing Assignments: 1- Written responses to readings and lectures by small groups of students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2- A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: - Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 - Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	M. Parravicini	
Contact Hours / Week x/x/x/x	Average between 4 and 6 hours per week, no education in week 1.9 and 1.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The program of the BES deals with present and future challenges in the field of architectural engineering, guiding students to evolve into environmentally conscious architects in this era of transitions and climate change. Each Building Engineering Studio (BES) forms an integral part of a related MSc1 Architectural Design Studio. The design theme set by the Architectural Studio also guides the content of the BES. The BES studios run parallel to the architectural design studios, each studio having its own assignment; each technical design project is specific and tailor-made to each studio. The main topic of the BES is the sustainable design of the technical aspects of a building through its different scales and domains (construction, climate and structure). Each design addresses specific sustainability topics through the conceptual integration of engineering into the architecture and its circularity. Through research on circularity, students move towards an elaborated position in their assignment that will guide the overall technical design process.	
Study Goals	By the end of this BES Master 1 course, students should be able to: LO1: Design a building, at Msc1 level, in mainline and on aspects which is coherent, significant, elaborated, correct, innovative and socially / environmentally responsible (sustainable, circular), by incorporating a building engineering concept and its elaboration within an architectural concept. LO2: Elaborate in mainline and in some detail the technical and material aspects of the design, including structure, climate, and construction, by following sustainable and circular design strategies. LO3: Demonstrate through a presentation the reasoned application of Building Engineering concepts and circular concepts into the design as a whole. LO4: Describe, analyze, criticize and present an existing building under the circular design point of view by using the CiDeA given template in a group work.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. For all BES studios the following methods are applied: Lectures on technical building design and materials. Analysis in small groups of a building according to the given template of the CiDeA (Circular Design Atlas) on its circularity aspects, presentation and group feedback. Design of a building through several iterations, by using technical drawings, study models, diagrams, and sketches, with individual feedback. Correct application and interpretation of technical details and solutions, with additional attention to innovation, by designing a fragment in detail, with individual feedback. At least one feedback moment by another BES mentor of the same studio, plus feedback at the mid-term presentation and after the final presentation.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand, the presentation is dependent on the theme and method of the studio. On the other hand, the presentation products have to show the content as formulated in the study goals and course content. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio. In general, every BES studio is assessed as follows: Process: in-class individual guidance and feedback. Products: design of a building through plans, sections, diagrams, details, models, and explanatory text. Mid-term and final presentations of the whole project through posters (with text, plans, sections, diagrams, details), models, and a coherent verbalization to demonstrate the correctness and the integration of the building engineering design (sustainable and circular) within the architectural design. Presentation (diagrams and text, by using the CiDeA chapters), reflecting on the design under the aspects of circularity. Weight of the design assessment: 100% at the final presentation. Research: group report and presentation of the CiDeA research using the given template, with Q&A and conclusions. Weight of the assessment for the CiDeA group work: it must be complete and presented to pass. Examiners: at the mid-term and final presentations, at least, assessment by the BES mentor and the Architecture design mentor. After the final presentation and before grading, all the products are examined by, at least, two BES mentors from the same studio. The maximum marking period is 10 working days. Feedback is given after the final presentation. Repair and retake are possible once in the same Academic Year for grades of 5.5 and lower. A retake implies a new short course on the same topic and a new examination. A repair is a revision to the existing assignment and can be discussed for tasks that are feasible within a short term and with limited effort. Retake and repair will be assessed with grades taking into account the advantage resulting from the extended time. For the assessment, the BES studio makes use of the EMMA rubrics that will be delivered to the students via Brightspace.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1DDS005	Design, Data and Society MSc1 Design Studio	10
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	MSc 1 (AR1DDS005): 8 hours per week starting from week 1.1 and ending in week 2.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected students have prior knowledge from completing a bachelors degree in architecture.	
Course Contents	In this MSc1 studio students will combine data and analysis within their design processes, in order to push the boundaries of the discipline of architecture to better address societal and environmental urgencies. The studio considers the building as a link between data and societal needs. This link allows for innovation and novel ideas for new buildings and typologies that are informed by todays technological, cultural and ecological context. The resulting design should inspire new ways to imagine buildings in the digital age, and also provide a practical and implementable solution given the context. Overall, the project will be rooted in data as a tool for designing, explaining, and measuring buildings and their impact. The design project, a medium sized building, will be located at a world class intellectual centre that spearheads new modes for thinking, living, and thriving in the age of technology. Under this umbrella concept, the studio will evolve a perspective on new building typologies. Examples include: The Brain Center, Clinic of the Future, and Living Laboratories. Location examples include: Cambridge University, Oxford University, Alan Turing Institute, Erasmus AI Institute. The MSc1 Studio operates in close collaboration with the Building Engineering Studio (BES) to promote new sustainable designs, materials, and construction. The BES instructors take responsibility for the technical, structural, and performative aspects of the design exercise, implementing a collaborative educational agenda focused on sustainability, energy efficiency, designing for disassembly, and adaptability to climate change. By working together, the Studios design instructors and the BES aim to develop building designs that are both innovative and environmentally responsible.	
Study Goals	Upon completion of the MSc1 Design Studio, the students should be able to: Analyse the historical, technological, cultural, behavioural, and environmental context to identify strategies to inform a new experimental building design. Design a medium sized building using social and environmental data, integrating architectural (functional, experiential, aesthetic) and building technology (structural, material, sustainability related) aspects. Visualise information and insights to effectively communicate architectural design qualities.	
Education Method	The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. The scientific approach will be combined with architectural narrative and visual representation to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. The Architectural Design Studio (AR1DDS005) and the Building Engineering Studio (AR1A080) are fully integrated to ensure the comprehensive development of each proposal. By combining these two areas of study, students will be equipped to translate their design concepts into practical, buildable solutions. Students will conduct research in groups, but will elaborate on their individual project for the final presentation and assessment. Each of these Studios runs parallel activities with the same objectives, including: Lectures Public discussions Field investigations / site-visits Formal and material design-oriented assignments in drawing and modelling Readings and writings	
Course Relations	Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues? The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas: CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives. ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort. RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges. Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them. The DDS MSc1 course is also a collaboration with the AR1A080 Building Engineering Studio	
Literature and Study Materials	Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace	
Assessment	The assessment consists of two parts: Analytical assignment for the research phase and Design examination of the design phase. Analytical assignment products: Research Report and Presentation (research products will include site analysis, material	

	resource analysis, and architectural reference analysis) Design assignment products: Architectural Drawing Set, Physical model, Presentation/Booklet (design representations will include concept schemes, site plan, floor plans, sections, elevations, critical details, collages/renders) The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). Detailed criteria for assessment will be communicated in the Studio Syllabus. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The maximum marking period is 10 working days
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	In Fall 2024, the MSc1 Design Studio focuses on the architecture of The Brain Center in the context of a large research institution. A site visit to Cambridge University in Great Britain is included in the program. Students should factor in potential additional expenses for printing, travel, and accommodation: ±300 euros per person, contingent upon the specific location and available options.
Period of Education	Q1 + Q2
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 AD

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. C. Wagenaar	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Ir. Y.B.C. van Mil	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. R.J. Lee	
Instructor	V. Baptist	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: - Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to succesfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. In line with the precursor course AR1A066, the history thesis explores architectural practice as it has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. A history thesis can also study the ways in which historical investigation has shaped the course of architecture and architectural practice in global histories and historiographies, exploring also the ways in which these histories are constructed and narrated. The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 5-6000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. Is able to plan and develop a scientific research project. Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. Is able to evaluate, interpret and make proper reference to available sources. Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material--readings and documents on research and writing--is available on the course Brightspace page.	

Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5ects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory. Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance.	
Course Contents	The Architecture Theory Thesis Seminar 'Thinking | Reading | Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator is advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	75
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment)

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architecture electives (10 EC)

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	

Course Contents

This course aims to encourage students to investigate the relationship between environmental design and human behavior from a cross-cultural perspective, while recognizing their own perspectives. It adopts a transdisciplinary approach, incorporating architectural anthropology and utilizing various research methods such as visual ethnography, spatial analysis, and participant observation to explore the interactions among humans, non-humans, and the environment.

In the initial phase of the course, participants will embark on an auto-ethnographic exercise to kickstart their learning journey. Following this exercise, they will conduct fieldwork, review relevant bibliographic references, collect and analyze data to develop a preliminary research output. In the subsequent phase, students will collaborate in groups to synthesize the collected data, utilizing architectural drawings as a primary means of facilitating meaningful communication across diverse audiences.

Throughout the course, students will be encouraged to maintain a personal logbook to record their activities and reflections. This logbook should document each student's individual perspective on the challenges, opportunities, and relevance of employing ethnographic methods in architectural research.

Study Goals

By the end of this course students will be able to:
1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective;
2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods;
3. Elaborate research outputs with a synthesis of the field work, literature review and data analysis, using adequate written and visual media;
4. Use story-telling as a medium to elaborate multi-sensory narratives based on the results of the fieldwork, data collection and analysis;
5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences.
6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.

Education Method

The course Architectural Ethnography comprises group assignments and individual work.
The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce deliverables containing a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course.

The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area.

Literature and Study Materials

The course will use the following publications as main textbook references:

Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts.

Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto

Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126.

Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE.

Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.

Assessment

The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects.
The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment.
The summative assessment will be based on the following deliverables:

a) Analytical Assignment:
Ethnographic Research: Research Report (Group Work)

b) Practical Exercises:
Poster: Synthesis of (Quantitative) Data Collection (Group Work)
Ethnographic Research: Narrative and Visual Synthesis (Group Work)

c) Writing Assignment:
Critical Reflection (Individual Work)

For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c).

The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.

Retake and repair regulations
In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year.
Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or

	excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An Open Intersectional Archive	5
Course Coordinator	Dr.ir. D. van den van den Heuvel	
Instructor	Dr.ir. D. van den van den Heuvel	
Responsible for assignments	Dr.ir. D. van den van den Heuvel	
Co-responsible for assignments	Dr. A. Campos Uribe	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	

Course Contents

This course advances critical tools for the analysis and projection of affordable collective housing design, examining the relevance and applicability of intersectionality theory to housing. The course looks into global tendencies of modernist housing production, yet with a focus on Dutch traditions. Methodologically, it investigates these traditions from the critical lens of 'global intersectionality'. The course combines analyses of case studies, archival studies and reading literature, followed by the development of a critical intersectional archive, organised in the form of a collectively-curated exhibition.

The course starts with an introduction to the Dutch national collection of architecture and urban planning at the Nieuwe Instituut in Rotterdam, and a discussion of intersectionality theory including its relevance and applicability to housing design.

Students will then develop analyses of selected case studies, revealing the historical and environmental circumstances of the project in relation to its cultural, material and architectural features as follows:
- Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerged, including institutional and governance frameworks and policy-making processes;
- They will investigate how each case study and its designers are embedded in the field of housing histories and cultural production, and how this relates to particular design features (typo-morphology, composition, structure, details, materialisation, technology);
- Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labour, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife.

In the last phase of the course, students will develop a critical intersectional archive that will be organised in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analysed case studies through research-based documentation, including graphic and textual outputs.

Study Goals

Upon completion of the course the student is able to:

Analyse and synthesise the main generative design components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs;
Understand the concept of intersectionality and its relation with the architecture of housing;
Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of 'global intersectionality' as a critical lens;
Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research;
Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design;
Present a curatorial design strategy to an audience of experts and non-experts.

Education Method

The course Housing Studies consists of three interrelated parts:

Part 1 - Concepts and Theory: This part is based on literature review. Each session will be organised on the model of flipped classroom through lectures and peer-review assignments.
Part 2 - Analysis: This phase will be based on the application of theoretical and historical research through morphological and typological design analysis of case studies. Students produce critical drawings based on collected archival materials. This phase will be based on tutorial sessions.
Part 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.

Course Relations

The course Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012), and broadly relevant to other housing design studios.

Literature and Study Materials

Readings and selected case studies will be presented in the course syllabus, which is updated with each edition.

Assessment

The evaluation methods in the course Housing Studies comprise a combination of two assessments:

- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material (drawing) towards the constitution of an intersectional archive.
- Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment.
- The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.

Period of Education

Spring Semester
Third Quarter: Week 3.1 - week 3.10

Concept Schedule

Friday Morning

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Pereira Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Pereira Roders	
Co-responsible for assignments	Y. Zhang	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.	
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.	
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Book chapters, journal articles and other lecture materials.	
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simply this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The course is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 12 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.	

	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	Total minimum number of participants: 12
Maximum number of participants	Total maximum number of participants: 40 Non-BK students: 10
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5

Course Coordinator	Prof.dr.ing. C.M. Hein
Course Coordinator	J.M.K.K. Hanna
Instructor	Dr. P. De Martino
Instructor	J.M.K.K. Hanna
Responsible for assignments	Prof.dr.ing. C.M. Hein
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8
Education Period	3
Start Education	3
Exam Period	none
Course Language	English
Course Contents	History matters to design, yes, but in what way? This course explores prominent themes in the history of architecture and urban planning in relation to contemporary challenges. Different ways of investigation (mapping, archival research, etc.) are used to discuss and question the relationship with current scientific and practical disciplines as well as the design discipline. The course aims to activate the past for the design of the future. It does this through a focus on critical spaces, such as (but not limited to) port cities in times of transition. Previous case studies have included Beirut, Naples, Rotterdam, Venice, Dunkirk, Rabat/Sale, Casablanca, Tangier. Check the catalogue for this year's theme or contact c.m.hein@tudelft.nl.
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.
Education Method	Lectures, Discussions, Tutorials, Workshops, Fieldtrips
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	3rd Quarter
Concept Schedule	Friday
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Study materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided in the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The porftolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment For questions contact a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Tuesday afternoon	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	24	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter 3	
Concept Schedule	Tuesday afternoon (Group 1, and 2) or Friday morning (Group 3)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Remarks	The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Friday	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. The exercise brings together ambitions of composition research and evocative, analytical modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and analytical exploration of a project. use and construct models; digital, graphical or physical models representing design issues. formulate and defend findings and conclusions, orally and in writing. contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contriburing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	Dr. Dipl.-Ing. H.H. Bier	
Responsible for assignments	Dr. Dipl.-Ing. H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxdelft.nl/2015/04/tedxdelft-events-tedxdelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqsLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwwMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNlIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Contact Hours / Week x/x/x/x	average of 4 hours per week, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	**Technoledge Design Informatics**	**5**
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Instructor	T.P. Bennebroek	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Y. Söylev	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course) On the scale of 10, the student achieves a sufficient result with the final grade of 6. If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSC1	
Summary	The Form and Inspiration course consists of an artistic journey from a personal inspiration to articulated, detailed form modelled (physically) on large scale. Students will thoroughly explore a chosen Inspiration through carious artistic expressions, abstractions and translations in order to reinterpret and express an architectural Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. This is a process of attentive observation and physical making. Your aim for this course is, to develop a thorough understanding of architectural form and its expression through artistic exploration, rooted in a personal fascination.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches and models, Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course without additional guidance. The repair task will be communicated on the day of the presentations.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students:	

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Special Information	for more information you can contact the responsible instructor or course coordinator.
Period of Education	2e semester, 10 weeks in quarter 3
Concept Schedule	Thursday afternoon
Used Materials	Various materials for drawing, painting, collaging, photography and modelling. The student is responsible for these materials, based on their specific artistic exploration and thematics.
Leerstoel	Formstudies
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	W.C. Yung	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvements of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by settingthe design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Responsible for assignments	Prof.dr. T.G. Vrachliotis	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Co-responsible for assignments	Ir. F. Geerts	
Co-responsible for assignments	N. Sanaan Bensi	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a ficto-critical short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0915	Online Digital Portfolio	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 94 hours	
Computer Use	Laptop required	
Literature and Study Materials	- The Brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Elective	Yes	
Period of Education	Semester	
Concept Schedule	Wednesday, lectures 10:45 - 12:30 Wednesday, workshops, group of max 15. students every two hours	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0918	Adaptive Architecture for Public Buildings: Theories and Practices	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	The course is also open to non-architecture majors.	
Expected prior knowledge	Software for design, graphics, data representation, and presentation (Rhinoceros, Autocad, Adobe Suite, Revit) Scientific research skills	
Course Contents	This seminar course will present the different theories and operative strategies informing architectural reuse, by focusing attention on the transformation of contemporary public buildings. In particular, the course will investigate notions of form, vocabulary, legibility deriving from the dialogue between old and new structures. What does transforming existing structures entail in terms of language and recognizability? How can old and new interact visually and formally? What are the strategies describing their relationship? (continuity, contrast, juxtaposition, fragmentation, etc.) How can reuse connect to climate adaptation? How did the act of transforming existing buildings evolve over the centuries? The course will address these questions within the general context of circularity. Students will be introduced to the most relevant positions on the topic, and will explore the consequences that terms such as retrofit, upcycle, adaptability, disassembly, etc. can have in the territory of architectural design. They will study historic precedents of reuse, current trends and also future directions. Students will not only be introduced to the current debate on reuse; they will also experiment with cutting-edge strategies that can contribute to the design of healthy living environments and public buildings. The course will consist of lectures and conversations with invited guests, who will introduce the students to a multiplicity of approaches. Students will react to those inputs by applying their design strategy to a concrete case study a small-scale project of adaptive reuse.	
Study Goals	Upon successful completion of the course, students will be able to: - understand the relationship between circularity and the contemporary architecture discourse; - recognize and evaluate the aesthetic potential of sustainable design; - investigate design techniques related to the adaptive reuse of existing buildings; - explore the relationship between old and new buildings in terms of legibility, identity, juxtaposi¬tion, assemblage.	
Education Method	The seminar runs several activities, including: Lectures & discussions Literature review Design assignments (collective work) Writing assignments (individual) Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0919	Exteriorless: The Design of the Ordinary	5
Course Coordinator	S. Corbo	
Responsible for assignments	S. Corbo	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Contemporary cities characterize themselves for the presence of building types which, despite the diversity of functions they accommodate, share a same condition: their floor plans are typically generic, their exteriors hermetic, their interiors vast. These spaces are warehouses, department stores, fulfillment hubs, data centers. Generally considered as conventional and banal structures, these spaces are in reality extremely relevant for the role they play in the daily functioning of our cities. This seminar course will investigate the nature of these types, as well as their spatial qualities and their material components, in order to ideate design strategies that can turn them into meaningful and expressive structures public buildings open to the city, capable to act as collective magnets and to generate new urban dynamics. By exploring how these structures can progressively be opened up, to become active part of the urban fabric, students will learn new design techniques: via different exercises, they will turn neutral structural grids into interior landscapes; they will engage in biophilic operations of green design; and, finally, they will use multi-sensory elements to transform existing typologies. What design strategies can we implement to turn the generic and the ordinary into the meaningful and the diverse? How can we give these spaces back to the cities, by injecting them with new functions and activities? How can existing and new functions hybridize? The course will consist of lectures, readings, and conversations with invited guests from fields other than architecture. The course will be divided in two parts; the first part will be research-oriented, in which students will select and investigate one specific spatial type; in the second part, students will apply the outcome of their research into the (re)definition of an existing type via light, colors, nature, sound, technology, etc.	
Study Goals	By completing the course, the students will be able: to situate the emergence of new spatial types in relation to the current phase of capitalist development; to understand the role of these spatial byproducts in the making of the built environment and their inner logic; to comprehend the design potential of generic structures; to develop design strategies aimed to envision public, inclusive and diverse buildings, which can act as urban catalysts; to experiment with design techniques such as density, hybridization, assemblage, collision.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar runs several activities, including: Lectures and discussions Literature review Research assignments Design assignments Presentations & reviews The class will be divided in groups depending on the number of participating students. In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, physical models, material samples, etc.; Writing assignments and analyses. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	4 hours per week, starting in week 3.1 and ending in week 3.10, no education in week 3.9	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process On the scale of 10, the student achieves a sufficient result with the final grade of 6 If graded below 6 (fail) a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Archtecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3,2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Dr. P. De Martino	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2.Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3.Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4.Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	Dr. M. Turrin	
Instructor	Dr.ir. M.C. Lugten	
Responsible for assignments	Dr. S. Brancart	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.	
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.	
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 90.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Expected prior knowledge

The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval.

MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080).

Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.

Course Contents

Architectural and Urban Design in Times of Transitions.
Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies.

In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism.

The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.

Study Goals

At the end of this course, the student:
 knows key literature and recent research on people, movement and public space
 understands main theories on people, movement and public space
 applies these theories in analysing real urban settings
 evaluates critically on these theories
 creates presentations analysing the subject on an academic level.

And, the student:
 understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city
 evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning
 creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation.

Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.

Education Method

The course consists of interactive studio work and lectures.

Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory.

Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches.

Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared!

Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class.

*) the discussion groups ideally consist of four/five members, who divide topics and peer each other.

Literature and Study Materials

Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.

Assessment

Studio work 80% - Lectures 20%

Assessment of studio work:
Analyses and design, presented in drawing form with written commentary and a model.

Assessment of lectures:

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	T. Maso	
Instructor	T. Maso	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	

Summary

The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future.

Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factorys findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions.

The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means.
A systematic, parametric exploration of parts of the design is an integral part of the research approach.

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO.

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

For more information about our previous studios, please visit:
https://thewhyfactory.com
https://thewhyfactory.com/education/

Course Contents

MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factorys agenda.

During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)

Study Goals

- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.

- Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal.

There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios:

1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester.

2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc..

3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration.

4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors.

5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development.

6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.

Education Method

Number of studio hours: 80
Number of self study hours: 332

STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO

During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group.

The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program.

Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality.

Applications Program:In the applications program model cities both are tested in real cities. The different models become

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	12
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15

Course Coordinator	Dr.ing. M. Bilow
Responsible for assignments	Dr.ing. M. Bilow
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the cursiosity of the students.
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Special Information	The maximum marking period is 10 work days.
Period of Education	Quarter 4
Concept Schedule	Tuesday and Thursday - details may vary
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extend can architecture provide an inclusive and healthy living environment for all. To understand the needs of the people within the selected environment (understanding and analyzing), the elective starts with a short phase of human-centered research (visual anthropology, participation and/or interviews) and the evaluation of the outcomes. Then, design work starts, done individually or in groups of two students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society(more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). At the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the (sub-)urban environment? We will explore the question by looking at one chosen neighborhood as a multi-domain structure and by working on different scales. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. We will discuss new ideas for an inclusive living environment.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods. - Studio specific study goal 2: The student is able to understand, apply, analyze and evaluate the research outcomes of the potential multiple user groups and their demands - Studio specific study goal 3: The student is able to translate and discuss research outcomes into design. In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. demonstrate sufficient insight in and knowledge of the design process position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location will be done; in-depth research on location - preferably combined with a stay at the location or several come-back days (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be send to the students before the course starts. Here the student finds literature suggestions as well.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product until the end of the course. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway or in consultation with the tutors, at the end of the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered and uploaded on Brightspace. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the course; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Course enrolment via Brightspace only is not considered a valid registration	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15

Course Coordinator	Dr. L. Cipriani
Responsible for assignments	Dr. L. Cipriani
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.
Concept Schedule	Wednesday, Friday
Minimum number of participants	15
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0897	Van Gezel tot Meester	20
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Instructor	Dr.ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a mattter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explcit and training meaningful actions and skills. Both are main subject in this MSc 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand. Be aware: course is in Dutch, because of the internship in the BSC first year	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design	
Education Method	In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistent teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio. *) In case of specific circumstances, the internship can be replaced by other ways to explore design education	
Assessment	Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises,lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	12 hours/week, no education in week 4.9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization and/or facing the challenges of climate change. To support the development of the project, participants in this course develop social-spatial analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphologies and typologies, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design manifesto / strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 7. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 8. Produce accurate and meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts. 9. Collaborate with peers in group work, contribute with personal inputs to the development of shared design goals and engages in experimentation and exploration of solutions for housing design.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, bi-weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan must be based on a clear design hypothesis, which should entail a coherent narrative framing the knowledge acquired from the first phase into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a residential project, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (during the tutorial sessions), and at the end of each phase. The summative feedback will be based on the assessment of the process of each individual student and an individual dossier. The individual dossier (format to be decided) should include the deliverables of group and individual work, distributed in the following phases: PHASE 1_Foundations Assignment 1. Fieldwork / Contextual Research (group of 4-5 students). Assignment 2. Problem Statement and Design Manifesto (Individual work). PHASE 2_Project Assignment 3. Urban Strategy / Masterplan (Group of 2-3 students). Assignment 4. Housing Project (Individual Work). Assignment 5. Project Description and Reflection (Individual Work). The following processes will be part of the course management and organisation: - Different group composition for assignments 1 and 3 - Peer Evaluation will be used in Assignment 1. - Self-Evaluation will be used in Assignment 3. - The use of AI in Assignment 5 is not allowed to produce content, but it is allowed to support improving clarity and preventing language issues such as typos and errors. Submission of the document will be made to Brightspace and a plagiarism check will be performed by Ouriginal. The final grade will be calculated with the following weight: Group Work: 30% Individual Work: 70%	

	In week 4.5, students will receive formative feedback on the results of the work produced for Assignments 1-2. In week 4.8, each student will receive formative feedback on the results of the work produced for Assignments 3-4. At the end of week 4.10 each student should compile a personal dossier including all the final versions of the work developed for Assignments 1-5. The Final grade will be determined by three components (Design, Reflection and Presentation, Study and Process), according to the following weight distribution, assignments, and relation with Learning Objectives: A DESIGN (70%). Related Assignments: Assignments 2, 3, 4 Related Learning Objectives: LO 2, LO 3, LO 4, LO5, LO 6 B REFLECTION AND PRESENTATION (20%). Related Assignments: Assignments 1, 2, 5 Related Learning Objectives: LO 1, LO 2, LO 7, LO8 C STUDY AND PROGRESS (10%). Related Assignments: All Assignments Related Learning Objectives: LO9 Students are approved with a grade of 6 or above. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Terms There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic). Principles & grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a grade of 6. 3. The retake and repair rules apply to all grades and partial grades registered in Osiris, as well as to partial grades not registered in Osiris that must be completed satisfactorily. Only partial grades of individual assignments are eligible to compensate insufficient grades, and be considered for the final grade after a retake of repair. Deadlines The repair should be completed by the student a maximum of two weeks after the deadline of the last examination / final presentation. The retake should be completed until the last week of the academic year.
Remarks	To participate in this studio, there will be a required field trip to the project's site lasting approximately two weeks during the Spring semester (mid-April to early May). The estimated cost of the field trip ranges from 1,000.00 to 1,500.00. Each participant is responsible for covering this expense and/or exploring opportunities for securing funding or sponsorships for the trip. If a visa is required, it is the student's responsibility to initiate the application process with the appropriate authorities.
Period of Education	The course is offered in the Spring semester, Q4
Concept Schedule	Tuesday and Friday
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	S. Corbo	
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays climate change, environmental pollution, spatial segregation, invasive privatization, etc. More specifically, the Studio focuses on architectures ability to resist shocks and adapt to future changes. In other words, the Studio tackles resilience as a tool as well as a design goal. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Topics such as climate-adaptability, design for disassembly, and flexibility will be explored. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; focus on the qualitative aspects of multiplicity in society and design; design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; confront questions of flexibility, adaptability and hybridization when developing a comprehensive design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: lectures field trips readings, writings and public discussion experimental research design tutorials specific exercises challenging innovative thinking In joining this course, the students agree to potentially have their work published on the Departments website, on social media and in other printed forms.	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Written statements and analyses. The design proposal is individual. The assessment process implies: in-class participation on a weekly base with public discussion of in-between results; Midterm and Final Reviews; The final grade reflects: Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The maximum marking period is 10 working days.	

Special Information	
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	The student is able to present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student is able to elaborate on the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. The student is able to develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (initiative, in-class discussion) (10 %) - Final Exam (clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	Y. Söylev	
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Instructor	Y. Söylev	
Instructor	I. Zaid	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2025.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking and reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. The maximum marking period is 10 working days* * In accordance with Article 1.26 of the Teaching and Examination Regulations Bachelor, and Article 1.27 of the Teaching and Examination Regulations Master, the marking period for each course should be published in the Study Guide. Results need to be determined by the examiner as soon as possible but no later than 10 working days after the examination. Only with permission from the Director of Education the marking period can be extended to a maximum of 15 working days. Standard condition for retake and repair: The student has achieved a sufficient result on scale 1 to 10 with 6. The student has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The retake and repair options students have for all (partial) examinations; this need to be in accordance with the Retake and Repair Regulations of the Board of Examiners. There is a difference between retake and repair. 1. A retake is an entirely new examination or assignment. 2. A repair is a revision or addition to the existing assignment (with the same topic).	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday and Thursday	
Minimum number of participants	12	

Maximum number of participants	40
Course evaluation	Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Leerstoel	Architectural Design Crossovers	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DDS005	Inhabiting Data: People, Objects, Spaces	15
Course Coordinator	A.E. Rout	
Course Coordinator	Prof.dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Contact Hours / Week x/x/x/x	MSc 2 (AR2DDS005): 10-12 h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	

Course Contents

The Design, Data and Society (DDS) MSc2 Design Studio Inhabiting Data challenges students to rethink and re-evaluate cultural techniques and architectural design methodologies including data analytics and AI perspectives, through an experimental design assignment. Students will design small or medium scale conceptual design projects that explore the relationships between people, actions, objects and spaces in the data society. Design projects will connect to the broader cultural, technological, environmental and infrastructural networks of society that are intrinsically bound to architectural expressions. The studios position is that these networks and their associated processes inform patterns of human life, transforming the spaces humans inhabit: from the most intimate living/working spaces, to larger social and industrial production spaces. By (re)examining and (re)constructing relationships between people, actions, objects and spaces through the perspective of data, architects can address current social and ecological challenges, contribute to understanding of the world, and promote positive change through architectural design. Each year, the studio will delve into the concept of "inhabiting data" by examining various types of spaces. These may include living and working environments, logistics and delivery spaces, public spaces, healing environments, and more.

The aim of the course is to bridge the gap between abstract concepts like data, information and AI and their application to architectural design. This entails development of new knowledge perspectives on design information and representation, spatial perception, design imagination, aesthetics, subjectivity, design ethics and the role of the architect in the age of data and AI.

Study Goals

By the end of the MSc2 Studio, students should be able to:

Explain how data collection, analysis, and interpretation informs the design outcome, considering both human and technological (AI) perspectives
Design an innovative building at a conceptual design development level, considering multiple spatial scales (site, building, fragment) and multiple design aspects (spatial, functional, aesthetic, cultural)
Visualise key design and data-related features using effective narrative and visual storytelling methods

Education Method

The Design, Data and Society Group (DDS) uses a scientific approach to data collection and analytics as a starting point to describe social processes and their connection to the built environment. Students will learn basic fundamental theories, methods and techniques of data collection and analysis. The scientific approach will be combined with architectural narrative and visual storytelling methods to convey information and insights in a way that they relate to spatial, functional, experiential and aesthetic qualities of architecture. Students are encouraged to experiment with drawings and visual representation and storytelling techniques, and use mixed-media methods to present their ideas. Studio will include reading seminars discussing theoretical and historical frameworks and interdisciplinary lectures, involving data and AI experts, computer scientists and other experts from industry, depending on the specific theme. The design products (posters, booklets) will be developed iteratively, through weekly assignments. Students will conduct research in thematic groups, while the design projects will be developed individually. In summary, studio activities include:

Design tutorials (including individual design progress feedback and group discussions)
Seminars (including lectures, workshops and reading discussions)
Excursion (study trip including the site visit)

Course Relations

Design, Data, and Society (DDS) Group critically explores the technological conditions under which architecture in the age of global digital infrastructures emerges. Data has operationalised our knowledge about space and time, people, materials, buildings, and cities, and has become a medium of integration - converging models, media, and methods from different disciplines and scales. What does it mean to design in a society that seeks a balance between the datafication of all areas of life, and urgent societal issues?

The Design, Data, and Society (DDS) education program enhances students abilities to address societal and ecological challenges. It equips students with skills and knowledge in design data literacy, connecting their design choices to social and environmental impacts, and heightening their understanding of broader techno-cultural processes. DDS teaches theoretical as well as practical skills in three primary areas:

CULTURE AND ETHICS, enabling students to re-think and re-evaluate established cultural techniques in architecture - reading, writing, modelling, drawing, seeing, searching from data analytics and AI perspectives.
ARCHITECTURAL QUALITIES, enabling students to connect data and AI to designs that elevate the Qualities of Architecture: sensory, spatial, visual, experiential, cultural, organisational, aesthetic, material, climate, and comfort.
RESOURCES AND TECHNOLOGIES, enabling students to apply mixed-methods techniques to evaluate, apply and derive meaning from societal and resources data in order to inform design processes and challenges.

Through the development of design projects in DDS studios, students explore the link between conceptual language of architectural objectives and qualities, and metrics for measuring, evaluating and improving them.

Literature and Study Materials

Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.

Assessment

Students achievement of learning goals will be assessed through Design examination, based on specific design products. Examples of final design products include:

Posters (including architectural drawings and other visuals)
Booklets and/or digital presentations
Short animations (such as GIF animations)

Some aspects of evaluation include: clarity and elaboration of the key design idea, creativity in visual storytelling, and effective summarization of the project narrative. Detailed criteria for assessment of individual products will be communicated in the Studio Syllabus. The midterm formative assessment will take place halfway through the studio program (not graded) while the final summative assessment (graded) will take place in the last week of the studio. Guests critics will be invited for the midterm and final presentations. The maximum marking period is 10 working days.

	Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The MSc2 Studio may take place either within the Netherlands or internationally. To ascertain the locations for this year's site visits, kindly reach out to the studio coordinator. Students should also factor in potential additional expenses for printing, travel, and accommodation, which could range from 50 to 150 euros per person, contingent upon the specific location and available options.
Period of Education	Quarter 4 (spring semester)
Concept Schedule	Tuesdays and Thursdays (mornings, and occasionally afternoons)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. S.M. Witteman	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	T.W. Kupers	
Instructor	Ir. S.M. Witteman	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Public Building and Housing Design - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The Studio High-Rise Culture will attempts to address those questions by setting up a collaboration between the Groups of Public Building and Architecture & Dwelling. Therefore tutoring sessions will be performed by members of both groups. Regular Studio sessions will be integrated by the contribution of external guests and lecturers. The Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Students will develop design strategies according to an interdisciplinary perspective, which blends different tools physical and digital modeling, advanced architectural representation, conventional drawings.	
Study Goals	Upon completion of the Design Studio, the students will be able to: convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc2 level; perform critical comparative research that results in a clearly formulated design hypothesis; demonstrate how urgent societal issues are addressed in the design project; demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; represent space in its complex interpenetration of people, architectures, technologies, materials; confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The section Public Building and Housing Design adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage in collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: lectures readings, writings and public discussion design tutorials / workshops specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: the position that is formulated with regard to the brief and its context; the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; aesthetic and technical/functional qualities; the elaboration throughout the respective scales; the quality of the presentation, the products and the argument; the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course. Repair regulations The student has achieved a sufficient result on a scale of 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,5 or lower. A repair has to be completed within 2 weeks after completion of the Studio. If a repair is done insufficiently, the student will be awarded the original grade (5,0 or 5,5). Retake regulations Students with a grade below 6,0 (5,5 or lower) will have the possibility of re-submitting the deliverables once over the course of that year. The retake submission deadline is set 1 week prior to the start of the new academic year.	

	The maximum marking period is 10 working days.
Period of Education	Half semester (Q4)
Concept Schedule	Friday
Leerstoel	Combined studio of the groups of Public Building and Architecture & Dwelling.
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020	**Urban Archipelago**	**15**
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. P. De Martino	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels and climate-change related spatial, social and cultural transformation. It rethinks the role of water as a foundation for the design of water-aware architectures and spaces. It investigates the spatial, social and cultural aspects of water embedded in infrastructures and buildings, in institutions and policies, in drawings, music, or writing. The studio posits that we need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth. Water is essential to human and non-human life through time and space. Architecture and urban design have a key role in bringing back water awareness and in developing future scenarios for living with water. The studio prompts students from Architecture, Urbanism and Landscape Architecture to develop strategies that connect architectural objects, urban settlements, landscapes with water systems. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. The Urban Archipelago course focuses on water systems and natural waterbodies as spaces of coexistence and conflicts. New scenarios and maritime mindsets are needed to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	The course helps students develop a comprehensive toolkit, including mapping (and counter-mapping), archival research, and vertical or sectional thinking. Students are urged to critically assess, test, challenge, and propose alternative tools that can enhance knowledge production about the sites and territories they engage with. Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable water systems, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign architecture and cities with a focus on water, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable territories.	
Education Method	Lectures, Seminars, Discussion, Studio design, Tutorials, Field trips.	
Assessment	Analytical assignment consisting of research and design components. Evaluation based on overall performance in the course. Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Tuesdays and Thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	A. Stanii	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions. Student work may be used for educational purposes, and students must self-notify the coordinator/ lecturer of the course if they do not wish to give permission to do so.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.	
Period of Education	One academic quarter, Q4	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Responsible for assignments	Dr. A.R. Thomas	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This studio will focus on the notion of 'play' in architecture and urban design. Students will design a space for 'play' on a vacant site in the city of Antwerp (in Belgium). The final design will be presented in the form of a short story graphic novel that follows three different users through the 'play' space.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Relate architecture to theoretical, historical, social and contextual frameworks; Formulate a clear position in a debate and defend this position through a well-founded argumentation; - Communicate design through an alternative, narrative method that presents the design in all its aspects: architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The course will consist of three parts: The first part of the course, Readings, Research and Debates will consist of group work and will comprise four weeks. During these four weeks, students will be given readings and will be asked to present a summary of these readings orally during the course, and engage in conversations/debates with other groups who have read other texts. During this first part of the course, students will also be invited to research and analyse reference projects. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the course, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment. The third and final part of the course, 'Designing Play' comprises five weeks and will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	Group presentations of the assigned readings: 10% Group presentations and submission of reference projects: 15% Contributions to and active participation in the debates: 15% Final design: 60% ASSESSMENT CRITERIA Group presentations of the assigned readings: Capacity to synthetize the argument of the paper Structure and clarity of the slides Clarity and conciseness of the oral presentation Group presentations and submission of reference projects: Relevance of the selected reference projects to the theme of the week Depth of the understanding/analysis of the reference projects Reliability of sources used for the reference projects (no internet sources please) Contributions to and active participation in the debates: Strength of opening arguments Ability to use the reference texts to formulate arguments and defend/oppose assigned positions vis-à-vis the statements in the debate Active participation in the debates The final design (and deliverable) will be assessed on: The critical position that is formulated regarding the brief The urban and architectural merits of the project The conceptual and aesthetic quality of graphic novel The consistency and coherence of the narrative (short story) elaborated within the graphic novel	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Course Coordinator	Ir. E.I. Ronner	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students are able to: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural theories and references into an argued position that applies to the design proposal. 3. Analyse and reflect on their own creative processes and those of his/her group 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Q4	
Concept Schedule	Monday afternoon and Thursday	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	

Summary

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.

Course Contents

This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.

Study Goals

After completion of the course the student is able to:

Overarching study goals
- Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment.
- Assess the role of entrepreneurship in changes in the architectural world and the built environment.
- Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge.
- Critically evaluate and reflect on entrepreneurial skills and plans.
- Design and deliver a compelling and succinct vision and business pitch.

Additional study goals for Variant: Develop your architectural firm (open to A en BT students)
- Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio.
- Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network.
- Develop a design vision to guide the future direction of their own design firm.
- Reveal the design vision in an actual design.
- Propose the next steps for establishing a self-owned design firm.

Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students)
- Assess building sector organisations, including their business models, revenue streams, and network.
- Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities.
- Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues.
- Defend their business plan in a convincing matter to industry experts and utilise expert input to improve.
- Architecture students will also develop a design as part of the impactful business model
- Propose the next steps for implementing the business model in an existing or new venture.

Education Method

The course's learning activities comprise:

- lectures: theory
- workshops
- self-study: developing vision and translation into entrepreneurial plan
- groupwork: peer reflection, and inspiration from the inside world
- guest lectures: inspiration from the outside world
- tutorials: to develop vision and entrepreneurial plan

Assessment

Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Explore Lab

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 Expl Lab

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3EX115	Explore Lab Graduation	55
Course Coordinator	Ir. R.R.J. van de Pas	
Course Coordinator	Dr.ir. E.J.G.C. van Dooren	
Responsible for assignments	Dr.ir. E.J.G.C. van Dooren	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Explorelab is a graduate studio to explore fascinations in an integrated, architectural research and design project. Originally a student initiative, the graduate studio allows students to define the content of their graduate year from their specific interests. With its existence, the Explorelab Studio brings topics, approaches and problems to the faculty that do not yet - or no longer - find a place in the workshops offered. It reflects the questions and interests held by students at any specific point in time. Each edition of Explorelab is therefore a unique educational experience shaped by the students, their fascinations and the choices they make. Specific to Explorelab is the degree of autonomy and responsibility students have in the organization of their thesis project. Unlike other graduate studios, the supervision team, project theme, research method(s) and project location are all determined by the student. This is first studied under the guidance of the supporting Explorelab team, whose main goal is to have students translate their fascination into a project that fits within the general criteria for graduation in Architecture at TU Delft. It is further developed, consolidated and discussed with the chosen teachers.	
Study Goals	The purpose of the graduate studio is to establish, develop and elaborate a research and design project within the chosen fascination and within the criteria of architectural graduation. More specifically for research, this means formulating, researching and answering a question formulated specifically within the fascination and in relation to the profession of an architect. For design, this means developing a position as a designer within the chosen fascination, studying and deploying appropriate knowledge and experience and experimenting to arrive at a design.	
Education Method	Students work individually on their thesis project, guided by a team of teachers. However, joint studio activities and collaborative work are advised. During graduation, the process established by the faculty is followed, with informal and formal assessment moments - P1 (research progress), P2 (research results and design proposal, go / no go, P3 (design progress), P4 (design result, go / no go) and P5 (public presentation and defense).	
Assessment	The assessment will be of both the research and design product. The student is free to determine the nature of the research and design process in discussion with their mentors, as long as the result is meeting the goals that are required for examination. The assessment focuses on some general terms. These include the degree of consistency, significance, correctness and elaboration. Grades are given for architectural design, technical building design, research and presentation. The whole results in a final grade.	
Enrolment / Application	For enrolment please contact coordinator, Roel van de Pas (R.R.J.vandePas@tudelft.nl). Some weeks before the general date of registration for courses and projects students get an email to inform students about the Explore Lab enrolment procedure. Only ca. 25-30 students can actually participate in the Explore Lab per period. Students will be accepted on presenting a strong fascination that cannot be explored in another studio and which meets the architectural graduation requirements. It is also essential that students have finished all prior courses and projects and to have collected sufficient credits.	
Period of Education	1 year	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Cross Domain City of the Future

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 CS

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3CS100	Graduation Studio Cross Domain City of the Future	55
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours/week (1.1-1.8 & 2.1-2.5) 8 hours/week (2.6-2.10, 3.1-3.10, 4.1-4.10)	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 & MSc2	
Course Contents	This Graduation Lab is a special thesis laboratory for students who would like to develop their own (design and/ or research) fascinations in a multidisciplinary setting. Students with different backgrounds and from different TU Delft MSc tracks will work on common challenges sharing insights, approaches and methodologies proper of their own disciplines. While doing that, students work together to enrich their own graduation pathway, setting up and developing workshops, lectures, excursions and visiting critics. The students of this Graduation Lab are responsible for the program and the agenda throughout the thesis period. Mutual collaboration and student' feedback are important means of education in this lab. The seminar Cross Domain City of the Future (AR3CS021), shared with students coming from other MSc tracks, is included in this course - no need to enroll for that. This studio adopts UNESCOs education framework for Sustainable Development Goals.	
Study Goals	The study goals are contingent upon and aligned with the graduation track of each student. For MSc Architecture students, the study goals are as follows. The student: - has skill in architectural design satisfying both aesthetic and technical/functional requirements - has appropriate knowledge of urban and spatial planning and associated techniques - has insight into the relationship between people and architectural constructions and between architectural constructions and their environment, as well as the need to gear architectural constructions and the spaces between them to human needs and standards - has appropriate knowledge of the industries, organisations and procedures that play a role in the conversion of designs into buildings and the incorporation of plans into urban and spatial planning - has appropriate knowledge of and insight into decision-making procedures and processes The graduation report demonstrates the students ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding Architectural ethics and his/her future role as architect. The individual graduation report should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which intercultural issues were addressed in the graduation project.	
Education Method	This is a student driven graduation laboratory. The educational method is therefore to be developed by the students in conversation with each other and the coordinators. The assumption is that studio instruction will be the primary teaching method. Students will guide their own studies and determine their own learning styles.	
Assessment	(P1) Interim mid-term evaluation (week 1.9/1.10) (P2) Go-No go assessment (weeks 2.9/2.10) (P3) Progress review (week 3.8/3.10) (P4) Go-No go assessment (week 4.4/4.5) (P5) Public final presentation and diploma ceremony (week 4.9/4.10/5.1) P2 is a Go-No go examination that determines the students ability to to proceed to the final part of the studio. P4 is a Go-No go examination that determines the student's ability to proceed to the public graduation presentation and ceremony (P5). The actual review weeks may be subject to change in accordance with the academic calendar. All the criteria for the evaluation are explained in the graduation manual and will determine the final mark obtained for the entire graduation studio.	
Period of Education	Fall Semester MSc3 (Q1, Q2), Spring Semester MSc4 (Q3, Q4)	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Resilient Rotterdam Graduation Studio - Veldacademie

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 Resilient Rotterdam Studio

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3RE100	Resilient Rotterdam Graduation Studio - Veldacademie	55
Course Coordinator	Prof.dr.ir. M.J. van Dorst	
Responsible for assignments	Prof.dr.ir. M.J. van Dorst	
Contact Hours / Week x/x/x/x	8 Hours per week	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A resilient city is prepared for the unpredictable future, this counts for the physical environment and the social environment. The Physical city consist of the designed or (wo)men made environment and the natural environment. The social environment consist of individuals, all the social layers (formal and informal), a cultural layer, governance, etcetera. Resilience is a dynamic concept. So, for a designer of a resilient city (architect, urban and landscape architecture) there are two aspects key. First the knowledge and now-how on how your design facilitates the social environment. Secondly, the awareness of the temporal aspects, as the city of today is not the city of tomorrow. In this graduation studio, designing for a resilient Rotterdam is very much content based by the follow elements: - It is a Veldacademie studio, in which students will be working on site within a local network. Getting close to reality and the willingness to make a change in the world is dominant in the approach. - This studio is part of the TU Delft, Erasmus University and Erasmus Medical Centre convergance program titled Resilient Delta. This network will be actively used so students can get in touch with researchers, the municipality and other actors that work on (social) resilience in Rotterdam. - The knowledge developed in the studio is brought into the debate about actual city development, through publications, events and conferences in which students play an active role. The studio is design driven, yet the complexity of social-spatial reality of Rotterdam also imply an in-depth research. The research will facilitate the design and has a multi-method approach, consisting of mapping, interviews, observations.A resilient city is prepared for the unpredictable future, this counts for the physical environment and the social environment. The Physical city consist of the designed or (wo)men made environment and the natural environment. The social environment consist of individuals, all the social layers (formal and informal), a cultural layer, governance, etcetera. Resilience is a dynamic concept. So, for a designer of a resilient city (architect, urban and landscape architecture) there are two aspects key. First the knowledge and now-how on how your design facilitates the social environment. Secondly, the awareness of the temporal aspects, as the city of today is not the city of tomorrow. In this graduation studio, designing for a resilient Rotterdam is very much content based by the follow elements: - It is a Veldacademie studio, in which students will be working on site within a local network. Getting close to reality and the willingness to make a change in the world is dominant in the approach. - This studio is part of the TU Delft, Erasmus University and Erasmus Medical Centre convergance program titled Resilient Delta. This network will be actively used so students can get in touch with researchers, the municipality and other actors that work on (social) resilience in Rotterdam. - The knowledge developed in the studio is brought into the debate about actual city development, through publications, events and conferences in which students play an active role. The studio is design driven, yet the complexity of social-spatial reality of Rotterdam also imply an in-depth research. The research will facilitate the design and has a multi-method approach, consisting of mapping, interviews, observations.	
Study Goals	By the end of this course: The student can do a social-spatial research for a complex assignment; He or she can present to and discuss with local professionals and citizens on what spatial interventions are needed to come to a resilient city; Gained knowledge is applied in a design in which an extensive functional and technical elaboration has been realized at all levels, from urban to detailing; The student delivers a design that lives up to the end terms of his or her MSc tracks.	
Education Method	There will be a set of lectures by local actors, a workshop on fieldwork and interactive sessions with different teachers involved (form TU Delft and the Veldacademie). Students work in an interdisciplinary group, on site	
Assessment	Design process and product will be assessed. Design process on the way how one does inquiry in combination with designing and to what extent the student is capable to incorporate new information. The product is assessed on the relation between design products and the given context (social and physical).	
Period of Education	Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Architectural Wood

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 and 4 Architectural Wood

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted at the time of P1.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: Problem statement and research questions; Definition of theoretical framework; Methodological positioning and description of research methods; Argument on relevance; Bibliographical references; Quality of writing; Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semsester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AW005	Architectural Wood Design Studio	55

Course Coordinator	Prof. A.H.C. de Rijke
Course Coordinator	Ir.arch. G. Koskamp
Instructor	Dr. S. Brancart
Responsible for assignments	Dr. S. Brancart
Contact Hours / Week x/x/x/x	8 hours per week, starting from week 1.1 and ending in week 1.8 (MSc3 - Q1) 8 hours per week, starting from week 2.1 and ending in week 2.8 (MSc3 - Q2) 8 hours per week, starting from week 3.1 and ending in week 3.8 (MSc4 - Q3) 8 hours per week, starting from week 4.1 and ending in week 4.8 (MSc4 - Q4) Next to this we schedule lectures, workshops and excursions.
Education Period	1 2 3 4
Start Education	1
Exam Period	none
Course Language	English

Required for

Pre-assignment - PLANT A TREE

Before officially commencing our studio work, let's engage in a simple yet impactful gesture: planting a tree. This assignment underscores the belief that initiating change begins with individual action and serves as an opportunity to reflect on the profound impact of our actions on the environment.

Action: Each student is encouraged to plant a tree before the official start of the studio (Take in mind the choice of species and the site).

Reflection: Alongside planting the tree, enrolled students will document and reflect on the significance of their action in the first day of studio along with a 5 minute presentation of themselves.

Consider the quote by Rabindranath Tagore:
The one who plants trees, knowing that he will never sit in their shade, has at least started to understand the meaning of life.

Expected prior knowledge

Students enrolling in this studio bring basic knowledge by first attending the MOOC massive open online course 'Sustainable Building with Timber' to find on the EdX platform: https://www.edx.org/learn/architecture/delft-university-of-technology-building-with-timber

Course Contents

Wood Studio TUD | Timber for Urban Density

As the planet grapples with carbon-induced climate change, cities grapple with the challenges of population growth and society grapples with inadequate space, clever low-carbon architecture and sustainable urban development is increasingly crucial.

Wood Studio TUD | Timber for Urban Density during MSc3/MSc4 embarks on an exploration of timber construction's potential for transforming cities. Our goal is to investigate how lateral design with timber can revitalise underutilised urban spaces while contributing to sustainable urban development. Topping-up existing buildings with timber and embracing amphibious architecture present promising solutions for revitalising urban areas, densifying urban fabric, and making alternative use of spaces like air and water. Situated in the metropolitan area of Amsterdam, our studio delves into specific themes of housing, health and well-being, timber tectonics, the anatomy of timber architecture, material physics, and bio-based material design. Through a hands-on approach students will gain practical experience and deepen their understanding of the benefits of learning through making; a creative engagement with materials, techniques and critical thinking skills.

The Wood Studio TUD | Timber for Urban Density seeks to define timber architecture futures; new building typologies aligned with modern bio-based building methods tested with/in urban applications. Students will begin by recalling key concepts related to timber construction and urban density, then analyse case studies to deepen their understanding.
Comparison with modernist construction tropes will be made, revealing how engineered timber can outperform rather than mimic concrete. Hands-on making activities, including 1:1 modelling of timber joints, will emphasise practical skills. Through critical analysis of existing urban fabric, students will identify sites for intervention and justify their selection. The final project challenges students to propose bold solutions addressing the pressing housing need. Propositions can include mixed-use, related or supporting programmes.

During our studio students embark on a series of excursions within NL/EU designed to deepen their understanding of wood and its role in past and future architecture and engineering. These excursions will provide firsthand experiences and insights into different stages of the timber lifecycle; from forestry to fabrication to building to rebuilding.

Study Goals

The Wood Studio TUD | Timber for Urban Density embrace the following goals based on:

The Copenhagen Lessons were presented at the conclusion of SUSTAINABLE FUTURES LEAVE NO ONE BEHIND, the UIA World Congress of Architects 2023, held in Copenhagen. The COPENHAGEN LESSONS build on the substantive new research and practice developed and brought together in the years up to and during the Congress, exploring architectures contributions to the UN Sustainable Development Goals. The lessons are formulated with the goal that they can serve as inspirational principles to build on, for a sustainable future. 10 Principles to build on:

1. Dignity and agency for all people is fundamental in architecture, there is no beauty in exclusion.
2. People at risk of being left behind must be accommodated first when we construct, plan, and develop the built environment.
3. Existing built structures must always be reused first.
4. No new development must erase green fields.
5. Natural ecosystems and food production must be sustained regardless of the build context.
6. No virgin mineral material must be used in construction, when reuse is possible.
7. No waste must be produced or left behind in construction.
8. When sourcing materials for construction, local renewable materials come first.
9. In everything we build, carbon capture must exceed carbon footprint.
10.When developing, planning, and constructing the built environment, every activity must have a positive impact on water ecosystems and clean water supply.

Education Method

In the 'Wood Studio TUD | Timber for Urban Density' curriculum, we aim to provide students with a structured approach to learning that encourages critical thinking, problem-solving, and creativity. Each level of the taxonomy represents a different cognitive skill set, allowing students to progress from foundational knowledge acquisition to more complex application and synthesis of concepts.

Remembering: Students will begin by recalling key concepts related to timber construction and urban density. This includes understanding basic terminology, historical contexts, and fundamental principles of timber construction and urban development.

	Understanding: Through in-depth examination of real-world case-studies, students will gain a deeper understanding of the complexities and nuances of timber construction and urban density. This involves identifying key components, analysing design strategies, and interpreting the impact of various interventions on the built environment. Applying: In our studio, we emphasise hands-on learning experiences from the outset, incorporating tactile exploration and modelling with wood as a fundamental aspect of understanding and applying concepts. Students will take the gathered information and create 1:1 conceptual models of studied joints from the case studies, emphasising hands-on learning. By engaging directly with materials and physical prototypes, students gain a deeper understanding of construction techniques, material properties, and spatial relationships. Analysis: Next, students critcallyl analyse the existing urban fabric, identifying issues that arise and selecting a site for further analysis. Through site visits, observation and documentation, students investigate the context, challenges and opportunities present within the existing built environment. This involves identifying organisational patterns, assessing spatial qualities, and evaluating the social, economic and environmental factors influencing the site. Evaluating: In this stage, students must justify why their chosen site is in need of or would benefit from an architectural intervention. Drawing upon their analysis, students will evaluate the site's strengths, weaknesses, opportunities and threats, considering its potential for transformation and improvement. They will critically assess the site's relevance, significance, and suitability for intervention, articulating their rationale for selecting it as a focus for their design exploration. Creating: Finally, students synthesise their knowledge and imagination into a project and new vision for the site. They proceed to assemble, construct, design, and develop their ideas into tangible outcomes. This involves generating innovative design solutions, conceptualising spatial interventions, and articulating proposals through drawings, models, and presentations. Students will engage in iterative design processes, refining their ideas based on feedback, critique, and reflection, ultimately producing comprehensive design proposals that address the complex challenges of timber construction and urban density.
Literature and Study Materials	inspiration you can find here: https://bk-wood.nl/category/research-and-innovation/
Books	books on timber construction you can find a summary here: https://bk-wood.nl/category/literature/
Assessment	Submission Requirements P1 Review Week 1.10 Pre-design research prospectus that illustrates site conditions, hypothetical design intent, discursive basis, outline program, and various supporting materials (drawings, models, etc.). P2 Review/Exam Week 2.9-2.10; pass/fail P1 submission plus schematic design package including drawings & models (Project Design), design journal/brief (Theory and Delineation), and completed Graduation Plan with concise problem statement and design objectives as per facultys graduation regulations. The P2 assessment will be proportionally weighed among the three branches of assignments. P3 Review Week 3.7 P2 submission plus evidence of progress from schematic design illustrating the direction and substance of design development. P4 Review/Exam Week 4.4-4.6; Pass/Fail P3 submission plus development in Technical Building Design, materialization, and representative construction details. Completed Graduation Report as per instructors specifications: The Graduation Report should demonstrate the students understanding of and adherence to professional responsibility, ethical awareness, scientific analysis, creativity, sensible decision-making, and argumentation skills. P5 Final Presentation & Graduation Further improvements and any additional materials to P4 submission in spoken presentation, drawings, and models as specified in P4. More detailed requirements will be distributed at the appropriate time as per each P(resentation). Assessment Methods Make: Students will be evaluated on their ability to translate their research findings into tangible outcomes. This includes the creation of prototypes, models, or design solutions that reflect their understanding of the research topic and its implications. Research: A significant portion of the assessment will focus on the quality and depth of research conducted. This includes the clarity of the research problem statement, the rigor of the research methods employed, and the thoroughness of the literature review. Understanding: Finally, the assessment will evaluate students' overall understanding of the subject matter. This encompasses not only their grasp of theoretical concepts but also their ability to critically analyze and synthesize information, demonstrating a nuanced understanding of the topic and its broader implications. Assessment Criteria 1. The student is able to critically analyze and evaluate timber construction techniques in the context of urban density challenges, demonstrating advanced comprehension and critical thinking skills by synthesizing information from case studies and site analysis. 2. The student is able to apply theoretical knowledge of timber construction to practical scenarios, demonstrating proficiency in conceptual models and hands-on construction techniques, thereby bridging theoretical understanding with real world application. 3. The student is able to synthesize information from case studies and site analyses to develop innovative design proposals supportive of or providing housing, showcasing creativity and problem-solving abilities through the creation of comprehensive design solutions. 4. The student is able to justify design decisions and interventions based on social, environmental, and economic considerations, effectively communicating their rationale and demonstrating ethical reasoning in the development of sustainable urban solutions. 5. Reflection Benefits Future Potential

	Furthermore and for the final grade compound, we refer to the Graduation Manual, Master of Science Architecture, Urbanism and Building Sciences. The final grade is the average, or may deviate from the average depending on the extent to which the whole does (or not) exceed the sum of its parts, or due other exceptional qualities of the work. For retake and repair we refer to the regulations of the board of examiners of the faculty.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	2024/2025 MSc3 starts in the fall semester. end of August 2024 MSc4 starts in the spring semester, starting February 2025 see academic graduation calendar 2024/2025
Leerstoel	Chair of Timber Architecture
Minimum number of participants	16 graduation students
Maximum number of participants	20 graduation students
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

variant Building Technology

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 Building Technology

AR1B011	Bucky Lab Design	10
Course Coordinator	Dr.ing. M. Bilow	
Course Coordinator	Ir. P.G. Teeuw	
Instructor	N.A. Remmerswaal	
Responsible for assignments	Dr.ing. M. Bilow	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The focus of the semester is an innovative building construction/product or facade design/product for the built environment, this may be a part of a building, a pavillion or a facade. The task is a building component or a facade product in which all the important technical and architectural aspects of a building are integrated. The course is hosted by the chair of Building Product Innovation - therefore the focus will be on architectural Product Design - this is not a general Building Construction course. The first three weeks students individually research and analyse the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. After the concept and design & engineering phase we will set up our mobile workshop and will build the concepts in a full scale prototype. The course will be finalised by a review of the prototype - possible improvements and changes will be summarised in the report as also the overall process of the semester. The final product will be presented in a public presentation.	
Study Goals	The student has a basic understanding of the field of science of Building Technology. The student is able to design a building component understanding the relation between design, society, realisation, materialisation and functioning. The student is able to test and evaluate the design on functioning and performances. The student is competent in collaborating with fellow students. The student has gained the skills to build concept and presentation models in oder to help visualising their concept and ideas to a non expert audience. The student is able to visualise a complex system in an infografic.	
Education Method	We follow the methods of Design by research - as also Problem-based learning (PBL) which is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We stimulate to experiment, brainstorm and out of the box thinking. Failure is always an option in order to find the best possible solution to the problem. We strive for a high pace of idea creation in the first phase, try everything in order to be sure that the chosen concept is really the best.	
Computer Use	The students are asked to use a wide range of software on their personal computers.	
Assessment	Individual report of innovative concept and reports in team of 3-4 students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The building / making of a full scale 1:1 prototype during the building weeks. Design 80 % Model 20% Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment (or get an oral exam) within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days. First three weeks innovative concepts individualy, remaining period teams of 3 to 4 students collaborating to final design. We will ask to sketch by hand till the elevator pitch / mid term presentation. We will work in the mobile Bucky Lab workshop under guidance of experts. Safety is key therefore we ask for personal safety shoes that at least suffice to a S1 level - which consist of a toe cap. For future use on building sites we highly recommend S2 level which are waterproof and therefore better suited for any kind of building site inspections.	
Period of Education	Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1B022	Climate Design	5
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Instructor	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Dr. L.C.M. Itard	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Dr.ir. A.M. Eijkelenboom	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	6 hours per week in week 1.1/1.3/1.5/1.7 4 hours per week in week 1.2/1.4/1.6/1.8 3 hours in week 1.10 (exam)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basically the Bachelor Bouwkunde is the starting point for this course and we expect students to remember (or being able to catch up) most of what was taught on building physics, building services and sustainable design in this program. We aim to make self-assessments for students available at the start or during the programme, which will direct them to study material to catch up on the basic knowledge on climate design. Please note: make sure that your are informed in advance about the expected prior knowledge. The main topics as basic knowledge are addressed below: - Knowledge of Physics and Mathematics at Dutch pre-university diploma level (VWO natuur en techniek) or equivalent: A-Level Mathematics and A- Level Physics. - Basic knowledge of Sustainable Design, Climate Design, Indoor Environment, Building Physics and Building Services: o Sustainable design. Refer to: Jón Kristinsson, Integrated Sustainable Design, Delft/Deventer 2012. This book will (soon) be available on www.kristinssonarchitecten.nl o indoor air quality, heating, ventilating and air-conditioning systems. Refer to: All you need to know about air conditioning the how and why of heating, ventilating and air conditioning in a nutshell by P.M. Bluyssen (Delft Academic Press) and All you need to know about indoor air A simple guide for educating yourself to improve your indoor environment by P.M. Bluyssen (Delft Academic Press). o Daylight, thermal comfort and sound. Refer to: "Building Physics" by A.C. van der Linden (2nd edition, English), chapters 4, 5 and 10. This book will be available on https://klimapedia.nl/leerbundel/basics-of-building-physics/. o Heat and moisture transfer, ventilation and solar radiation. Refer to: "Building Physics" by A.C. van der Linden (2nd edition, English), chapters 1, 2, 6 and 7. This book will be available on https://klimapedia.nl/leerbundel/basics-of-building-physics/.	
Course Contents	This course will give a full overview of health, comfort, building physics, HVAC systems and sustainable climate design necessary for Building Technology students. The course focusses on content knowledge primarily, linking the research-focused Research and Methodology course, the design&build-focused Bucky Lab Design course and the more specialised climate design courses offered during the MSc2 and on.	
Study Goals	The student will be able to: - describe and explain the design performance indicators for sustainable, healthy, and comfortable buildings and urban environments; - explain the various types of energy systems at building, neighbourhood and city level including their effects on a sustainable development; - optimise the energy performance and Indoor Environmental Quality (IEQ) of buildings and building designs; - make a conceptual design of HVAC systems in order op optimise energy and IEQ performances; accounting for the interactions and relations between occupant, indoor environment, energy systems and building.	
Education Method	As didactic means this course will combine lectures, demonstrations, simulations, exercises, lab tests, as well as excursions. Students work individually and sometimes in groups.	
Literature and Study Materials	Literature and Study Materials will be made known prior to the start of the course in Brightspace.	
Assessment	Written examination. One retake opportunity in the academic year for the written examination.	
Period of Education	Quarter 1 of the first semester	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1B023	Sustainable Architectural Materials and Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Ir. P.G. Teeuw	
Instructor	Dr.ir. F.A. Veer	
Responsible for assignments	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	5 hours a week from week 1.1 - 1.7 3 hours week 1.9 (exam)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of physics and chemistry at Dutch pre-university diploma level (VWO natuur en techniek), equivalent to: - A-Level Chemistry - Core Text Fourth Edition 4th Edition - A Level Physics Fourth Edition 4th Edition Knowledge of Structural Mechanics: Statics and equilibrium of forces, centroids and moment of inertia, reaction forces of statically indeterminate beams, bending and shear force diagrams, stresses due to normal force, bending moments and shear forces, deformations of beams, buckling of columns. Refer to: "Mechanics of Materials" by J.M. Gere (9th edition, SI edition, English). Please note: make sure that your are informed in advance about the expected prior knowledge.	
Course Contents	The Sustainable Architectural Materials & Structures course equips students with the skills to develop efficient structures at conceptual stage, make appropriate materials choices and design structural elements and structural systems with these materials. This course also provides a foundation for the more advanced courses on materials and structures in subsequent parts of the BT master programme. It consists of a series of lectures and associated workshops in order to: - Cover the basic principles of practical design of typical architectural structures, with applications across a range of commonly-used materials. - Establish links between materials science and structural mechanics (taught in bachelor courses) towards the practical goal of designing safe and sustainable structural systems. - Demonstrate the importance of appropriate design decisions on materials choice and structural form at concept stage and the use of simple effective tools to generate efficient design options. - Provide the knowledge and tools required to evaluate the safety and sustainability of structural materials and systems.	
Study Goals	On successful completion of this course students should be able to: - Understanding the principles of conceptual structural design and fundamental material science. - Create viable early-design-stage options of structural forms and material choices for given realistic scenarios. - Applying simple hand calculations and simple supporting software to support building design choices on materials and structural form. - Evaluate the efficiency and appropriateness of the different options of materials choices and structural forms for building design. - Design safe and sustainable structural elements and structural systems underpinned by more detailed calculations.	
Education Method	Lectures and associated workshops.	
Literature and Study Materials	Study material will be provided on Brightspace and www.restruct-group.tudelft.nl	
Assessment	Written examination One retake opportunity in the academic year	
Period of Education	Quarter 1 of semester 1	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1B024	Introduction to Computational Design	5
Course Coordinator	Dr. S. Aut	
Course Coordinator	Ir. P.G. Teeuw	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Instructor	Dr. A. Rafiee	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	6 hours per week starting from week 2.1 and ending in 2.4. (half morning & afternoon) 6 hours per week starting from week 2.7 and ending in 2.9. (half morning & afternoon) 2 hours on week 2.10 (morning)	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to introduce the uses of computational design methods within an integrated workflow. It is a hands-on design course in which the students develop skills by working on a design assignment. The course introduces the students to a series of practical workshops each of which focuses on a specific application such as gathering environmental data, developing spatial designs, implementing performance analyses and digital fabrication. They are then asked to integrate all of these applications within a comprehensive workflow to design, analyze, prototype and present a small scale architectural product. The course content is composed of the following topics: 1. Introduction to Computational Design (Computational Spatial Geometry, Algorithmic Design, Visual Programming). 2. Data and Information Models (Spatial Data Acquisition, Database and Information Models). 3. Performance Based Computational Design (Using algorithms for Performance Analysis, Design Iterations towards Rationalization). 4. Digital Materiality (CAD/CAM, Physical Computation).	
Study Goals	1. Understand the uses of digital technologies in the design and production of the built environment. 2. Understand spatial data acquisition techniques. 3. Use database and information models for design development and analysis. 4. Formulate algorithms to generate solutions to architectural design and building engineering problems. 5. Create visual programs for design development, analyses and fabrication.	
Education Method	Workshops: The students will work on the design assignments in groups and under the supervision of the tutors. Lectures: The students will join lectures which will be given on the campus. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (30 %) Design Examination - Final (50 %) Participation (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 1, Quarter 2	
Minimum number of participants	NA	
Maximum number of participants	NA	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1B032	Research and Methodology	5
Course Coordinator	T. Konstantinou	
Course Coordinator	Ir. P.G. Teeuw	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Responsible for assignments	T. Konstantinou	
Contact Hours / Week x/x/x/x	4 hours a week in week 2.1 / 2.4 / 2.7 till 2.10 8 hours a week in week 2.3 / 2.3	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	In this course students become acquainted with research methods applied in Building Technology, which deal with the issues of building design and research, not as competing activities but rather as complementary. Research is a way to generate new insights and new concepts for design in a systematic way, explore new realms and finding answers beyond the state of the art. It is the cornerstone on which all progress in our modern society is founded. Research methods show us the way how to conduct research in a systematic and reliable way. In this course you will learn how to execute a technical-scientific research framework for experimental research, simulations, surveys, case studies and research through design; and how to consistently and coherently report the results of such research. In the first part, the students learn about different research methods and apply some of them in practical exercises. Moreover, they learn to consistently and coherently report the results of such research. During the second half of this course, based on the Bucky Lab Design project and/or the other MSc1 courses and/or the student's own fascination, the students individually conduct a literature study on a chosen topic, resulting in a scientific (review) paper.	
Study Goals	After successfully completing this course the student should be able to: - explain scientific integrity and outline ethical issues when doing research. - execute and report on a technical-scientific research framework for experimental research, simulations, surveys, case studies and research by design; - analyse a selected topic, in a consistent, coherent and well-structured review paper, with proper referencing; - interpret the complex relationship between research and design as two aspects within building technology;	
Education Method	Lectures, practical workshops, group tutorings and self study.	
Literature and Study Materials	Literature and Study Materials will be made known prior to the start of the course in Brightspace.	
Assessment	Writing assignment: Review paper, report of the experiment. Repairs: A final mark of 6.0 or higher is needed for successfully completing this course. Each separate part should be graded at least with a 5.0. In case a partial grade is insufficient, students get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. Retakes: In case students fail the course, the course can be taken again in the next academic year.	
Period of Education	Quarter 2 of semester 1	
Concept Schedule	The First 4 weeks, before the building weeks of the Bucky Lab Design course, will be about research methods, the last 4 weeks will be the paper writing.	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 Building Technology

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Building Technology electives (choose 3 courses)

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	- The student is able to design a bridge that demonstrates symbiosis in contextual, architectural, structural and functional design. - The student is able to integrate spatial, historical and cultural context in the design of a bridge. - The student is able to structurally elaborate the conceptual design of a bridge/civil structure. - The student is able to present (on paper and oral) how landscape-, urban-, architectural- and structural-design are integrated in the design approach.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers, Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed.	
Assessment	Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Dr.ir. F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	Dr.ir. F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Responsible for assignments	Dr.ir. F. Oikonomopoulou	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Responsible for assignments	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. General planning: The content of the course program is split around two main parts: Part A (analysis) and Part B (design / research) . Both Parts A and B consist of several lectures, excursions to construction site and product manufacturers and several assignments. Content-wise these parts are interrelated during the course.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possbility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Prof.dr.ing. A.C. Boerstra	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Dr. R.M.J. Bokel	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenics (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Franscisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 or 2 .Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Thursday afternoon (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of	24	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0138	Technoledge Design Informatics	5
Course Coordinator	Dr. S. Aut	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Aut	
Responsible for assignments	Dr. S. Aut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5

Course Coordinator	Dr. B. Lubelli
Instructor	Dr. B. Lubelli
Instructor	S. Naldini
Instructor	T.P. Bennebroek
Responsible for assignments	Dr. B. Lubelli
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.
Period of Education	Q3
Concept Schedule	Wednesday afternoon
Minimum number of participants	15
Maximum number of participants	60
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Responsible for assignments	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: Identify key parameters for making building products circular, Correlate the key parameters to reason complex domain interdependencies, Design a circular product or circular product concept by prioritizing key parameters and relations, Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about developments in facade technology and innovation- or research opportunities. The course consists of a capita selecta of recent buildings of which we invite different involved experts to discus the different relevant disciplines. Lectures: We will invite experts from the TU, the industry and engineering firms to lecture about buildings they have been involved in. The students are required to analyse the buildings and prepare questions prior to the lectures. Excursions: We organize excursions to construction site and/or product manufacturers in cooperation with the course Technoledge facade design. For these excursions students will have to pay a nominal fee.	
Study Goals	The student is capable of understanding technical developments and reflecting on façade designs in relation to the architecture, can formulate relevant bill of requirements based on critical evaluation of the provided information, is able to understand (façade) designs and concepts in terms of building physical, structural and constructional aspects and translate this into well balanced design decisions Is able to design an adaptable or flexible building based on a bill of requirements	
Education Method	design assignment, lectures, plenary discussions, excursions, all work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Assessment	report and design assignment. All work will be groupwork. The individual contributions will be evaluated by the student-teams which will lead to individual grades.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	10 weeks in q3	
Concept Schedule	Tuesday morning and Friday morning. The excursions (combined with Technoledge Facade Design) may be in the afternoon)	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	Dr. M. Turrin	
Instructor	Dr. M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	Dr. M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0917	Timber Architecture Design	5
Course Coordinator	Prof. A.H.C. de Rijke	
Course Coordinator	P. Eigenraam	
Course Coordinator	Dr. S. Brancart	
Responsible for assignments	P. Eigenraam	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is advised that students have a basic knowledge of: - Materials and material properties - Structural design and mechanics - Computational modelling (e.g. Grasshopper)	
Course Contents	Stimulated by the ongoing climate crisis, the construction sector is simultaneously rediscovering and exploring new avenues for the use of timber and other biobased materials. Aside from their potential to reduce the environmental impact of construction, biobased materials possess a range of different properties that can add value to the design, manufacturing, construction and use of a building. A good understanding of these properties, the ways to design with and apply the materials, and the environmental impacts associated with them, is crucial to achieve those values. This course focuses on the use of timber and other biobased materials in load-bearing applications. Through lectures from experts in the fields and a site visit, students learn about the properties and application potential of different biobased materials, from their origin, processing and manufacturing to the different construction methods through which they can be applied. Through workshops they learn to apply (new) computational methods to create high performing and sustainable biobased structures. Finally, working in small groups, students apply the acquired knowledge and skills in a small design project. The project combines aspects of design, analysis and making. The course will provide students with the knowledge and skills to weigh design choices on their environmental impact and critically reflect on the application of timber and biobased materials. Throughout the course, students will submit small reports that synthesise the results and insights from the workshops. They present their design project in a final presentation at the end of the course.	
Study Goals	After successful completion of this course, you should be able to: - Describe the available biobased construction methods for load-bearing structures and their environmental impact - Develop computational models for bio-based construction methods - Design and assess the environmental impact of different bio-based construction alternatives	
Education Method	Lectures, workshops, design tutoring	
Assessment	Reporting and presentation Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	3rd quarter	
Concept Schedule	Wednesday	
Maximum number of participants	32 (24 Building Technology students + 8 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5

Course Coordinator	mr.dr. H.D. Ploeger
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom
Instructor	Prof.dr.ir. P.J.M. van Oosterom
Instructor	mr.dr. H.D. Ploeger
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom
Contact Hours / Week x/x/x/x	4
Education Period	3
Start Education	3
Exam Period	3 4
Course Language	English
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)
Assessment	Written exam (3 hours) and formative assignments. The exam result will determine the final score, but assignments need to be sufficient (>=5,75). The main purpose of the assignments is to support the understanding of the theory via hands-on experience. The exam is the only summative assessment and determines your grade for 100%. However, students need to pass at least three of the four formative assignments in order to pass the course. In case of non-passed assignments, in week 9 of the course, students have the opportunity to repair these non-passed assignments.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 3.
Concept Schedule	Tuesday 3e/4e hour + Friday 1e/2e hour (all with reservations)
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

AR0139	MEGA	15

Course Coordinator	Prof.dr. M. Overend
Course Coordinator	Dr. A. Luna Navarro
Course Coordinator	Dr. S. Brancart
Course Coordinator	Dr. M. Turrin
Instructor	Dr.ir. M.C. Lugten
Responsible for assignments	Dr. S. Brancart
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 4
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 90.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude mussen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflections, and presentations. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a Pass. (A retake can be done in the next year) The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15

Course Coordinator	Prof.dr.ing. U. Pottgiesser
Course Coordinator	Ir. W.L.E.C. Meijers
Course Coordinator	Prof.ir. W. de Jonge
Instructor	Ir. A.C. de Ridder
Instructor	Ir. W. Willers
Instructor	Ir. A.W. Hermkens
Instructor	Ir. W.L.E.C. Meijers
Responsible for assignments	Ir. W.L.E.C. Meijers
Contact Hours / Week x/x/x/x	12
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Expected prior knowledge	It is expected that students have the knowledge from a Master 1 design course and the Building Engineering Studios (AR1A080).
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is considered, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urban context, construction, and interior, related to architectural history and current societal questions like sustainability, climate design, ecological system, or circular materials. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by 'Learning from others' for Heritage & Architecture, by theory, and reference material is guiding for the architecture of the re-design. In the beginning of the course in small groups students research related questions to the proposed subjects for the transformation design. The design process is supported by lectures and workshops and building visits. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material, and architectural-technical design. The design choices are based on an understanding in relation to cultural value, theory, and current societal issues.
Study Goals	Upon completion of the design studio, the student is able to: -demonstrate argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method, and composition of a design proposal for a cultural-historical context. -position the project within a particular theoretical, historical, social, or contextual framework, emphasizing moral sensibility, analysis, creativity, and judgment skills regarding architectural ethics. -develop a consistent and coherent design process, making informed and well-argued decisions, utilizing appropriate analogue and digital tools for drawing, model making, and responding to feedback from tutors and peers. -develop an architectural idea for the project based on the brief, site specifications, and precedent research, and subsequently create a coherent, elaborated, and integrated design considering technical aspects, material aspects, and aspects of use. -present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.
Education Method	The design studio incorporates individual and group tutorials, thematic exercises, excursions for site visits, lectures, and seminars to inform the subject. Coaching sessions during educational weeks are customised for the specific design project. Group work is initiated at the start of the course for researching studio subjects, leading to individual design development towards the end of the course.
Literature and Study Materials	Literature and Study Materials will be available one week prior to the start of the course in Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022
Assessment	Presentations will be held during the quarter. A final presentation will be at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process will be presented both visual and oral formats. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process Principles and grades 1. All grades of 5.5 or lower are eligible for retake or repair. No distinction will be made between 5 - 5.5 on the one hand and lower than 5 on the other; excluding a second examination in advance based on a grade lower than 5 is not possible. 2. A successful repair will only be assessed with a 6. The maximum marking period is 10 working days.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Important note - for MSc courses only: If your course is open to non-BK students (i.e. students from other TUD programmes or guest students from other universities), you also need to copy and paste the following default text: Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	The course is open for MSc students of the track Architecture and of the track Building Technology.
Period of Education	The course takes place in the second quarter of the Spring semester.
Concept Schedule	On Wednesday or Friday throughout the quarter. During the first 3 weeks, there are additional studio days depending on the organisation of building visits, lectures, and excursions.
Leerstoel	Heritage & Design and Heritage & Technology
Minimum number of participants	12
Maximum number of participants	60 in total
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.	
Course Contents	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.	
Study Goals	After completion of the course the student is able to: Overarching study goals - Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment. - Assess the role of entrepreneurship in changes in the architectural world and the built environment. - Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge. - Critically evaluate and reflect on entrepreneurial skills and plans. - Design and deliver a compelling and succinct vision and business pitch. Additional study goals for Variant: Develop your architectural firm (open to A en BT students) - Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio. - Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network. - Develop a design vision to guide the future direction of their own design firm. - Reveal the design vision in an actual design. - Propose the next steps for establishing a self-owned design firm. Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students) - Assess building sector organisations, including their business models, revenue streams, and network. - Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities. - Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues. - Defend their business plan in a convincing matter to industry experts and utilise expert input to improve. - Architecture students will also develop a design as part of the impactful business model - Propose the next steps for implementing the business model in an existing or new venture.	
Education Method	The course's learning activities comprise: - lectures: theory - workshops - self-study: developing vision and translation into entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop vision and entrepreneurial plan	
Assessment	Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may	

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 Building Technology

AR3B025	Building Technology Graduation Studio	15
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Responsible for assignments	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	1 hour per week	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 Building Technology (Bucky Lab Design and MSc1 courses), MSc2 Building Technology electives, MSc2 studio, CORE or User-centred Sustainability Studio.	
Course Contents	The Building Technology Graduation Studio continues the build-up of Bucky Lab Design and the BT MSc1 courses, the MSc2 Building Technology electicve courses, the MSc2 design studio's, and CORE or User-centred Sustainability Studio to the student's own graduation project in the Building Technology master track. In the graduation studio students work (preferably in groups) on their individual graduation project comprising a preparative technical-scientific study (until the P2, this first part) and design, design by research or research by design (after the P2, the second part of the graduation studio). The studio intends to be in strong coherence with themes from the research programme of the AE+T department, so students can benefit from and contribute to the research activities by staff of the AE+T department. While the graduation project's subject can relate to any of the disciplines represented by the Building Technology master track, emphasis lies on sustainability-related topics of structural design, façade & product design, climate design and design informatics. These are interlinked. The student will be assigned to two supervisors from different 'colours' of AE+T (Structural Design / Façade & Product Design / Climate Design / Design Informatics), the student projects content as well as design and research is reflected by these two supervisors. Limited consultations with other staff members might be arranged if thats desired thematically.	
Study Goals	The student: - is capable of delivering innovative contributions toward the development of sustainable structural, façade, product, climate and computational design, and to technical-scientific research in these areas; - has insight in the profession and competences of structural, façade, product, climate or computational designers and their societal and professional role and ethical responsibilities in sustainable development of the built environment; - is able to present process and result in a clear and systematic way, reflect on process and work, and argument in a scientific way.	
Education Method	Supervised self study, with possibly intermediate studio workshops	
Assessment	P1 First ideas of the graduation project, conceptual research framework, first literature study results P2 Definite research framework, literature and desktop research results, outline for the design research See the current Graduation Manual AUBS for more details and for the assessment criteria.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams The graduation studio is not open for non-BK students.	
Special Information	On set conditions, Building Technology students have the possibility to carry out their graduation research project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found at https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/forms. The agreements can be signed at the secretariat of Education and Student Affairs.	
Period of Education	Quarter	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory Choice (choose 1 project)

AR3B012	CORE	15
Course Coordinator	Dr. S. Aut	
Course Coordinator	Ir. P.G. Teeuw	
Responsible for assignments	Dr. S. Aut	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	CORE (COmputational REpertoire for Architectural Design and Engineering) is a design studio course. It primarily focuses on Design Computation in an architectural and structural design context, with emphasis on building technologies. It addresses various scales within the built environment, ranging from urban to building products, by paying attention to environmental, social, cultural, and ethical aspects. The course setup involves and integrates 5 modules as; 1. Design Studio 2. Programming 3. Materials & Structures 4. Design of Construction 5. Tools, Methods, and Inspirations	
Study Goals	LO1: Integrate architectural design, structural design and computer science disciplines to develop planning and design solutions for the built environment. LO2: Apply computer programming towards developing solutions for spatial design, planning, analysis, and construction design. LO3: Analyze an urban design and planning strategy to address societal needs and sustainability demands in the given context. LO4: Design an architectural project with construction solutions, addressing the needs on different scales ranging from urban to product levels. LO5: Create a structural design by analyzing and optimizing the material and structural performances.	
Education Method	The studio provides: 1. Reviews, debates, and feedback on design development. 2. Lectures, workshops, and consultations on computer programming. 3. Lectures, workshops, and consultations on structural design. 4. Consultations on construction design. 5. Lectures and debates on various tools and methods, as well as inspirational topics. The students work in teams to design and submit an architectural project which addresses the 5 tracks of the studio.	
Assessment	The course includes formative (ungraded) and summative (graded) assessments. Formative assessments provide feedback on the work in progress. They help the students prepare for the final assessment. They are performed during: 1. Design Studio meetings (as feedback on design development and the design of construction) 2. Proposal presentation (as feedback on the proposed concept project) 3. Mid-term presentation (as feedback on the comprehensive project, prior to final presentation) 4. Programming workshops (as feedback on programming skills) 5. Structural design workshops (as feedback on topics related to structural design and materials) The only summative assessment is performed on the final presentation and the submission at the end of the course. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	

AR3B015	User-centred Sustainability Studio	15
Course Coordinator	Dr. C.L. Martin	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Responsible for assignments	Dr. C.L. Martin	
Contact Hours / Week x/x/x/x	Approximately: 6 hours per week in week 1.1 / 1.2 / 1.5 till 1.9 40 hours a week in week 1.3 / 1.4 (or simular in quarter 3, week 3.1 and so on)	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The successful completion of MSc1 Bucky Lab, BT MSc1 courses and MSc2 EXTREME technology or MEGA (or approved equivalent) is required before entry onto the User-centred Sustainability Studio can be approved. In addition, Building Technology students are strongly advised to have completed two elective courses (2 x 5 ECTS) of Technoledge in the first quarter of the MSc2.	
Parts	3 Key elements of the 15 ects course: 1. Briefing: Group site analysis, background research & readings. 2. Intervention: Onsite workshop. 2-week intensive collaborative group studio. 3. Elaboration: Group finish, detail realization & report submission.	
Course Contents	After having focused on technology in the Bucky Lab and on integrated design in EXTREME / MEGA, the aims and objectives of the User-centred Sustainability Studio are as follows: Aims: To meet societal challenges and inequalities in the urban built environment by promoting well-being, health and carbon neutral lifestyles in an era of unprecedented global uncertainty. By engaging with diverse city stakeholders, real urban contexts, global trends and future scenarios, within student groups develop design solutions that are greater than the sum of their technological parts. Ensuring that societal value is revealed at various city scales. Objectives: To collaborate effectively in trans-disciplinary groups of students and city stakeholders. To analyse the urban context and local characteristics (climatic, historic, socio-cultural and technical) of a innovative intervention assignment and to communicate the process in a suitable/effective format i.e. text, diagrams, maps, images and film etc... To develop a position in relation to future urban conditions and sustainability. To technologically respond to local circumstances with contextually appropriate interventions / engagement initiatives.	
Course Contents Continuation	Elaborate description of what students will learn (attainment): 1. Takes account of the temporal and the social context. Students to be competent in acquiring local knowledge of context, daily rhythms and rituals, lifestyle, aspirations and be competent in working alongside local inhabitants to reveal the needs of communities. The student has a critical attitude in what it means to innovate sustainably. 2. Is competent in co-operating and communicating. Students to be competent in co-operating with local stakeholder and other group members, gaining trust and feedback. Students to be competent in bridging the communication and knowledge gap between local city stakeholders and the scientific community. 3. Possesses basic intellectual skills. Students to be competent in contributing to discussions on how sustainable strategies are multifaceted and are needed at every scale. 4. Is competent in conducting research. Students to be competent in methodological research into what, why and for whom you are designing in an era of climate change. 5. Is competent in designing and/or innovating. 6. Has a scientific approach. Students to be competent in using appropriate theories, methods and/or (modelling) techniques to critically investigate and analyse existing, newly proposed and self-formulated projects.	
Study Goals	The student: - is able to design a coherent, significant, elaborated, contextually responsive and innovative urban intervention - on MSc 3 level - is able to develop a group proposition in relation to the local characteristics, sustainable futures and development, societal and ethical context. - is able to underpin the design decisions by scientific research at MSc3 level - Is able to collaborate effectively in trans-disciplinary groups of students and city stakeholders in order to communicate coherently often complex and integrated propositions to peers and local stakeholders.	
Education Method	Educational Method The User-centred Sustainability Studio aim is to meet societal & environmental challenges of urban neighbourhoods through contextual responsiveness, meaningful engagement & technological knowledge. Students are to collaborate effectively in groups to communicate sustainable innovative interventions to an audience of participating experts and local stakeholders. Studies begin by analysing the local urban context (climatic, historic, socio-cultural and technical) and global theory relating to sustainable city development. Group work is then to be elaborated within those groups to emphasize a position relating to future urban conditions, form and sustainability. The onsite workshop (Intervention) can in some cases be based abroad, as a consequence students are expected to financially contribute to this excursion. In the acquisition of self-funding a level of pro-activeness is required (amounting to around 500 euro maximum).	
Course Relations	The User-centred Sustainability Studio Studio forms an integrated part of the Building Technology master track and is aligned with previously completed consecutive courses (MSc1 Bucky Lab, MT MSc1 courses, MSc2 Technoledge and MSc2 EXTREME technology/MEGA) and with the MSc3/4 Building Technology Graduation Studio that immediately follows.	
Assessment	Design examination: In regards to the learning goals, site data is to be group collated, then group elaborated to form a design argument / city	

	engagement initiative. This process facilitating the differentiation of grades. Grades will be based on the following: 1. Quality and extent of the local analysis (in a preparative presentation with text, maps and images). 2. Quality of design investigation, design iterations, communication, final product/initiative and critical reflection. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5 students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. In case students fail (following the repair) the course can be taken in the next semester.
Special Information	The maximum marking period is 15 work days.
Period of Education	The course lasts for one full quarter (9 weeks). The combination of the User-centred Sustainability Studio work with other course material is strongly discouraged.
Minimum number of participants	No (However, in cases where it is deemed that insufficient student numbers are available an alternative programme may be used).
Maximum number of participants	40 (first semester); no (second semester).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 4 Building Technology

AR4B025	Building Technology Graduation Studio	30
Course Coordinator	Prof.dr. M. Overend	
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Responsible for assignments	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	1 hour per week	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR3B025	
Course Contents	The Building Technology Graduation Studio continues the build-up of Bucky Lab Design and the BT MSc1 courses, the MSc2 Building Technology electicve courses, the MSc2 design studio's, and CORE or User-centred Sustainability Studio to the student's own graduation project in the Building Technology master track. In the graduation studio students work (in groups) on their individual graduation project comprising a preparative technical-scientific study (until the P2) and design, design by research or research by design (after the P2, this second part of the graduation studio). The studio intends to be in strong coherence with themes from the research programme of the AE+T department, so students can benefit from and contribute to the research activities by staff of the AE+T department. While the graduation project's subject can relate to any of the disciplines represented by the Building Technology master track, emphasis lies on sustainability-related topics of structural design, façade & product design, climate design and design informatics. These are interlinked. The student will be assigned to two supervisors from different 'colours' of AE+T (Structural Design / Façade & Product Design / Climate Design / Design Informatics), the student projects content as well as design and research is reflected by these two supervisors. Limited consultations with other staff members might be arranged if thats desired thematically.	
Study Goals	The student: - is capable of delivering innovative contributions toward the development of sustainable structural, façade, product, climate and computational design, and to technical-scientific research in these areas; - has insight in the profession and competences of structural, façade, product, climate or computational designers and their societal and professional role and ethical responsibilities in sustainable development of the built environment; - is able to present process and result in a clear and systematic way, reflect on process and work, and argument in a scientific way.	
Education Method	Supervised self study	
Assessment	P3 First design, design by research or research by design results, conceptual thesis report, plan for the remaining graduation timespan, draft reflection P4 Final design, design by research or research by design results, draft final thesis report P5 Final presentation of the design, design by research or research by design, final thesis report See the current Graduation Manual AUBS for more details and for the assessment criteria.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Once enrolled for AR3B025 (start of the graduation) students automatically continue their graduation in AR4B025	
Special Information	On set conditions, Building Technology students have the possibility to carry out their graduation research project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found at https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/forms. The agreements can be signed at the secretariat of Education and Student Affairs.	
Period of Education	Semester	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

variant Management in the Built Environment

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 Management in the Built Environment

AR1MBE015	Research Methods 1	5
Course Coordinator	K. Qian	
Course Coordinator	Dr. J.S.C.M. Hoekstra	
Instructor	Dr. J.S.C.M. Hoekstra	
Instructor	Y. Chen	
Instructor	K. Qian	
Instructor	Dr. A. Ersoy	
Instructor	Prof.dr. P.W.C. Chan	
Instructor	Dr. V. Danivska	
Responsible for assignments	Dr. J.S.C.M. Hoekstra	
Contact Hours / Week x/x/x/x	On average 4 to 6 hours per week, starting in week 1.1. and ending in week 1.10	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Research Methods 2 (AR3MBE010) and Research Methods 3 (AR0185)	
Expected prior knowledge	Bachelor in Architecture and the Built Environment or equivalent.	
Course Contents	In this course, students acquire methodological knowledge and practice their research design skills through designing their own research proposal on a relevant MBE topic. In this proposal, there needs to be a methodological fit between the research design components such as the research question(s), logic(s) of inquiry, theoretical framework, and methods of data collection. A data management plan and a reflection on research ethics are also included in the research proposal.	
Study Goals	After successfully completing the course, the student 1. is able to identify and compose research design components in the field of MBE; 2. is able to explain the concept of methodological fit; 3. is able to compose a basic mixed method research design in the field of MBE; 4. is able to identify typical ethical issues in research and plan a proper response; and 5. is able to plan research data management in terms of organization, storage, preservation, and sharing of collected data.	
Education Method	Lectures on the different components of a research design Interactive Workshops on various elements of the research process Working on individual assignments, under supervision of a tutor.	
Literature and Study Materials	To be announced on Brightspace. In previous years, the following book was used: Blaikie, N. and Priest, J. (2019) Designing Social research, The Logic of Anticipation, 3rd edition, Polity Press, Croydon, UK	
Assessment	Written assignments (100% of the final grade). If the final grade is lower than 5,5, there is a repair possibility in Q2. The maximum grade after a repair is 6.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Tags	Ethics Research Methods	
Period of Education	1	
Minimum number of participants	25	
Maximum number of participants	120	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1MBE020	Design and Construction Management	10
Course Coordinator	P.B. Vermey-Tassa	
Instructor	Dr. J.L. Heintz	
Instructor	Prof.dr. P.W.C. Chan	
Instructor	Dr. D.M. Hall	
Responsible for assignments	Dr. J.L. Heintz	
Contact Hours / Week x/x/x/x	35 hours per quarter / 5 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Master Management in the built environment.	
Expected prior knowledge	Bachelor Architecture or equivalent	
Summary	Contemporary buildings require the participation of many different design professions, suppliers of materials and systems, construction companies working together to provide both better service and better product quality in the face of increasingly complex and demanding clients and society. Steering a project through this complex web of relationships and requirements has become a profession in its own right - Design and Construction Management. This course covers the knowledge and application of project management techniques, general management skills in the context of project initiation, architectural design, tendering and construction. It covers both the systemic side of construction management and the social economic context. Working in small groups, students will study a wide range of readings on these subjects and create a summary representing the knowledge content of the course. This knowledge will be contextualised through case-based exercises in which students will have to develop management responses to typical situations in the design and construction processes. A combination of peer evaluation, reports and a final exam will ensure that each student receives individual recognition for their work.	
Course Contents	The Design and Construction Management course will cover the range of knowledge and skills required for overseeing the management of projects using literature and cases. Strategies about Scope definition; Project lifecycle (Timeline); Project delivery proposal; Quality management plan; Stakeholder involvement plan; Risk management plan and others will be researched, discussed and created.	
Study Goals	The student can: - Evaluate a real estate project of a given case and create a well justified project proposal, defined as related strategies. - Explain the related position and challenges of the project manager; - Assess her/his own strengths and weaknesses as the project manager of the case at hand.	
Education Method	Students work in a matrix organisation of study and workshop groups. Study groups The study groups will function to amplify the students' ability to read and study a large range of literature relevant to the aims of the course. Each week, the students will be assigned a series of texts; each student must take a share of the texts and prepare a summary. The students then share their summaries and discuss the texts with each other in the study group. The students will then participate in the construction of a combined summary in which students will consolidate their summaries on each of the assigned texts. Workshops Workshops can take the form of role-playing or other exercises carried out by the group. The purpose of the workshop is to study the task and to determine the content of the deliverable. Cases The course will be structured around one or more cases upon which the assigned tasks will be based. Exam An open-question exam (report) on the three study goals.	
Literature and Study Materials	Principal text: - Winch, G. (2010) Managing Construction Projects. An information processing approach. Blackwell Science. Oxford. - Additional texts: scientific publications and other texts made available via Brightspace.	
Assessment	Evaluation will be based on: - weekly performance: 40% - reflection: 10% - exam: 50% The first two grades form one half of the final grade and the exam the other half. The student passes if the average of the two grades is 6.0. A partial grade is at least 5.0.	
Permitted Materials during Tests	None.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	The maximum grading period is 10 working days.	
Remarks	None	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1MBE025	**Building Economics**	5
Course Coordinator	Ir. E.H.M. Geurts	
Course Coordinator	Ir. H.W. de Wolff	
Responsible for assignments	Ir. E.H.M. Geurts	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course requires some understanding of working with MS Excel. Extra instruction materials are provided, if prior experience is not in place. But students should expect to have a higher workload if no prior experience with MS Excel.	
Course Contents	This course is titled Building Economics (5ects) and deals with relevant financial and economic concepts that are essential for understanding the built environment. The course focusses on real estate markets and includes: real estate financial modelling, investment decisions in real estate, the relationship between real estate markets and the macro-economy, urban economic models and patterns and the relation of real estate markets and financialisation. The course objectives are focused on the development of both knowledge and (financial modelling and debating) skills and takes the format of a blended learning course design whereby online and on-campus study activities are combined.	
Study Goals	At the end of this course students are able to: relate real estate market characteristics to observed market dynamics; apply the basic principles of urban economics in real estate market analysis; assess the financial and economic impact of real estate development following from investment analysis; apply basic formulas and procedures for converting future cash flows and risk patterns to present value; evaluate the financial feasibility of (public and private) real estate investment decisions; debate the role of finance (and in particular financialisation) in real estate and the built environment.	
Education Method	This course consists of a combination of: - Lectures - (Group) assignments - Workshops - Self-paced video tutorials combined with individual assignments The course is designed in a blended format whereby education is delivered online and on-campus. In particular the financial modelling component is self-guided and self-paced.	
Literature and Study Materials	Literature and Study Materials list is given with the course outline on Brightspace.	
Assessment	Three assessment methods are employed that together determine the total weighted average grade of the course: Written examination [weight 50%] - retake next quarter Computer test of calculation assignments/ [weight 35%] - retake next quarter Financialisation (group) assignment [weight 15%] - repair option in case of fail. [Repair grade is a maximum 6 and needs to be completed within an indicated time slot after release of final grade] The maximum marking period is 10 working days.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Second quarter of semester 1 Master Management in the Built Environment	
Concept Schedule	The course contact hours are scheduled on 2 days a week. The self-study materials are released module by module and it is strongly advised to actively follow the course during the quarter and to not postpone studies until the exam week. Please note that there are exams in week 6 (just before Christmas break) and 10 and compulsory group work on financialisation in between. Retakes of the exams take place in Quarter 3. The group work is a repair assignment within a fixed time slot after the release of the grades.	
Minimum number of participants	15	
Maximum number of participants	All students enrolled in the MBE track are accepted. In addition there is space for maximum 15 students from other tracks or non-BK students.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1MBE030	Real Estate Management	10

Course Coordinator Ir. H.J.M. Vande Putte

Course Coordinator Dr.ing. G.A. van Bortel

Responsible for assignments Ir. H.J.M. Vande Putte

Education Period 2

Start Education 2

Exam Period 2
3

Course Language English

Summary

Real estate management is the ongoing process of aligning the built environment and the needs of users. This alignment happens at all scales of the built environment, for all types of users, and for all real estate aspects such as location, cost, function, time and quality. Real estate management is a particular type of portfolio or asset management and addresses operating the portfolio and assets as well as adapting them. This course studies real estate management from a systemic point of view and focuses mainly on the strategic management level.

Course Contents

The course first introduces the basics of real estate management through an overview of the past, current and future real estate delivery models and their potential to fulfil user needs and meet the socio-economic and ecological challenges. The studied models range from self-provision to acquisition via cooperation, markets and state initiatives. The models are situated within the main management, social, economic and organisational theories. The course then zooms in on how two particular types of users, namely organisations and households, are currently provided with the real estate they need.

Parallel to this theory immersion, students engage in a project assignment wherein they first assess the performance of the current real estate provision to organisations or households and thereafter develop a strategy to improve this provision in the future.

The first type of users that is studied in detail organisations ranges from small to large organisations across all industries and services, and from the private to the public sector. The management of real estate for organisations is called corporate real estate (CRE) management. CRE managers provide the built assets that organisations use to house their business. They aim to maximise their organisation's stakeholder and shareholder value. CRE management should be distinguished from the approach of a real estate investor or developer. The CRE managers key performance indicators of the are the safety, reliability, suitability, sufficiency and efficiency of the accommodation provided, not financial return. This course studies how CRE strategies and operations respond to organisational expectations and how they can continue to do so in the future.

The second type of users that is studied in detail are households. These range from individuals to families, communities and organised groups, from the low- and middle-income to the high-income, at the local, the national and global levels. The real estate management that addresses households is commonly called Housing management. Acquiring the right home is a crucial aspect of a households aspirations, albeit not simply a matter of individual choice as the housing market is an imperfect market. Especially in larger cities housing production and pricing mechanisms limit the possibilities of (groups of) users to attain their goals and hinder social cohesion and urban competitiveness. As a consequence, the delivery of housing in developed economies became highly institutionalised, and different types of government interventions were introduced. Housing managers need to improve the environmental sustainability of the housing stock, which constitutes the largest investment challenge in the built environment. New housing delivery models are emerging to better realise the right to housing for all in the future.

Study Goals

1. Understand the key delivery models of real estate, for the wide variety of users and types of real estate, and in more depth how CRE and Housing provision can be aligned with the expectations of organisations and households respectively.
2. Apply the above knowledge through the development and execution of a performance analysis of real estate portfolios of different kinds and scale levels, and through the evidence-based design of a future strategy for these portfolios
3. Critically evaluate these delivery models from the point of view of the key challenges for the future: their implications for the end-users, society and eco-system.
4. Analyse current theories on their specificity and create building blocks for a generic real estate management theory.

Education Method

- Project-based learning, where the theory and the project assignment run parallel from the first week, and theory topics are chosen and presented (as much as possible) to support the assignment.
- Intermediate assessments to help students synchronise theory acquisition and application, and developing their writing skills.
- Project assignment, wherein a strategy is developed for a real-life CRE or housing portfolio; this assignment is a real-life case and is provided by a portfolio owner; the assignment is developed partly in groups of 3 or 4 students and partly individually; it is executed and tutored in weekly studio sessions.
- Learning activities comprise lectures, expert talks, site visits to companies and projects, seminars, studio work with feedback, peer assessments, and tutorials.

Note: the one day excursion in week 1 is a compulsory part of the program of which the (limited) costs need to be covered by the student.

Literature and Study Materials

The literature and study materials will be announced on Brightspace. They consist of:
- Reader material for the theory part
- Reader material for the project assignment part
- Slides and literature from the lectures that support both.

Assessment

The assessment consists of two parts:
1. Theory exam (individual work), consisting of several assessments: 60% of the grade
2. Project assignment report (partly group work and partly individual): 40% of the grade

To pass the course, the student must obtain a grade of 5 or higher for each part and a grade of 6 or higher for the weighted average of the two parts.

Retakes:
In case of a retake of one or several parts during the same academic year, the grades of 5 or higher will be kept and combined with the grades of the retaken parts.
The retake of the theory exam is a single session exam of 3 hours and takes place in week 3.10.
Retake of the project assignment report comes in the form of a reparation without tutor support. The maximum grade for the reparation is 6/10. The deadline is the end of week 3.5.

Enrolment / Application

Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see:
https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams

Additional enrolment information for non-BK students:

Course enrolment via Brightspace only is not considered a valid registration
Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below)
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Period of Education Semester 1, quarter 2 (weeks 2.1 à 2.10)

Concept Schedule	The course contact hours are scheduled on 2 or 3 days a week. Intermediate assessments and project assignment deadlines are spread over the quarter. Group work sessions should be joined. The course therefore needs to be actively followed during the entire quarter and studies cannot be postponed until the exam week.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 Management in the Built Environment

AR2MBE011	Building Law	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr. F.A.M. Hobma	
Instructor	Ing. P. de Jong	
Instructor	Prof.mr.dr. E.M. Bruggeman	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	2 hours per week starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	MSc track Management in the Built Environment. This course cannot be chosen as an elective by students Construction Management and Engineering.	
Expected prior knowledge	Bachelor of Science	
Course Contents	Building Law covers topics which are essential for understanding of the built environment. Every building project has certain needs: (a) capacity. This means that the building initiator needs designs, engineering, building materials and labor. The building initiator will use the legal instrument of contracts to arrange this. In this context, rules for procurement are relevant. These rules require that sometimes competition between suppliers of goods and services is required. (b) Land. This means that the building initiator has to buy or lease land on which the project is realised. Contracts, pre-emption right and expropriation may be used to get access to land. (c) Permission. The building initiator needs permits and plan approvals before the work can commence. Permit procedures make sure that the building project is tested against relevant norms and criteria. On top of this, the building initiator needs to make sure that conflicts are avoided as much as possible. The fields that are taught in this course are: Building contract law, Procurement law, Property law, Planning law, Environmental law and Public private partnerships. It includes private law and public law, from the building level to the level of urban development.	
Study Goals	Students are able to: 1. contrast the different building contract models. 2. evaluate the strengths and weaknesses of different contract models. 3. describe the essential rules regarding procurement. 4. address legal aspects of real estate development projects from the European and Dutch Planning and Environmental legislation perspective. 5. discuss the different forms of cooperation between public and private sectors regarding buildings and urban development.	
Education Method	Lectures, assignments and group discussions	
Literature and Study Materials	- A Practical Guide to Dutch Building Contracts - F.A.M. Hobma and P. Jong, An Instrumental Approach to Planning and Development Law in the Netherlands - Uniform Administrative Conditions for the Execution of Works and Technical Services	
Assessment	Two assessment methods are employed that together determine the total weighted average grade of the course: Written theory exam [weight 70%] Law assignments [weight 30%] There is no minimum grade for the assignments. Therefore there is no retake for the assignments.	
Permitted Materials during Tests	Open book exam	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	The third quarter of the academic year	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MBE030	Urban Development Management	10
Course Coordinator	Dr. A. Ersoy	
Course Coordinator	Prof.dr. W.K. Korthals Altes	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Urban Development Management is concerned with managing the decisions of the many stakeholders involved in the development of urban areas. The course of Urban Development Management focuses on strategies for complex spatial development projects in contemporary cities and regions. These strategies include multi-actor and multi-level design, planning and governance activities, and lead to rounds of collaborative decision-making about organising and (re)financing long-term, sustainable urban development efforts.	
Study Goals	Study goals: Understanding and analysing the multi-actor and multi-level design, planning and governance activities that shape urban development project strategies in cities and regions. Identifying the institutional conditions that constrain or promote participatory processes and sustainable outcomes in urban development practices. Assessing the competing societal values and objectives apparent in contemporary urban development, and reflecting on how these are synthesised into area-based projects. Developing strategic proposals for multi-level and multi actor urban development projects, taking into account societal values and participatory processes, and making use of legal or policy instruments, organisational structures, and financing arrangements.	
Education Method	This course uses a combination of: - Lectures - Online materials - Assignments - Workshops	
Assessment	Written exam (50%) and a portfolio of assignments (50%). The course adheres to Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Master in retake and repair regulations	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC)

ARFW1501	Entrepreneurship in the Built Environment	15
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Ir. H.J.M. Vande Putte	
Instructor	Ir. A.C. Bergsma	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Dr. M. Bos-de Vos	
Instructor	Dr. D.M. Hall	
Instructor	Dr.ir. M.U.J. Peeters	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organizations within the built environment operate and function from an inside perspective. In the course, students will develop an entrepreneurial attitude and learn about the real-world dynamics between different types of built environment organizations and their respective roles in fostering innovation. Students are challenged to create a business proposition based on their understanding of the built environment. To enable this, the course provides students with the required knowledge and skills to apply all the core entrepreneurship principles, such as legal, financial, and business modelling. Furthermore, the course contains track-specific elective components based on personal interest and track requirements. This means that A and BT students can choose to explore the nuances of starting their architectural design studio. While MBE, A, and BT students can create and defend a business proposal for an innovative venture active in the Built Environment. For architecture students both variants contain a design component. The course contains ample opportunities to interact with practice, and students learn to use these opportunities to articulate their vision and business propositions effectively.	
Course Contents	This course is designed for Architecture (A), Management in the Built Environment (MBE), and Building Technology (BT) students eager to (1) explore opportunities for starting their venture in the built environment (such as an architectural firm, development company, tech startup, etc.) and/or (2) discover how organisations within the built environment operate and function from an inside perspective. This course allows students to analyse how organisations (such as architectural studios, developers, contractors, and industrial construction companies) in the built environment operate, how they change in time, and how they can contribute to innovation. In the course students delve into the position of these companies in the built environment and learn how companies align their business goals with their surroundings. Moreover, the course provides students with the unique opportunity to analyse existing organisations and go beyond this. Students learn to identify how they can contribute to taking on challenges in the built environment using their personal skills, interests, and characteristics. To do so, students will develop an entrepreneurial attitude and investigate the opportunities for a (design) venture in the built environment. Students will focus on identifying the kind of impactful work they aspire and exploring how they can bring about meaningful change. The course is suitable for students of the tracks Management in the Built Environment (MBE), Architecture (A) and Building Technology (BT). The course contains shared components focusing on fundamental elements of entrepreneurship, including theory about legal, financial, business modelling, market forces and organisational issues. In addition to the shared components, students can choose from one of two variants that match their interests and the track-related conditions (tracks MBE, A and BT). For example, A and BT students can choose to explore the nuances of starting their architectural design studio. MBE , A, and BT students can create and defend a business proposal for an existing or new, innovative construction industry venture. For Architecture students the course also contains a design component.	
Study Goals	After completion of the course the student is able to: Overarching study goals - Identify opportunities for entrepreneurship by analysing the relations between different organisations in the built environment. - Assess the role of entrepreneurship in changes in the architectural world and the built environment. - Craft a vision and use this to create a coherent business plan that combines business modelling, finance, legal, and market knowledge. - Critically evaluate and reflect on entrepreneurial skills and plans. - Design and deliver a compelling and succinct vision and business pitch. Additional study goals for Variant: Develop your architectural firm (open to A en BT students) - Assess the influence of their personal characteristics on entrepreneurial opportunities and formulate their motivations for launching an architectural design studio. - Assess the viability of starting a self-owned company, leveraging passion, knowledge, skills, and network. - Develop a design vision to guide the future direction of their own design firm. - Reveal the design vision in an actual design. - Propose the next steps for establishing a self-owned design firm. Additional study goals for Variant: Develop your impactful business model (open to A, MBE and BT students) - Assess building sector organisations, including their business models, revenue streams, and network. - Identify opportunities for innovative business activities in the construction industry and evaluate the viability of these opportunities. - Debate the value of innovative business activities and reflect on relevant social, scientific, and ethical issues. - Defend their business plan in a convincing matter to industry experts and utilise expert input to improve. - Architecture students will also develop a design as part of the impactful business model - Propose the next steps for implementing the business model in an existing or new venture.	
Education Method	The course's learning activities comprise: - lectures: theory - workshops - self-study: developing vision and translation into entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop vision and entrepreneurial plan	
Assessment	Individual and group report and pitch, including (design) vision, entrepreneurial plan, and personal reflection. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may	

	only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 85.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1502	Peri-Urban Transformations: Analysis, Development, Design	15
Course Coordinator	T.N. Broekmans	
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	15 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this new elective course the interaction between spatial design, area and project development tools and social, economic, political, and environmental issues is at stake. Students from the MBE, Landscape Architecture and Urbanism tracks and the MSc CME Design & Integration Domain (from the coupled course Transformative Design), collaborate on the transformation of a peripheral urban area in the Netherlands. The pressure on these urban fringes is both urban and rural and characterised by urban sprawl, fragmented agricultural or industrial activity, remnants of natural and landscape values. Working together in groups you will develop a master plan and elaborate individually a more detailed, imaginable transformation proposal. Students confront spatial qualities, opportunities, limitations and challenges of the chosen area with societal needs, transition goals, ownership issues and stakeholder claims. You design and define proposals on different scales and complexities including area, portfolio and project level. You will apply development strategies, feasibility criteria and research by design activities. This course is organised as a multidisciplinary experience combining approaches, knowledge and skills from the viewpoints of management, urban design and landscape development and is therefore representative for area development practice in the real world. This collaboration provides insight in the complementarity of viewpoints, data, tools, vocabulary, methods and fun.	
Study Goals	(1) students can analyse the characteristics, conditions, stakeholders, opportunities and potentialities of peri-urban transformation areas (using a theoretical framework) (2) students can develop a tentative spatial perspective (e.g. spatial, social, economic, political, environmental) in a peri-urban context including primary notions of phasing: from initiative to implementation (masterplan version 0.5) (3) students can elaborate a crucial, domain-specific aspect or component of the masterplan like a portfolio, structure, block typology or building (4) students can synthesise varied input into an integrated, substantiated master plan for a peri-urban transformation area (masterplan version 1.0) (5) students can reflect on the implications of the domain-specific theory on the application of the masterplan and its elaborations	
Education Method	In this course four main educational methods are used: (1) Lectures (2) Studio work in groups (2) Studio work individually (4) Plenary presentations and discussions	
Assessment	This course has two assessments (1) group: atlas and integrated masterplan (2) individual: elaboration proposal and reflection report A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4.	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 75.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 Management in the Built Environment

AR3MBE010	Research Methods 2	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	Research Methods 1 AR1MBE015	
Course Contents	The Research Methods 2 (RM2) course of Management in the Built Environment (MBE) offers training in qualitative research methodologies. The course begins by revisiting the fundamentals of the qualitative research cycle. Subsequent modules go into various qualitative methods such as interviews and observations. Participants will learn essential techniques for analyzing qualitative data and presenting their findings. Each module comprises a lecture focusing on the specific method followed by sessions dedicated to practical exercises. Students work on both group and individual assignments This course is specifically designed for Master students who have previously completed Research Methods 1 at MBE. MBE Students who want to learn more about statistical research methods should participate in the elective Research Methods 3 course (RM3).	
Study Goals	The student: understands the basic principles of qualitative research and its relevance to the social sciences. learns essential qualitative research methods commonly used in social science studies. develops basic skills in analyzing qualitative data and drawing preliminary conclusions. gains familiarity with the practical application of qualitative research techniques through hands-on exercises. develops skills for effectively presenting qualitative research findings.	
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.	
Computer Use	Students will learn to use Atlas Ti, a software for analyzing qualitative data. This software is provided by the TU Delft.	
Literature and Study Materials	Literature and Study Materials will be available 2 weeks prior to the start of the course in Brightspace.	
Assessment	The grade is based on group (40%) and an individual (60%) assignments.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	The course runs in quarter 1 and 3.	
Minimum number of participants	There is no minimum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3MBE100	MSc 3 Graduation Laboratory Management in the Built Environment	10

Course Coordinator	Dr. M.H. Arkesteijn
Course Coordinator	Prof.dr.ir. H.J. Visscher
Responsible for assignments	Prof.dr.ir. H.J. Visscher
Education Period	1 2 3 4
Start Education	1 3
Exam Period	none
Course Language	English
Course Contents	AR3MBE100 covers the first part of the graduation process, from entry in the MBE Graduation Lab up to and including the P2. Students passing the P2 proceed automatically to the second part of the graduation process, course AR4R010, where they complete their thesis and graduate from the Master of Science programme. IN AR3MBE100 each student selects a topic and supervisors for their thesis research, and develops a full research proposal for their activities after the P2.
Study Goals	The purpose of AR3MBE100 is that each student develops a promising, substantiated and feasible research proposal that can be implemented in AR4R010. To this end, it is strongly recommended that all exploratory and preparatory work is completed before the P2: literature and market research, acquaintance with available cases, agreements with external organizations (including internships) etc.
Education Method	Introductory lectures, followed by individual supervision by the mentor and group guidance in the graduation labs.
Assessment	- P1: problem statement - P2: full proposal
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Q1 and Q2 or Q3 and Q4
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Free electives 15 EC (MBE or other)

AR0117	Didactic coaching skills for architecture and the built environment	5

Course Coordinator	Dr.ir. S. Zijlstra
Course Coordinator	Dr. M.H. Arkesteijn
Instructor	Dr. H.J.F.M. Boumeester
Instructor	Dr.ir. R.M. Rooij
Instructor	Dr. M.H. Arkesteijn
Responsible for assignments	Dr. M.H. Arkesteijn
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.
Education Period	1
Start Education	1
Exam Period	1
Course Language	English
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in de surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brighstpace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade for the exam is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Minimum number of participants	15	
Maximum number of participants	-	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participate in the Case Studies Methods course Research Methods 2 and want to learn about Applied Statistical Methods should participate in the elective Research Methods 3 course (RM3). The aim of RM3 is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by an additional analysis. There will be several statistical approaches available. In the RM2 course only basic procedures will be mastered. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design. One then will use the (already)collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences.	
Study Goals	The student: - is able to perform several basic statistical approaches in SPSS; - is able to properly interpret the resulting output in SPSS; - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question.	
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.	
Literature and Study Materials	- Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: SagePublications Ltd, ISBN 9781526419521	
Assessment	- The mark will be based on the evaluation of a final SPSS assignment and an individual or logistic regression assignment.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Research methods 3 runs in Quarter 1 and 3.	
Minimum number of participants	At least 6. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel this elective course.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	D.K. Czischke	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Q1	
Concept Schedule	Monday	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 4 Management in the Built Environment

AR4R010	MSc 4 Graduation Laboratory Management in the Built Environment	30

Course Coordinator	Dr. M.H. Arkesteijn
Course Coordinator	Prof.dr.ir. H.J. Visscher
Responsible for assignments	Prof.dr.ir. H.J. Visscher
Contact Hours / Week x/x/x/x	25 hours per semester
Education Period	1 2 3 4
Start Education	1 3
Exam Period	none
Course Language	English
Required for	Final graduation
Expected prior knowledge	Master 1, 2 and 3 of the MBE track, Master Architecture, Urbanism and Building Sciences.
Parts	P3, P4 and P5
Course Contents	AR4R010 is the second and final part of the graduation process in the Master track MBE. In it, students work towards the completion of their graduation thesis under supervision by the mentor team appointed in AR3MBE100.
Study Goals	Based on the P2 results in AR3R010 and under supervision by their mentors, students individually conduct research towards the completion of their graduation project, in a way that demonstrates their ability to meet the final learning goals of the MBE Master track, of the Faculty of Architecture & the Built Environment and of Delft University of Technology.
Education Method	Individual research towards completion of a Master thesis, under supervision by two or three mentors.
Literature and Study Materials	See Brightspace
Assessment	- P3: halfway through the semester and before submitting the P4 request, each student delivers an interim presentation and a draft graduation report. - P4: at the end of the semester, each student delivers a presentation of the completed project and a draft final graduation report. - P5: having passed the P4 assessment, each student finalises their final graduation report and defends it in the final examination session.
Exam Hours	- P3: scheduled by each student individually - P4 and P5: scheduled centrally by the Faculty of Architecture & the Built Environment, upon request by the student
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Once enrolled in AR3MBE100 (start of the graduation) students automatically continue their graduation in AR4R010
Special Information	The maximum marking period is 10 work days. MBE students have the option to carry out their graduation research at an appropriate external organisation. Students who wish to do so are required to sign a standard internship agreement, as stipulated at http://studenten.tudelft.nl/en/students/faculty-specific/architecture/forms/.
Period of Education	Semester
Concept Schedule	See Brightspace and https://www.tudelft.nl/en/student/faculties/a-be-student-portal/education/academic-graduation-calendar
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Cross Domain City of the Future

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 Cross Domain City of the Future

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory for MBE students

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Compulsory

AR3CS021	Seminar Cross Domain City of the Future	5
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. T.A. Daamen	
Course Coordinator	Dr.ir. J.H. Baggen	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	4 hours/week (1.1-1.8 & 2.1-2.5)	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 & MSc2	
Course Contents	The Cross Domain City of the Future seminar focuses on the challenges linked to the transformation of our cities, specifically to: - analyze and understand which factors, tendencies, (design research) approaches, technological innovations, development, and governance models will play a role in shaping the future of our cities. - study relevant examples and/or best practices around the world and compare them with the city-study areas of the graduation studio. - collect data into research books to be shared among the participating students. - argument and develop the investigations (research positions) into a written article. Multiple aspects / disciplinary perspectives regarding the (research design) project will be presented by the involved researchers in interactive sessions. A.o., the following topics will be discussed: transportation, connectivity, new mobility concepts, housing challenges, climate change mitigation, cultural planning, health and private, public and semi-public spaces, circular economy, politics, resiliency, and adaptation strategies. The seminar challenges students to develop critical and comparative investigations focusing on specific insights and positions to be determined within the (research) framework of the studio/students' group.	
Study Goals	- Become aware of different (design) research methods in order to inform own graduation process. - Acquire necessary research skills in order to set up and develop a research article - Explore different inquiry methodologies and research approaches. - Work in a collaborative way within a multidisciplinary group and probe different constraints to define own (design) research approach.	
Education Method	Workshops/tutorials given by the involved tutors and/or invited guests/experts	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	- Writing Assignment - Analytical Assignment Next to the participation in the seminar, the assessment of the course will enclose a writing assignment: Option 1, a position paper Option 2, a report of the thematic seminar sessions related to a personal research standpoint. The assessment criteria will be: -Is the paper or report coherent/concise, regarding structure and clearness of style? -Is the language adequate? -How did the paper or report follow the review comments? Problem field and research question: -Has a research question / standpoint been clearly defined? -Has the question / standpoint been developed beyond its initial formulation? -Does the paper / report acknowledge a state of the art, regarding this question? Societal and scientific relevance -Is the paper /report relevant to the studio theme City of the Future from a societal point of view? -Is the paper / report relevant to the studio theme City of the Future from a scientific point of view? -Has a position been taken, in relation to this state of the art? Theoretical and operational frameworks -How is the paper / report connected with the graduation topic? -Are the sources pertinent, and well used and is the use of reference and own words balanced? -Does the paper / report follow a clear approach / methodology and shows scientific quality? On the scale of 10, the student achieves a sufficient result with the final grade of 6; If failed in the first attempt student are offered the possibility of a repair. The repair has to be completed within 2 weeks after the completion of the course and will be graded by either O (fail) or V (pass).	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	1-2	
Concept Schedule	Friday afternoon	
Leerstoel	Architectural Design Crossovers	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3CS081	Graduation Cross Domain City of the Future	20
Responsible Instructor	Dr.ir. R. Cavallo	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Co-responsible for assignments	Y. Chen	
Co-responsible for assignments	Dr.ir. T.A. Daamen	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This Graduation Lab is a special thesis laboratory for students that would like to develop their own (design and/ or research) fascinations in a multidisciplinary setting. Students with different backgrounds and from different TU Delft MSc tracks will work on common challenges sharing insights, approaches and methodologies proper of their own disciplines. While doing that, students work together to enrich their own graduation pathway, setting up and developing workshops, lectures, excursions and visiting critics. The students of this Graduation Lab are responsible for the program and the agenda through the thesis period. They are expected to work together as much as they possibly can, because mutual critic and collaboration is one of the important means of education in this lab. Students of the MBE track complete the first two assessments (P1 and P2) by selecting a research subject and mentor team; conducting literature and market research; developing a problem statement, objectives and goals, and an approach to solving the problem and reaching their goals. This course is including a student driven workshop module. Students are asked to develop a collective educational program addressing themes relevant for all participants. These activities may take the form of or include workshops, lecture series, visiting critics, excursions, charettes, etc.	
Study Goals	Problem analysis The student has knowledge and understanding of research approaches and methods for translating a subject with scientific and societal relevance into a problem analysis, problem statement, research objectives and research questions in a critical and grounded manner. Literature review The student is familiar with fundamental and recent literature in the area of MBE, and is able to conduct a comprehensive, in-depth literature review that retrieves literature relevant to their graduation project, through which they can substantiate research hypotheses and approaches. Synthesis The student has a creative, innovative and investigative approach to solving the selected problems. The student is capable of identifying relevant knowledge in their own and related areas, acquired in part through the literature review, and systematically utilising it for the definition of a coherent theoretical framework and conceptual models for their research. Methodology The student is capable of selecting appropriate research methods in a transparent and substantiated manner and of applying these in a scientifically and ethically responsible manner. Acceptability and relevance The student is capable of evaluating their own process, products and performance in relation to current scientific and professional knowledge. The student is able to formulate clear conclusions and recommendations for further research and application, and through these demonstrate that their work meets the standards of scientific research and contributes to the solution of societal problems. Time management The student is aware of the requirements for interim and end products, has sufficient time management skills to make a realistic estimate of activities and the amount of time needed for each of them and, on the basis of these, can produce a reliable working plan. Reporting and communication The student is capable of producing informative written reports, suitable to a scientific and professional audience, that provide a structured, coherent, consistent, precise and insightful account of their research process and products. The student is capable of delivering oral presentations of their work in an informative and engaging manner and at an appropriate scientific level, using valid arguments in discussing their subject. The student is open to constructive criticism and is willing to learn from feedback and comments.	
Education Method	This is a student driven graduation laboratory. The educational method is therefore to be developed by the students in conversation with each other and the coordinators. The assumption is that studio instruction will be the primary teaching method. Students will guide their own studies and determine their own learning styles.	
Assessment	- P1: halfway through the course, each student presents the progress of their work in the form of a preliminary report (draft research proposal and plan for the thesis) - P2: at the end of the course, each student presents a final research proposal and plan for the completion of their thesis	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Fall Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Free electives/Research Methods 2

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 4 Cross Domain City of the Future

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

For MBE Students

AR4CS030	Cross Domain City of the Future	30
Course Coordinator	Dr.ir. T.A. Daamen	
Responsible for assignments	Dr.ir. T.A. Daamen	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This Graduation Lab is a special thesis laboratory for students that would like to develop their own (design and/ or research) fascinations in a multidisciplinary setting. Students with different backgrounds and from different TU Delft MSc tracks will work on common challenges sharing insights, approaches and methodologies proper of their own disciplines. While doing that, students work together to enrich their own graduation pathway, setting up and developing workshops, lectures, excursions and visiting critics. The students of this Graduation Lab are responsible for the program and the agenda through the thesis period. They are expected to work together as much as they possibly can, because mutual critic and collaboration is one of the important means of education in this lab.	
Study Goals	Based on the P2 results and supervised by the mentor team, students individually conduct research towards completion of their graduation project, in a way that demonstrates their ability to meet the final learning goals of the MBE Master track, as well those of the Faculty of Architecture & the Built Environment and of Delft University of Technology.	
Education Method	Individual research towards completion of a Master thesis, under supervision by two mentors.	
Assessment	- P3: halfway through the semester each student delivers an interim presentation and a draft graduation report. - P4: at the end of the semester each student delivers a presentation of the completed project and a draft final graduation report. - P5: having passed the P4 exam, each student finalises their final graduation report and defends it in the final examination session.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Once enrolled in AR3CS081 (start of the graduation) students automatically continue their graduation in AR4CS030	
Period of Education	Spring Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

variant Landscape Architecture

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 Landscape Architecture

AR1LA011	Architecture and Landscape: Design Studio	10
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	F.T. Toni	
Instructor	Dr.ir. I. Bobbink	
Instructor	Dr.ir. S.I. de Wit	
Instructor	D. Piccinini	
Instructor	F.T. Toni	
Responsible for assignments	Prof.ir. E.A.J. Luiten	
Contact Hours / Week x/x/x/x	12 average	
Education Period	1	
Start Education	1	
Exam Period	1 5	
Course Language	English	
Summary	The design studio Architecture and Landscape develops knowledge, skills and stances in landscape architectural research and design via an assignment for a garden-building ensemble in a semi-natural environment. With this assignment the studio takes an in-depth look at physical landscape space and human experience; delves into site and context from a (garden) design perspective; elaborates elementary landscapes dynamics; and develops skills in design composition and design elaboration at the domain scale, with an emphasis on planting design.	
Course Contents	The studio focusses on the garden (domain) scale of landscape architecture, as the most condensed form of (the ontological complex) landscape. The design project takes an in-depth look at physical landscape space, notions of context and human experience, landscapes dynamics and change, and planting design. The site has delineated boundaries, a specific scope of transformation and clearly defined ownership and operation. Integrating site and context (geographical, historical, cultural etc.) into the design forms a central motif for the assignment. Knowledge, skills and stances are developed in interpretations of a given landscape, working with landscape processes, conceptual design thinking, elaborating a landscape architectural (domain) composition, articulating the relationship between concept, composition and design details, planting design and ethical dimensions of landscape architecture.	
Study Goals	By the end of the course, you should be able to: - identify landscape qualities by making use of phenomenological analysis (human perception); - design a landscape architectural composition and a planting scheme, which is the expression of the sensory qualities of the landscape (scale, materiality, movement) and landscape processes (relationships between environmental factors and resulting vegetation over time); - develop a coherent and clear relationship between concept, composition and detail; - demonstrate a variety of presentation and communication techniques, expressing human experiences (scale, materiality, time, movement); - reflect on your design, involving ethical considerations as an expression of theoretical perspectives on landscape, landscape architecture and the relation between humans and nature.	
Education Method	Studio work Workshops Instruction sessions Excursions Fieldwork	
Course Relations	The quarter Architecture and Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking. Four perspectives on the analysis and design of (urban) landscapes form the backbone of the master track Landscape Architecture: landscape perception, landscape as palimpsest, landscape as scale continuum and landscape as ecological, economic and social process.	
Literature and Study Materials	- Steenbergen, C.M. and Reh, W., 'Architecture and Landscape - the Design Experiment of the Great European Gardens and Landscapes', Basel, Boston, Berlin, Birkhauser, 2003. - Dee, C. (2001), Form and fabric in landscape architecture; a visual introduction. Londen, New York: Spon Press. - Bell, S. (2004), Elements of visual design in the landscape. New York: Spon Press.	
Assessment	2D drawings and supporting visuals Physical/digital 3D-model Oral presentations Written reflection The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Retake takes place during the summer break or during the new academical year.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1LA021	Architecture and Landscape: Theory, Method and Critical Thinking	5

Course Coordinator	Prof.ir. E.A.J. Luiten
Course Coordinator	Dr.ir. S.I. de Wit
Instructor	F.T. Toni
Responsible for assignments	Dr.ir. S.I. de Wit
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.7.
Education Period	1
Start Education	1
Exam Period	1 5
Course Language	English
Summary	The Theory, Method and Critical Thinking course Architecture and Landscape introduces landscape architectural compositional principles and their application and focuses on the architectural form of plants and planting schemes over time. The course presents tools, theories, methods and techniques related to the landscape architecture discipline.
Course Contents	In this course different knowledge fields are introduced, revolving around landscape architectural composition. Phenomenology, environmental psychology, compositional theory and planting knowledge will feed the students insight in landscape architectural composition. Seminal landscape architectonical objects will be presented and discussed, with the emphasis on the discovery and study of the complex compositional rules with which a design is built up and the relationship between composition and perception. Theories, concepts and design aspects are brought into a wider scope addressing analytical and compositional techniques and typological research. Furthermore, the role of planting will be explored and discussed in terms of space and time dynamics. The emphasis is on basic knowledge of assortment and its application in landscape design in order to define and organize space into imaginative compositions that evolve through time. Crucial is the understanding of the relationship of planting with spaces and objects, with their natural surroundings (habitat) and their evolution in time. Field trips and literature study will provide the material for studying and documenting examples of several species and their arrangements in order to discover their spatial and environmental characteristics.
Study Goals	After this course, you should be able to: apply aesthetic landscape experience (phenomenology) as framework/tool/technique in the analysis of a designed landscape; demonstrate basic ecological knowledge/insight in habitus-habitat relationships (relationships between environmental factors and resulting vegetation); interpret the formal qualities of woody plants and arrangements and their spatial-experiential characteristics from a landscape architectural perspective; demonstrate an understanding of the compositional repertoire of landscape design in order to effectively describe, interpret and evaluate designed landscapes; demonstrate analytical drawing and writing techniques that reveal spatial and experiential qualities of planting compositions in relation to their landscape context.
Education Method	Lecture series Field work Discussion sessions You will work in groups.
Course Relations	The quarter Architecture and Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking. Four perspectives on the analysis and design of (urban) landscapes form the backbone of the master track Landscape Architecture: landscape perception, landscape as palimpsest, landscape as scale continuum and landscape as ecological, economic and social process.
Literature and Study Materials	- Dooren, N. van (2018). The landscape of critique; The state of critique in landscape architecture and its future challenges. SPOOL Vol. 5 No.1 Landscape Metropolis #4, Criticising practice - practicing criticism, 13-32. doi: 10.7480/spool.2018.1.1936. (available on Brightspace) - Kaplan, R. and Kaplan, S (2005). Preference, restoration and meaningful action in the context of nearby nature. In Barlett. P.F. (ed.) Urban Places; reconnecting with nature. MIT press. (available on Brightspace) - Moore, G.T. (1979). Environment-Behavior Studies, in Snyder, J.C. and Catanese, A.J. (eds.) Introduction to Architecture. McGraw-Hill, 46-71. (available on Brightspace) - Steenbergen, C.M. and Reh, W. (2003). Architecture and Landscape; the Design Experiment of the Great European Gardens and Landscapes. Birkhäuser. - Steenbergen, C.M. (2008) Composing Landscapes: Analysis, Typology and Experiments for Design. Birkhäuser. - Wohrle, R.E. and Wohrle, H.J., (2008). Designing with plants. Birkhäuser.
Assessment	Plan critique including an analysis in drawings and words and critical interpretation with arguments based on the offered knowledge fields. When a specific part of the plan critique is missing or insufficient, leading to a grade of 5,5 or lower, you will get the opportunity to repair. A repair needs to be handed within 2 weeks after receipt of the grade, and will result in a 6. When the plan critique as a whole is insufficient you will have the chance to retake the assignment: a plan critique on a different case. This will take place during the Summer period, to be finalised before the start of the following academic year. During this period three discussion moments with your tutor may be planned.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter
Leerstoel	Landscape Architecture
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1LA051	Dutch Landscape: Design Studio	10
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. I. Bobbink	
Instructor	Dr.ir. I. Bobbink	
Instructor	Prof.dr.ing. S. Nijhuis	
Instructor	D. Piccinini	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	12 hours / week average	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Summary	The design studio Dutch Landscape emphasizes on exploration, understanding and modification of landscape patterns, conditions, structures and their spatial expression. The assignment asks for interventions in a rural area in which the expression of the balance between land and water plays an important role, fitting the accommodation of a range of claims (agriculture, recreation, living, nature development) of the given site.	
Course Contents	In the design course you formulate your own assignment within a defined challenge, in which different scales and perspectives come together: water issues and the social structure of a rural society and ongoing spatial claims (like farming, energy production, recreation, nature development) on the scale of a polder landscape. The aim of the design is to explore the transition of landscape as a means of adapting and giving expression to existing landscape patterns, leading to new landscape architectural (spatial and experiential) compositions in which beauty, landscape processes, functionality and inclusiveness (of people, animals and plants) coincide. You will design through scales from concept to detail, with an understanding of the area of influence, area of effect and area of control. The design is based on a landscape analysis, understanding the spatial structure and patterns, which should lead to a spatial composition for an imaginable future, allowing for and anticipating on social and ecological processes. You learn how to design through time: from past relevant periods to new conditions for a desired future, according to the assignment that is defined and developed by yourself.	
Study Goals	By the end of this course, the student should be able to: - understand and interpret the Dutch polder landscape, its water management, physical physical patterns, and spatial formation; - understand the interdependence of hydrological systems, soil conditions, land use and vegetation characteristics; - apply landscape architectural design tools through scales: from ditch to horizon by employing options and alternatives; - understand and integrate landscape engineering techniques and ecological knowledge related to plant succession, water/land gradients and maintenance in the design; - create self-explanatory drawings expressing systems, scale ranges, texture, and materials of an integrative rural landscape proposal.	
Education Method	Studio work (working in pairs) Workshops Lectures Excursion	
Course Relations	The quarter Dutch Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking.	
Literature and Study Materials	- Bobbink, I. en Loen S. (2013) 'Water inSight' http://repository.tudelft.nl/view/ir/uuid:e1af985b-7f72-4a55-9c07-2fc0f4c7e4f1/ - Reh, W. en Steenbergen C. (2007) Zee van Land, Stichting Uitgeverij Noord-Holland - various editions of Dutch and European landscape architecture, Blauwdruk, Wageningen	
Assessment	booklet and/or posters physical models oral presentation written reflection on design	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1LA061	Dutch Landscape: Theory, Method and Critical Thinking	5
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Instructor	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. G.A. Verschuure-Stuip	
Contact Hours / Week x/x/x/x	8 hours / week average	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Summary	The Theory, Method and Critical Thinking course 'Dutch Landscape' provides insight in the interrelation of the different natural and societal forces, spatial and temporal qualities, materials and components of the cultivated predominantly rural landscape.	
Course Contents	The course addresses the structures, layers, and components of the Dutch landscape and combines the following knowledge fields: historical-geography, polder history, water management, morphology, landscape ecology (e.g., gradients, plant geography), environmental philosophy (visions to landscape and nature) and social geography. Different analytical techniques: cartography, photography, layer analysis, narratives, etc. will be used to reveal and represent the identity - which is: landscape forming processes and their topographic appearance - of a specific landscape. You learn how to critically evaluate your insights into the different aspects of landscapes in a descriptive well-illustrated document that takes the shape of a landscape biography. You need to elaborate on the outcome of the analyses as input for future transformations.	
Study Goals	By the end of this course, the student should be able to: identify and apply different perspectives on the interaction of natural and cultural processes and how they are expressed in rural landscapes; demonstrate a working knowledge of historical, social, ecological, and technical interpretative models of landscape; identify, valuate and represent landscape dynamics, conditions and logics, and their spatial manifestation in relation to land use; identify and criticise the landscape architectural contribution to rural land reclamation and civil engineering of land and water; describe and visualise the narrative of a specific landscape as a continuous process in time.	
Education Method	Lectures Workshops Seminars Excursions	
Course Relations	The quarter Dutch Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking. Four perspectives on the analysis and design of (urban) landscapes form the backbone of the master track Landscape Architecture: landscape perception, landscape as palimpsest, landscape as scale continuum and landscape as ecological, economic and social process.	
Literature and Study Materials	- Bell, S. (1999) Landscape. Pattern, Perception and Process. Taylor & Francis. - Cronon, W. (1999) Uncommon Ground: Rethinking the Human Place in Nature. W. W. Norton & Company. - Odum, E., Barrett, G. (2005) Landscape Ecology chapter 8, in Fundamentals of Ecology, 5th edition, p.374-411. - Ven, G.P. van de (2004) Man-made Lowlands. History of water management and land reclamation in the Netherlands. Matrijs. - Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press.	
Assessment	Presentation of a landscape as a living archive, in text and visually.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quater	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 Landscape Architecture

AR2LA011	Urban Landscape: Design Studio	10
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	J.R.T. van der Velde	
Instructor	J.R.T. van der Velde	
Instructor	Dr.ir. N.M.J.D. Tillie	
Instructor	Dr. L. Cipriani	
Responsible for assignments	Prof.ir. E.A.J. Luiten	
Contact Hours / Week x/x/x/x	8 hours/week average	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Summary	The design studio Urban Landscape explores the role of landscape design in the context of a metropolitanising region. The studio explores the relation between systems and places and the social interaction of people in the public realm. In this design studio students work on a comprehensive spatial proposal that supports the sustainability of an urbanized landscape under transformation.	
Course Contents	Urban Landscape: design studio In the course two specific dimensions of urban landscape architecture will be elaborated and integrated: the city as an earth-life landscape system and the city as a socio-spatial environment. The project outcome includes a spatial vision addressing the sustainable metabolism and systemic form of a specific urban landscape. Next to that you are to develop a design proposal for a particular urban landscape component (green-blue site, specific pattern or structure), addressing socio-spatial and physical aspects and criteria. A well-defined and thoroughly elaborated landscape architectural intervention in the balance between privatized and public space) is expected. You have to illustrate how the proposal contributes to large-scale and long-term urban sustainability goals and how social behaviour influences the design and responds to the result.	
Study Goals	By the end of this course, you should be able to: - identify and interpret an urban territory as an earth-life landscape system and as a social-spatial environment, using established analytical methods; - discover and define (diagnose) a landscape problematique for a mid-sized urban territory with respect to biodiversity, social inclusion and climate adaptation; - formulate and develop a landscape architectural assignment in an urban context referring to urban ecology and urban forestry perspectives; - design on various levels of scale and complexity, relating the whole and the component parts in a mutual and coherent relationship; - resolve and elaborate an urban landscape design (intervention) in spatial, material and technical terms, involving water management, soil mechanics, pavement, planting and micro-climate aspects.	
Education Method	Studio work Workshop Lectures Excursion Seminar	
Course Relations	The quarter Urban Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking.	
Literature and Study Materials	- Steenbergen, C.M., Reh, W. and Pouderoijen, M. (2011). Metropolitan Landscape Architecture: urban parks and landscapes. Bussum THOTH Publisher. - Sim, D. (2019). Soft City. Washington: Island press - Gehrels, H. et al, (2016). Designing green and blue infrastructure to support healthy urban living. https://www.researchgate.net/publication/308165682 - Henderson, K., Lock, K. & Ellis, H. (2017). The art of building a Garden City, Designing communities for the 21st Century, Newcastle upon Tyne: RIBA - McHarg, I.L. (1969). Design with nature. NewYork: John Wiley and Sons, inc.	
Assessment	Posters or booklet Oral presentation Physical (and digital) 3D-model Written reflection on design	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2LA021	Urban Landscape: Theory, Method and Critical Thinking	5
Course Coordinator	Prof.ir. E.A.J. Luiten	
Instructor	Prof.ir. E.A.J. Luiten	
Instructor	J.R.T. van der Velde	
Responsible for assignments	Prof.ir. E.A.J. Luiten	
Contact Hours / Week x/x/x/x	40 hours	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Summary	The Urban Landscape: Theory, Method, and Critical Thinking course is dedicated to the complex relationship between city and landscape; it introduces and explains open spaces and the basic concepts of urban green-blue elements and structures as the backbone for urban development and regeneration on various scales.	
Course Contents	Historical periods, cultural and political contexts, geographical features, and critical projects have influenced the complex relationship between city and landscape, culture, and nature. The spectrum of theories, concepts, approaches, and projects on this topic forms the backdrop for this course. The lectures provide different lenses covering historical and theoretical aspects focusing on contemporary concepts and approaches on the landscape as the medium and model to guide and transform urban development.	
Study Goals	By the end of this course, you should be able to: - identify and elaborate on theories, concepts, and approaches to urban landscapes/landscape urbanism; - identify and elaborate on spatial design and its social and ecological influences and ethical implications in the context of urban landscape/landscape urbanism; - position landscape architecture concerning other spatial design disciplines like visual arts, architecture, civil engineering, ecology, and urbanism; - logically and critically reflect on a selected discourse verbally, interactively and in writing, based on main academic literature, applying the basic rules of academic writing.	
Education Method	Lecture series Workshop Seminar	
Course Relations	The quarter Urban Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking.	
Literature and Study Materials	Study literature will be delivered to students in class.	
Assessment	Research paper	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC or a combination of 10 + 5 EC)

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master AUBS students	
Expected prior knowledge	spatial design skills	
Summary	In the On Site elective students are invited to investigate, understand and intervene in an existing urban landscape. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the experiential potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop a deep understanding of the qualities of the existing. The aim of the elective is to explore the potential of interstitial spaces, as spaces that are outside the regulated spaces of society and as such allow for the unexpected and the unregulated, for alternative and surprising uses and meaning. The task for the designer is to make these spaces accessible, mentally, visually and physically, by means of inventive analysis and minimal intervention, and thus transform them into spaces of coexistence and unexpected encounters with other humans and non-humans. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	fieldwork studio work interactive lectures workshops constructing on site	
Assessment	Exhibition of situational encounters Conceptual design of green network On site physical intervention	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday and Wednesday	
Minimum number of participants	15	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15

Course Coordinator	Dr. L. Cipriani
Responsible for assignments	Dr. L. Cipriani
Contact Hours / Week x/x/x/x	0/0/0/4
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.
Concept Schedule	Wednesday, Friday
Minimum number of participants	15
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2FO010	The Delta Shelter	15
Course Coordinator	G. Coumans	
Course Coordinator	M.G. Vink	
Course Coordinator	D. Piccinini	
Course Coordinator	F.T. Toni	
Instructor	G. Coumans	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have working and thinking level of a MSC1 design course in Landscape Architecture or Architecture.	
Course Contents	The design studio Delta Shelter is a combined studio, offered by the Architecture group Form Studies and the Section of Landscape Architecture. Starting from a shared interest in the impact of human made interventions in their natural environments, the assignment is located in a Dutch Delta landscape. This specific condition, characterized by the dynamic environment on the border of water and land and the ongoing negotiation between human presence and natural forces, is leading for this elective project. The poetic qualities and natural forces of the Delta environment will be the starting point of this course; the relation between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of the human in this vulnerable yet strong environment is to be understood through artistic documentation and physical acts of making, sensing and experimentation. The vastness of the location and the constantly changing conditions require students to research the meaning of boundaries, movement and time. In the design assignment students study how inherently static architectural forms are temporal and affective experiences, part of natural rhythms and therewith objects to consider over time. What it means to feel sheltered in these particular conditions, is studied through the lived experience and physical structure depicted in physical models. On a reflective level, the studio deals with the role, impact and contribution of architecture and landscape architecture in such places. What are the necessary conditions to give a valuable contribution to this environment? Is it inevitable that an intervention is a disturbing factor, can it only be of temporary presence, or can human interventions contribute to the regeneration of these kinds of environments? Ecological, ethical and aesthetic questions are brought together in the consideration and initiation of built intervention in the Delta environment.	
Study Goals	Upon completion of the course the student is able to express sensitivity and understanding of the dynamics and temporality of the Delta environment through analogue documentation and artistic expression. present a coherent, significant, elaborated, correct and innovative design - on mainline and aspects. conduct design research and research-by-design by experiments on-site, using physical models and hand drawings as a tool throughout the design process. communicate by making active use of various scale models to present and develop the design in all its aspects; the architectural composition, the landscape architectonical composition, materialisation and integration of construction tools, expressing aspects of time and dynamism through scales. The student will be able to communicate his/her contemplations and reflect on the role and position of the spatial designer in this assignment.	
Education Method	Experimentation, observations and documentation 'in loco' (fieldwork) based on the Immersion research method will be used to disclose ephemeral, more subjective properties of the site. Lectures and design-tutoring in studio format with weekly assistance in groups as well on an individual basis. The project will be developed using physical scale models, hand sketches and text during all the phases of the project; the analysis, design and presentation. This method aims to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Literature and Study Materials	Syllabus	
Assessment	Assesment based on process, analysis, documentation, (re)presentation of and reflection on the end result. Presentation will be oral and is guided by a variety of physical models, drawings, photographs and text, produced during the course. The products should give a clear insight in spatial logics, structure, experience and the relation and meaning of the design towards its environment. The design is assessed on its coherency, relevance, precision, elegance and elaboration. Its relevance and coherence are determined by its' explicit position and relation to the dynamic Delta environment, the program and the embodied experience of the place. The student has achieved a sufficient result on a scale 1 to 10 with 6, can take a repair with a mark of 5 or 5,5 and failed with 4,5 or minor. Repair entails one clear assignment, given on the presentation day and has to be completed within 2 weeks after completion of the studio, without additional guidance.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	coordinator	
Remarks	The studio is open to both Landscape Architecture and Architecture students. The program will be similar and the groups mixed. The final technical development and details will be steered towards the students' specific track. Students will go on a guided, collective excursion within the Netherlands. The site visit and its fieldwork-exercises form a crucial part of the course. In Q3 students will be contacted to arrange bookings for the excursion.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Studio days on Tuesday and Thursday - with exceptions for the excursion and taking formal holidays in consideration.	
Used Materials	Model materials - for students account	

Leerstoel	Form Studies & Landscape Architecture
Minimum number of participants	15
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW0501	Applied Spatial Analysis for Sustainable Urban Development	5
Course Coordinator	C. Forgaci	
Course Coordinator	Dr. D. Cannatella	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Spatial analysis is a crucial part of projects carried out in urbanism and landscape architecture and a core application domain for the field of geomatics. After having encountered basic spatial analysis with short targeted workshops in the first three quarters of their MSc tracks, Urbanism and Landscape Architecture students following this intradisciplinary course work on an integrated, end-to-end analytical assignment, combining, advancing and extending their data-related skill- and tool-sets, while focusing on advanced workflows. The course offers the opportunity to MSc Urbanism Landscape Architecture, and Geomatics students to advance their technical skills required for scalable, automated and reproducible spatial analyses, practicing working in an intra-disciplinary setting with a real-world assignment. All students have the chance to learn skills from the other tracks and in the process strengthen their analytical skills for their upcoming thesis. The course will start with an introduction to multi-criteria decision analysis, typology construction and their corresponding workflows suitable for a sustainable spatial development challenge. The higher-level principles of scalability, automation and reproducibility, as well as their implementation through computational notebooks, are also introduced early in the course. The course emphasizes data visualization as a way to explore data iteratively, generate new knowledge and communicate project results. Students learn about different types of thematic analysis, such as spatial analysis, network analysis, and landscape analysis which they can integrate in their analytical workflows. To apply the workflows and analytical methods introduced throughout the quarter, students work in intra-disciplinary groups on a given location. The workflows presented in the course involve open-source tools using graphical user interfaces, such as QGIS, as well as programming in R and/or Python for spatial analysis. In the era of open data, the course emphasizes the importance of leveraging accessible and shared datasets to enhance the quality and depth of spatial analyses. In addition, the course underscores the significance of transparent and collaborative research practices by incorporating a dedicated segment on publishing datasets, workflows, and results. MSc Urbanism, Landscape Architecture and Geomatics students will be given the opportunity to contribute to the broader academic community through the dissemination of their work.	
Study Goals	After following this course, students will be able to: 1. Construct analytical workflows in the context of spatial design and planning. 2. Employ spatial data visualization both for exploratory data analysis and for the presentation of their final project results. 3. Conduct integrated spatial analysis to understand and describe the complexity of urban environments and to inform spatial design and planning decisions targeting their transformations. 4. Adopt open science practices, including FAIR data publication, reproducible workflows, and usage of open-source software. 5. Report on analytical findings as part of a multidisciplinary team.	
Education Method	The course has a blended learning set-up that makes use of online learning material developed or curated as part of the Spatial Data & GIS education platform of the Department of Urbanism. During contact hours, the focus will be on lectures introducing key concepts and hands-on workshops addressing the methodological and technical challenges of the workflows and methods introduced in the course. Self-study time will involve both individual and group work, with an emphasis on the latter. In addition to the online software tutorials, peer troubleshooting will be facilitated.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Students keep track of their progress and present their final products in a group report set up as a computational notebook in a version control system. This allows them to reflect on their process throughout the quarter and the instructors to assess individual contributions in detail. Peer feedback as an essential component of the course is employed in the midterm and the thematic workshops. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on one day part per week; on Tuesday or Friday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of	For this course the maximum number of participants is 30.	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW0502	**Plantscapes**	**5**
Course Coordinator	J.R.T. van der Velde	
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Responsible for assignments	Dr.ir. N.M.J.D. Tillie	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	During the first weeks, we build the FOUNDATION of knowledge/skills regarding planting in the built environment. After this, we work with DIMENSIONS OF DESIGN, build upon the foundation and focus on planting strategies for gardens. Following, we strengthening PLANTING ENGINEERING strategies to improve ecosystem services. Finally, we integrate the knowledge gained in the past weeks into the design of (parts of) a park, while focussing on DRAWING REPRESENTATION.	
Study Goals	By the end of this course, the student should be able to: - recognize typologies that suit specific planting strategies. - recognize planting potential. - design with plants for climate adaptation, biodiversity and aesthetics. - develop his/her own library with knowledge about a variety of plants, trees and shrubs.	
Education Method	Studio work Workshops Instruction sessions Excursions Fieldwork	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	2D drawings and supporting visuals Physical/digital 3D-model Oral presentations Written reflection A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on two day parts per week; on Tuesday and Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable	
Leerstoel	Landscape Architecture	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1001	Integrated Metabolic Design for Sustainable Neighborhoods, Cities and Regions	10
Course Coordinator	Dipl.-Ing. U.D. Hackauf	
Course Coordinator	Dr. A. Wandl	
Responsible for assignments	Dipl.-Ing. U.D. Hackauf	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In the course, students address sustainability challenges of society metabolism based on a systemic understanding, analysis and design: through different spatial scales (from urban block and street to neighbourhood, city and region) through different disciplines of engineering, design and governance through interaction with different institutions and stakeholders A theoretical foundation of society's metabolism and the methods used to improve it is introduced, discussed and applied through a methodological, multiscale, iterative cycle of analysis, design and assessment. Presentations are used to share findings with stakeholders and gather feedback for further design iterations.	
Study Goals	After the course, the student can: Apply the concept of societal metabolism to analyse integrated sustainability challenges in a specific context develop a multiscale design proposal based on their analysis. Evaluate their proposal in the context of Dutch policymaking. Communicate and discuss proposals with stakeholders in the concerned context. Assess the effectiveness of design variants of environmental technologies through scales.	
Education Method	systemic design, lectures, group work, stakeholder workshops and self-study	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The final assessment has two parts: (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) a written reflection reacting to the feedback received on the poster presentation. Each individual week finishes with a formative assessment, either a review by the course instructors or a peer review between the students. This way, the students get weekly formative assessments. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	4	
Concept Schedule	The sessions are scheduled on two different day parts per week, most likely Monday, Wednesday and Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 36.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1002	Heritage Landscape Design	10

Course Coordinator Dr.ir. G.A. Verschuure-Stuip

Responsible for assignments Dr.ir. G.A. Verschuure-Stuip

Contact Hours / Week x/x/x/x
Wk 1-6: 10 hrs per week
Wk7: 28 hrs per week
Wk 8-9: 8 hrs per week

Education Period 4

Start Education 4

Exam Period 4

Course Language English

Summary

The Heritage Landscape Design elective offers spatial landscape and/or urban design training in a challenge-based (heritage) transformation by participating in a design workshop (charette) in interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies to create a visionary spatial plan for a transformation site based on local needs.

Course Contents

Identity, continuity and transformation are essential notions of todays spatial design practice. In this course, you will learn how to analyze a given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site in contact with the public. The scale of the assignment can vary from a historical garden to an (urban) landscape.

Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities.

In preparation for the interactive workshop, you study literature on theory and methods of heritage transformation, landscape design, value assessment and participation with local communities in weekly assignments. Youll team up with students from different disciplines. You will debate theories on place-making. During the workshop you will work with experimental analysis visualization methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders.

In this course, the Landscape Architecture section wants to strengthen the interaction between landscape transformations and the role of (local) stakeholders in a multidisciplinary setting.

Study Goals

By the end of this course, you should be able to:
- systematically identify, visualize and value the main characteristics of the (cultural) landscape design project as a continuous process in time;
- debate and apply methods and tools on landscape transformation, place-making and heritage management;
- use methods and tools to create a visionary transformation plan and present it to a larger audience in different visualization methods;
- understand the processes of creating plans and the role of spatial designers in citizen engagement, facilitating dialogue in participation processes;
- analyse and communicate effectively on complex design (transformation) ideas within a multidisciplinary setting with and for local stakeholders.

Education Method

Lectures and assignments
Literature study
Three-day design workshop in the Netherlands

Literature and Study Materials

Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press.

Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; Tijdschrift voor Economische en Sociale Geograﬁe 2014, Vol. 105, No. 5, pp. 622629.

Manual: Erfgoed en participatie (in prep.)

Other recommended literature will be mentioned on the specific Brightspace page at least one week prior to the start of the course.

Assessment

A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.
In case of a final grade below 6,0 the student needs to reenroll for the course.

Enrolment / Application

Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see:
https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams

Additional enrolment information for non-BK students:

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Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below)
The course can be taken only after confirmation of placement in BIS by BK-enrolment

Remarks The maximum marking period is 15 working days.

Period of Education Quarter 4.

Concept Schedule

The sessions are scheduled on two different day parts per week, most likely Monday, Wednesday and Thursday mornings or afternoons.
The actual schedule will be available via Bright Space > My Timetable.

Minimum number of participants For this course the minimum number of participants is 15.

Maximum number of participants For this course the maximum number of participants is 30.

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1502	**Peri-Urban Transformations: Analysis, Development, Design**	**15**
Course Coordinator	T.N. Broekmans	
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	15 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this new elective course the interaction between spatial design, area and project development tools and social, economic, political, and environmental issues is at stake. Students from the MBE, Landscape Architecture and Urbanism tracks and the MSc CME Design & Integration Domain (from the coupled course Transformative Design), collaborate on the transformation of a peripheral urban area in the Netherlands. The pressure on these urban fringes is both urban and rural and characterised by urban sprawl, fragmented agricultural or industrial activity, remnants of natural and landscape values. Working together in groups you will develop a master plan and elaborate individually a more detailed, imaginable transformation proposal. Students confront spatial qualities, opportunities, limitations and challenges of the chosen area with societal needs, transition goals, ownership issues and stakeholder claims. You design and define proposals on different scales and complexities including area, portfolio and project level. You will apply development strategies, feasibility criteria and research by design activities. This course is organised as a multidisciplinary experience combining approaches, knowledge and skills from the viewpoints of management, urban design and landscape development and is therefore representative for area development practice in the real world. This collaboration provides insight in the complementarity of viewpoints, data, tools, vocabulary, methods and fun.	
Study Goals	(1) students can analyse the characteristics, conditions, stakeholders, opportunities and potentialities of peri-urban transformation areas (using a theoretical framework) (2) students can develop a tentative spatial perspective (e.g. spatial, social, economic, political, environmental) in a peri-urban context including primary notions of phasing: from initiative to implementation (masterplan version 0.5) (3) students can elaborate a crucial, domain-specific aspect or component of the masterplan like a portfolio, structure, block typology or building (4) students can synthesise varied input into an integrated, substantiated master plan for a peri-urban transformation area (masterplan version 1.0) (5) students can reflect on the implications of the domain-specific theory on the application of the masterplan and its elaborations	
Education Method	In this course four main educational methods are used: (1) Lectures (2) Studio work in groups (2) Studio work individually (4) Plenary presentations and discussions	
Assessment	This course has two assessments (1) group: atlas and integrated masterplan (2) individual: elaboration proposal and reflection report A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4.	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 75.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 Landscape Architecture

AR3LA011	Landscape Architecture Analysis and Visualisation	5
Course Coordinator	Dr. L. Cipriani	
Course Coordinator	Dr.ir. I. Bobbink	
Instructor	Dr.ir. I. Bobbink	
Instructor	Dr.ir. S.I. de Wit	
Instructor	Dr. D. Cannatella	
Instructor	Dr. L. Cipriani	
Responsible for assignments	Dr.ir. I. Bobbink	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The course aims to provide students with a platform where (1) they can get to know digital and analog tools as vehicles for research and design in the field of landscape architecture and (2) enable students to experiment with alternative new approaches to thinking and communicating about spatial design issues. Workshops will be organized according to the four spatial dimensions: maps (2D), sections and models (3D), and time (4D).	
Course Contents	To represent the landscape means to understand, interpret, and transform it. It refers to a process rather than a completed product. Representation entails cartographic, and model explorations, activities that construct and visually communicate spatial and temporal knowledge and design. Representation helps landscape architects to acquire new or latent information, which is the basis for generating spatial knowledge through time. Furthermore, it is a creative act that helps the process of communication of design possibilities. This set of workshops intends to focus on representation emphasizing the four dimensions of space that are all particularly relevant for landscape architects: maps, sections, models, and time. Workshops will include learning analog and digital tools to enable landscape designers to use different approaches to think and communicate about spatial design issues. The course consists of three separate workshops where you get to know landscape architecture analysis and visualization methods in a practical setting: 1. Thinking with Maps: GIS in Landscape Architecture 2. Thinking with Sections and Models 3. Drawing Time	
Study Goals	By the end of this course, you should be able to: - apply and present different relevant design-related analysis and visualization methods and techniques in landscape architecture; - identify the potentials and limitations of the relevant design-related analysis and visualization methods and techniques in landscape architecture; - select and use suitable design-related analysis and visualization methods and techniques in a particular research and design context (e.g., graduation project).	
Education Method	Three thematic workshops with an introductory lecture followed by a short assignment focusing on applying a particular analysis and visualization method or technique. The hands-on assignments are elaborated in groups of at least two students and guided by an expert in the field.	
Literature and Study Materials	Provided during the workshops.	
Assessment	Students can pass or fail. This decision will be based on the students presence, active participation in the workshops, and outcome. After each workshop, the student must hand in a report that presents the outcome and consists of the material produced in the workshop.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3LA020	Research Methodology in Landscape Architecture	5
Course Coordinator	Dr.ir. I. Bobbink	
Course Coordinator	Dr.ir. S.I. de Wit	
Instructor	Dr.ir. I. Bobbink	
Instructor	Dr.ir. S.I. de Wit	
Instructor	Ir. M. Veras Morais	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	45 hours per quarter	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	English	
Summary	The course provides you with a framework of academic knowledge and skills for understanding, designing and conducting design research in landscape architecture. To this end, the course presents you with an overview of the most prevailing theoretical themes and perspectives in landscape architecture, and of the discipline's proven research strategies, methods, tools and techniques. Additionally, the course familiarizes you with the academic, formal and structural requirements of a landscape architectural research design in the context of the elaboration of your graduation plan.	
Course Contents	Research skills are not only crucial to landscape architecture researchers working in an academic setting, but are also becoming increasingly important to landscape architectural professionals as the complexity of contexts and challenges landscape architects are tackling is increasing at fast pace. This course provides you with (a) an in-depth understanding of the most prevailing theoretical perspectives and themes in the field of landscape architecture and landscape architecture research, (b) an overview of the disciplines most prevailing design research methods, strategies, tools and techniques, (c) important theoretical clues and a practical aptitude for developing a critical academic attitude towards design research in landscape architecture. The course furthermore teaches you (d) how to construct a research design that meets the disciplines academic, formal and structural requirements. The course is organized in three parts: 1. Theory in landscape architecture - introduction to theory in landscape architecture; - landscape architecture as a design process; - form, meaning and experience in landscape architecture; - society, language and the representation of landscape; - ecological design and the aesthetics of sustainability; - landscape as an integrator of site, place and region. 2. Research strategies, methods, tools and techniques in landscape architecture - introduction to research strategies and related methods, tools and techniques: description, modelling, experimentation, classification, interpretation, evaluation & diagnosis, engaged action, design projection, logical systems. 3. Composing a research proposal - building blocks of a research proposal, and their interrelation: theoretical framework, problem statement, research question, research strategy, methods, tools and techniques, relevance of the proposed research; - academic, formal and structural requirements of a research proposal.	
Study Goals	By the end of this course, students should be able to: - identify landscape architecture as an academic design discipline with its own theories, methods and techniques; - critically discuss the following, major theoretical themes in landscape architecture theory: (i) Landscape architecture as a design process, (ii) Form, meaning and experience in landscape architecture, (iii) Society, language and the representation of landscape, (iv) Ecological design and the aesthetics of sustainability, (v) Landscape as an integrator of site, place and region; - compare different research strategies in landscape architecture and their related methods, tools and techniques, among others: description, modelling, experimentation, classification, interpretation, evaluation & diagnosis, engaged action, design projection and logical systems; - critically reflect on different landscape architectural methods of analysis and design - build up a coherent, feasible and effective research plan for their graduation project, complying with the academic, formal and structural requirements of a scientific research proposal.	
Education Method	The course consists of lectures and seminars. The seminars follow the flipped classroom model, where you will be asked to prepare for the seminar sessions in forehand, by means of writing and presentation assignments.	
Literature and Study Materials	Groat, L. and Wang, D. (2002) Architectural Research Methods. John Wiley and Sons. Swaffield, S. (Ed.) (2002) Theory in Landscape Architecture: A Reader. University of Pennsylvania Press. Swaffield S. and Deming M.E. (2011) Research strategies in landscape architecture: mapping the terrain. Journal of Landscape Architecture, 6(1), 34-45. - and others depending on your individual research topic	
Assessment	Presentations and written assignments.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3LA031	Graduation Studio Landscape Architecture: Flowscapes	20
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. I. Bobbink	
Instructor	Dr.ir. I. Bobbink	
Instructor	Dr.ir. S.I. de Wit	
Instructor	Dr.ir. G.A. Verschuure-Stuip	
Instructor	Prof.ir. E.A.J. Luiten	
Instructor	Prof.dr.ing. S. Nijhuis	
Instructor	D. Piccinini	
Instructor	J.R.T. van der Velde	
Instructor	Dr.ir. N.M.J.D. Tillie	
Instructor	Dr. L. Cipriani	
Responsible for assignments	Dr.ir. I. Bobbink	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The Flowscapes studio focusses on specific context-related design projects, in which knowledge from spatial design, ecology, civil engineering and social sciences is synthesized into a coherent multi-scale landscape design proposal. Within the Flowscapes studio different labs provide specific research themes. Within the lab you cooperate with your lab-mates on research and define your own project. Spatial, societal and environmental issues are explored through the application of design research and research -by-design.	
Course Contents	Flowscapes interprets and elaborates landscapes as infrastructures and infrastructures as landscapes. Social, cultural and technological developments of our society demand a fundamental review of the planning and design of the landscape, in particular in relation to environmental issues. Urbanization, ecological crisis and climate change are international challenges; while the technical challenges may be considerable, the spatial and cultural challenges are by far the largest. Therefore, a renewed understanding of space-time condition of landscape and its capacity to adapt, transform and develop further offers promising opportunities to find new solutions to these problems. In this changing context the Flowscape graduation studio focusses on strategy development and design exploration of landscape compositions and systems. The emphasis is on the design of new topographies to create conditions for spatial development by employing landscape-based approaches, ecosystem services and landscape architecture design principles. Here landscape design is considered as a synthesizing activity that explores the dynamic between structure and process in various natural, cultural and urban settings, taking the specific qualities of the place (genius loci) as a starting point. By combining theoretical and functional knowledge, and by combining several scales and disciplines in a stimulating learning environment, the studio aims to prepare you for world-wide challenges related to our (urban) landscape.	
Study Goals	The thesis should prove that you are able to: - select and use suitable design-related research strategies and techniques in a particular context; - determine and design landscape architecture strategies and spatial interventions at multiple scales, which meet aesthetic, technical, ecological, functional and ethical requirements as proof of their academic knowledge; - develop and assess a landscape design assignment from an environmentally ethic perspective, focusing on moral responsibility towards the natural world. - integrate knowledge from other design disciplines and scientific fields; - write a report of an academic standard including method, theory, research and design; - can substantiate research and design decisions based on ethical and verifiable arguments; - present the work by combining oral, written and graphical media.	
Education Method	- studio meetings Flowscapes - lab meetings - lectures - seminars - workshops - writing - fieldwork	
Course Relations	The accompanying courses Research Methodology and Landscape Architecture Analysis and Visualisation will help to develop your skills and attitude in theory, methods and techniques in becoming a landscape architect.	
Literature and Study Materials	Please check the semester guide.	
Assessment	In this process of graduation visual thinking and communication are considered to be crucial. Drawings, mappings and models are used to reveal and create relationships, explore and elaborate landscape systems (in terms of geometry, quantity, velocity, force, trajectory) and for critical reflection. - products of design research and research-by-design (as defined in semester book) - written and illustrated report - oral presentation	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Semester + MSC4	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 4 Landscape Architecture

AR4LA010	Graduation Studio Landscape Architecture: Flowscapes	30
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. I. Bobbink	
Instructor	Dr.ir. I. Bobbink	
Instructor	Dr.ir. S.I. de Wit	
Instructor	Dr.ir. G.A. Verschuure-Stuip	
Instructor	Prof.ir. E.A.J. Luiten	
Instructor	Prof.dr.ing. S. Nijhuis	
Instructor	D. Piccinini	
Instructor	J.R.T. van der Velde	
Instructor	Dr.ir. N.M.J.D. Tillie	
Instructor	Dr. L. Cipriani	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	135 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The Flowscapes studio focusses on specific context-related design projects, in which knowledge from spatial design, ecology, civil engineering and social sciences is synthesised into a coherent multi-scale design proposal. Within the Flowscapes studio different labs provide specific research themes. Within the lab you cooperate with your lab-mates on research and define your own project. Spatial, societal and environmental issues are explored through the application of design research and research-by-design.	
Course Contents	Flowscapes is about designing landscapes as infrastructures and infrastructures as landscapes. Social, cultural and technological developments of our society are demanding a fundamental review of the planning and design of its landscapes, in particular in relation to environmental issues and sustainability. Urbanization, ecological crisis and climate change are international challenges; while the technical challenges may be considerable, the spatial and cultural challenges are by far the largest. Therefore, a renewed understanding of space-time condition of landscape and its potential for change offers promising opportunities to find new solutions to these problems. In this changing context the graduation studio focusses on strategy development and design exploration of landscape compositions and systems. The emphasis is on the design of new topographies to create conditions for spatial development by employing landscape-based approaches, ecosystem services and landscape architecture design principles. Here landscape design is considered as synthesising activity that explores the dynamic between structure and process in various natural, cultural and urban settings, taking the specifics of the place (genius loci) as a starting point. By combining theoretical and functional knowledge, and by combining several scales and disciplines in a stimulating learning environment, the studio aims to prepare you for world-wide challenges related to our (urban)landscape.	
Study Goals	The thesis should prove that you are able to: - select and use suitable design-related research strategies and techniques in a particular context; - determine and design landscape architecture strategies and spatial interventions at multiple scales, which meet aesthetic, technical, ecological, functional and ethical requirements as proof of their academic knowledge; - develop and assess a landscape design assignment from an environmental ethic perspective, focusing on moral responsibility towards the natural world. - integrate knowledge from other design disciplines and scientific fields; - write a report of an academic standard including method, theory, research and design; - substantiate research and design decisions based on ethical arguments; - present the work by combining oral, written and graphical media.	
Education Method	- studio meetings Flowscapes - lab meetings - lectures - seminars - workshops - writing - fieldwork	
Course Relations	The accompanying courses Research Methodology and Landscape Architecture Analysis and Visualisation will help to develop your skills and attitude in theory, methods and techniques in becoming a landscape architect.	
Assessment	- Products of Design Research and Research-by-Design (as defined in semester book) - Written report - Oral presentation	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Once enrolled in AR3LA031 (start of the graduation) students automatically continue their graduation in AR4LA010	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Leerstoel	section of Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

variant Urbanism

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 1 Urbanism

AR1U090	RandD Studio: Analysis and Design of Urban Form	10
Course Coordinator	B. Hausleitner	
Course Coordinator	T. Kuzniecow Bacchin	
Course Coordinator	L. Qu	
Responsible for assignments	B. Hausleitner	
Contact Hours / Week x/x/x/x	10 hours per week starting from week 1 and ending in week 9.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Summary	Dealing and designing with the growing complexity of a city requires a good understanding of the basic structure and key urban elements and the ways they work. Low Lands, the quarter 1 design studio of the Urbanism master track aims to develop students' critical reading, understanding and projecting of urban structure, form and functioning at different scales. Through a sequence of analysis and design exercises, students get familiar with the urban design language and the urban landscape's key structural components and fabrics. The studio approaches design with site sensitivity by emphasising the cultural values and environmental qualities of the specific place. Around the globe, Low Lands are seen historically as places with the highest degree of urbanisation. Their highly dynamic and fragile landscapes and high level of exposure to the effects of climate change question historically developed urban centres and the idea of the urban project. The studio takes the condition of the Western part of the Netherlands exemplary for such Low Lands to rethink forms and places of urbanisation and urbanity, the commons and public good. The Research and Design Studio puts the following interrelated lenses central: the landscape and site; the settlement structure and urban typo-morphology; the spatial condition of the urban landscape as place for coexistence of society and natural processes, informed by the relationship and dynamics among the main elements of the urban landscape. These perspectives on the design of the urban landscape will be introduced and discussed in the lectures and workshops on History and Theory of Urbanism (AR1U121). The content of the lectures is crucial input for the work in the Studio. By making analytical and conceptual maps, drawings and models, by comparing, combining, analysing, concluding and projecting, students will learn: to identify the main layers of the urban and their relational integration; to relate and design with multiple scales and layers; to understand the relation between urban composition and performance, and the fluidity of this relation; to experiment with urban concepts; to use design to uncover potentials and challenges in a project; to design for dealing with uncertainty via a re-composition of the main material layers; to (re)conceptualise the idea of the city. Students will build a body of knowledge by searching, selecting and using different sources and reflecting on their work in a theoretical framework of Urban Design. Students will be able to point out which urban elements and structures had, have and could have an influential role in shaping the Low Lands. Students will discover the structural qualities of the Low Lands, its permanence and fluidity, and the ruptures and discontinuities. During the entire studio, they will use drawings, models and other representational techniques to communicate their findings and explore and develop their projection of the future of the Low Lands, including long-term developments and short-term interventions.	
Course Contents	Content of Q1: Analysis and Design of City Form: Low Lands. Design Studio; Method workshops and seminars; All information is presented in the Quarter Guide available on Bright Space. Students will analyse and compare the structure, form, and space of three different Dutch places in this studio. These are related to the three main physical conditions in the Low Lands: coastal, riverine, and reclaimed land. Students work in groups of three-four with one tutor on one of the Low Land conditions. Each student will work individually on an atlas and one design perspective for one of the three places in the Low Lands. The atlas is based upon the diachronic and synchronic mapping of the functional and material layers across different scales from different perspectives. The 3-4 students working on one city per tutor group jointly synthesize their design perspectives in a structure plan, thereby formulating future forms of inhabitation in a designerly way. Lectures, debates, and workshops support this quarter's individual research and design assignment. The complementary workshops enhance skills in critical thinking, representation techniques, and the formation of narratives and introduce law guiding spatial plans. The students' different products (atlas, projection, exhibition) require different working attitudes. The different forms of education activate the learning process and advance the level of knowledge, skills and academic attitude.	
Study Goals	At the end of this course, the student is able to: recognize and understand the structural and spatial components of a city and their influential role in the urban (re)development. use analysis and experiment as a design tool and show the effects of structural interventions on multiple scales and layers. structure and shape a narrative on the actual condition and future potential of a city. (re-)conceptualize and shape the idea of the city in models and drawings. build a body of knowledge by searching, selecting and using a wide variety of sources and by reflecting their work to a theoretical framework in Urban Design.	
Education Method	Studio, lectures, and workshop instructions. Combination of mainly individual but also partly group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of	

	the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 1.
Concept Schedule	The studio sessions are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and Friday afternoons. In addition to these studio sessions lectures, excursions, workshops and seminars will be scheduled on other day-parts tuned with the schedule of the parallel course AR1U121. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1U100	RandD Studio: Designing Urban Environments	10

Course Coordinator
V. Muñoz Sanz

Course Coordinator
L. Iuorio

Course Coordinator
L. Qu

Responsible for assignments
V. Muñoz Sanz

Contact Hours / Week x/x/x/x
15 hours per week starting from week 5 and ending in week 9 (NB. Teaching and self-study activities are also scheduled in week 1, 2, 3 and 4).

Education Period
2

Start Education
2

Exam Period
2

Course Language
English

Summary
The object of the R&D studio is the design of sustainable urban environments. The focus is on the scale of the urban project, exploring the agency of urban design in addressing urgent societal and environmental challenges. Students work in groups and individually in the context of a particular environmental challenge faced by a specific urban area in the Netherlands. The goal is to improve socio-ecological inclusivity of the urban environment by developing an urban design project that embraces the understanding of the urban landscape as a complex and dynamic system. Thus special attention is given to the relation between design and engineering (parallel course AR1U131), and the application of knowledge gained during the first quarter.

Course Contents
The Department of Urbanism has set as its mission to advance, share and apply knowledge on how to adapt the built environment to societal and environmental changes and how to apply design, planning and engineering interventions to better satisfy human needs, and to achieve fairer, healthier and more resilient urban landscapes (The Delft Approach to Urbanism, 2021). Against this backdrop, the object of this R&D studio is the design of sustainable urban environments, exploring the potential of spatial design to act as a catalyst for socio-ecological inclusivity.

The ultimate goal is to improve the socio-ecological performance of a given urban environment towards a fairer, healthier and more resilient future. For that, students should embrace the understanding of the urban landscape as a complex and dynamic system. That implies considering a variety of issues which shape and impact the built environment across multiple scales and temporalities. These range from urban climate adaptation and mitigation, landscape systems, energy, typomorphology, parcel structure, density, programme, access, and ownership, to situated knowledge, architectural qualities and sensorial experiences.

If the first quarter focused on analyzing and designing structures, forms and systems at the city scale, this studio combines the visionary with an ambition to create facts on the ground. Therefore, while insisting on the importance of trans-scalar relations, the focus is on the scale of the urban project. That is, students should question the agency of urban design in addressing urgent societal and environmental challenges through concrete ideas and projects on the ground. Specifically, students work in groups and individually in the context of a particular challenge faced by a specific urban area in the Netherlands.

Overall, this quarter puts the stress on:
1. Connecting research and design towards a comprehensive understanding of socio-ecological inclusive urban design;
2. The integration and collaboration across disciplines, specifically design and engineering;
3. Skills in spatial thinking and communication, emphasizing parallel processes and iterative feedback loops between analysis, design, and presentation.

Study Goals
At the end of this course, the student is able to explain:
sustainable development as a complex multidimensional-multiscale process consisting of environmental and liveability dimensions on street-, neighbourhood- and city-scale.
human dimensions in urban design related to urban design thinking, environmental psychology and urban complexity.
the characteristics of natural and human systems on both the neighbourhood and public space level.
contemporary urban transformation practices in a Dutch context.

The student is able to analyse:
the urban environment from an urban landscape, typo-morphological and open space perspective.
the environmental and liveability dimensions on the street and neighbourhood scale, in the context of the city.
the synergy between natural and human systems on the neighbourhood and public space level.
the overall socio-spatial-environmental performance of the urban environment in terms of liveability.

The student is able to design:
a strategy/vision on neighbourhood scale, in the context of the city.
a public space as a translation of the strategy/vision, by combining design and engineering, including natural and human systems on the neighbourhood and street-level.
context specific sustainability criteria to evaluate the overall performance of the strategy/vision and public space design in terms of liveability.

The student is able to evaluate:
the strategy/vision and public space design based on the overall goals and criteria that were set by the student in the beginning.

The student is able to demonstrate:
skills in writing, drawing and calculating as a means to generate design ideas and concepts.
skills in reporting and presenting to communicate to a variety of audiences.
skills in collaborating.

Education Method
Studio, lectures, workshops and instructions.
Combination of individual and group work.

Literature and Study Materials
Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.

Assessment
Oral examination plus design examination.

A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.
In case of a final grade below 6,0 the student needs to reenroll for the course.

Enrolment / Application
Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see:
https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams
Additional enrolment information for non-BK students:

	Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	(External) students can only enrol for this course when also enrolling for the AR1U131 course, since the AR1U100 and the AR1U131 course are intertwined.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 2.
Concept Schedule	The studio sessions are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and Friday afternoons. In addition to these studio sessions lectures, excursions, workshops and seminars will be scheduled on other day-parts tuned with the schedule of the parallel course AR1U131. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1U121	History and Theory of Urbanism	5
Course Coordinator	Dr.ir. G. Bracken	
Course Coordinator	B. Hausleitner	
Course Coordinator	T. Kuzniecow Bacchin	
Course Coordinator	L. Qu	
Responsible for assignments	T. Kuzniecow Bacchin	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Summary	The course gives an overview of the 'body of knowledge' of Urbanism. First, it explores the idea of the 'urban' and its major related concepts. Next, it analyses and determines the conditions of their emergence within a broader cultural context. Finally, it traces how these concepts have changed through time to enhance our present understanding of urbanised landscapes, cities, and their transformation. The programme combines urban history and theory by focusing on the persistences and changes of the 'urban' as a spatial, material, socio-economic, political, cultural, and ecological project. Students will learn how to bridge the gap between theory and design and between past and present by relating critical thinking to concrete urban plans and designs.	
Course Contents	The course provides a lecture series on the theory and history of urban design and urban planning. The aim of the course is: to foster awareness of the disciplines intrinsically historical nature, the evolution of which is motivated by concepts, interpretations, and interactions with social, political, cultural, economic, technological, geographical, and environmental processes. to teach students how to assess the historical background of urban plans and designs from different historical periods. to enable students to identify and explain the urban as form, structure, function and process, the relation between space, society, landscape, and site, and its composition, elements and material layers. to provide students with the ability to evaluate the theoretical concepts underlying urban interventions, the evolution of the discipline in terms of the goals it sets itself and their - alleged - impact. to enhance academic writing skills and to enable students to write about their discipline, urban phenomena, design and planning proposals, and theories.	
Study Goals	At the end of this course, the student: Has knowledge of the theoretical fundaments of the discipline of Urbanism. Is able to position historic urban designs and theories, and to relate them to the specific societal context of a historic period. Is able to take a critical position concerning different design and planning experiments and theories of Urbanism, based upon knowledge and arguments. Is able to position his/her own opinions, designs and plans in relation to the history and theory of Urbanism.	
Education Method	Lectures, self study (readings) and debates. Combination of mainly individual but also group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1.	
Concept Schedule	The studio sessions of AR1U090 are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and Friday afternoons. The AR1U121 course schedule is tuned with the AR1U090 course, thus not scheduled on Tuesdays and Fridays. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1U131	Sustainable Urban Engineering of Territory	5
Course Coordinator	Dr. F.L. Hooimeijer	
Course Coordinator	L. Qu	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	15 hours per week starting from week 1 and ending in week 4.	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Summary	This quarter focusses on the making or remaking of the city and this course more specially on the city as technical construction: the natural and man-made conditions of the urban landscape, in order to integrate all interest in an urban plan. The act of integration and design requires systemic knowledge on a wide range of subjects. In this course the city is seen as a technical construction, with the central question being how to reconnect this construction to the natural systems of the territory? Therefore, this course provides vital input for the parallel studio of Q2.	
Course Contents	In the master track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable urban development. This quarter focusses on the making or remaking of the city and this course more specially on the city as technical construction: the natural and man-made conditions of the urban landscape, in order to integrate all interest in an urban plan. The act of integration and design requires systemic knowledge on a wide range of subjects. In the course the city is seen as a technical construction, with the central question being how to reconnect this construction to the natural systems of the territory? In new to build areas, or green fields, there is a lot of freedom to engineer or not to engineer the natural system. One of the greatest urban challenges these days is urban renewal. This is a complex enterprise in itself because you deal with the existing urban use and fabric. On top of that we need to deal with trends like climate change and the energy transition. One dimension that is reintroduced in the urban project is the subsurface system, which plays a crucial role in water management, ecology and energy supply. The subsurface system was for a long time excluded from the urban planning and design process, because it was considered a technical aspect that was dealt with by civil engineers, not as part of the urban design. However, the subsurface sets the conditions with highest impact: it is more costly and takes more time to change a cable system then it is to build a building or construct a road. Especially the idea that the natural system has already been altered for urban use, and thus lost, is preventing innovative solutions that deal with climate change and the energy transition in urban renewal. In order to incorporate natural and technical conditions in urban plans, a better cooperation with civil engineers and a better understanding of what they do not know is crucial for urban designers. This is what the course aims at. To bring order in all the technical information we make use of the System Exploration Environment and Subsurface (SEES) that uses six functional layers with different dynamics, professional domains and knowledge fields: people, metabolism, occupation, public space, infrastructure and subsurface. The subsurface layer is ordered in (for the urban planner and designer) recognizable themes: water, energy, civil constructions and subsoil. Within the themes there is a logical order in the qualities of the subsurface (www.ruimtexmilieu.nl). SEES connects the subsurface information with the urban surface in order to inspire and set clear boundaries for the development of the urban surface. It is used for analysing potential problems, chances, demands and supports a creative interaction early in the process of urban planning.	
Study Goals	At the end of this course, students are able to: Explain what for them is sustainable development and how technology is integrated. Evaluate the skills of an urban designer in relation to sustainability. Explain the city as a hybrid system of natural and human characteristics both the surface and subsurface level. Analyse the synergy between natural and human systems on the surface and subsurface level. Identify new approaches in which the conditions and chances that are given by the natural system are re-introduced. Demonstrate and design spatial principles or an urban ensemble with the desired performances, by combining design and engineering, including natural and human systems on the surface and subsurface level to develop ethical standards considering sustainable development.	
Education Method	Interactive lectures and intensive workshops, full time three weeks starting in week 2 of the quarter. Mainly group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The assessment of the assignments is done using the rubric that is available in the quarter guide on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 2.	
Concept Schedule	The AR1U131 course is 15 days scheduled in the first 4 weeks of Q2. The AR1U131 course schedule is tuned with the AR1U100 course. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 2 Urbanism

AR2U086	RandD Studio: Spatial Strategies for the Global Metropolis	10
Course Coordinator	N. Katsikis	
Course Coordinator	V.E. Balz	
Course Coordinator	L. Qu	
Responsible for assignments	V.E. Balz	
Contact Hours / Week x/x/x/x	10 hours per week starting from week 1 and ending in week 9.	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Regional design is the core assignment of the R&D studio in this quarter. Students work in groups on regional analysis and design for a self-defined goal in the framework of sustainable development, focusing on a metropolitan region in the Netherlands. The rationale behind this assignment is the fact that the way global economic powers influence social, cultural and environmental development is best sensible at the regional level. Such global influence results in the inability to fully control spatial development. In this sense, regional design is seen as a tool for steering development in the right direction, which -as the exploration of plausible futures- promotes and debates solutions to problems in a given context. It is a reflection on prevailing spatial conditions, political agendas and planning regimes, meant to improve good (democratic) decision-making and to inform long-term strategic planning approaches to desirable spatial change.	
Course Contents	This quarter emphasizes on (1) a comprehensive, evidence-informed understanding of regional spatial structures and development trends, (2) an understanding of interrelations among design, planning and politics and (3) communication skills that are required in collaborative decisionmaking. During the R&D studio 'Spatial Strategies for the Global Metropolis' students use this knowledge and skills to conduct a regional design. The design process knows two products, notably (1) a spatial vision and (2) a development strategy. Products are interrelated. The vision represents a desirable spatial future; it serves as a guiding normative principle for the development strategy that sets out a path towards spatial change, by means of spatial interventions that are ordered over a time and associated with capacities of actors in development. The R&D studio 'Spatial Strategies for the Global Metropolis' is the core activity of this quarter. Students conduct a regional design in groups of 4-5 students. The thematic exercises of Spatial Development Strategies (SDS) are an integral part of the studio. Knowledge on regional design and planning approaches will be provided during lectures and applied during workshops. SDS assists in and steers studio work. The series Capita Selecta also adds to the studio. It provides students with knowledge about spatial planning and discusses relations between planning, governance and design on these grounds. Parallel to the R&D studio runs the course 'Research & Design Methodology for Urbanism'. The course focuses on a theoretical understanding of design, planning and research. Students learn to position their work in a theoretical debate and write a report on these grounds.	
Study Goals	At the end of this course, the student is able to: Understand the basic roles and instruments of strategic spatial planning in delivering public good, spatial quality and equality and sustainable regional spatial development. Understand the complexity, multiscalarity and uncertainty of regional spatial development; can consider the limitations that these conditions set to regional planning and design. Understand and critically reflect on roles and impacts of regional design in/on inclusive planning decision-making. Formulate and argue for a comprehensive regional vision, drawing on commonly shared values and norms, evident regional spatial development and appropriate planning principles. Convert a vision into a regional development strategy that is relevant and feasible in a given institutional context and robust in respect to uncertainties of long-term regional development; can estimate a fair distribution of costs and benefits among stakeholders involved. Justify a vision and development strategy conceptually, making use of theoretical notions and an understanding of how theory and practice interact. Use communication media that are effective in collaborative decision-making (visualize design proposals clearly, consistently and persuasively, using images and text); can engage in critical debate. Explain the ethical issues involved in the activity of planning and designing for people.	
Education Method	Studio, lectures, workshops and instructions. Mainly group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	(External) students can only enroll for this course when also enrolling for the AR2U088 course, since the AR2U086 and the AR2U088 course are intertwined. (External) students enrolling for this course should have developed design skills.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 3.	
Concept Schedule	The studio sessions of AR2U086 are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and Friday afternoons. The AR2U088 course schedule is tuned with the AR2U086 course, thus not scheduled on Tuesdays and Fridays. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	

participants	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2U088	Research and Design Methodology for Urbanism	5
Course Coordinator	Dr. M.M. Dabrowski	
Course Coordinator	L. Qu	
Course Coordinator	Dr. R.C. Rocco de Campos Pereira	
Responsible for assignments	Dr. R.C. Rocco de Campos Pereira	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	The EC of this course have to be obtained in order to be allowed to enrol for the P2 presentation of the MSc3.	
Summary	This course enables you to do academic research that will support and fundament your work in the parallel AR2U86 studio and in the Master 3 and 4 graduation project.	
Course Contents	The course Methodology for Urbanism runs parallel to the Q3 studio. It is one of the central elements of the quarter. It prepares you to do academic research that will support and provide a solid theoretical foundation for your work in the studio. The course has two components: 1. It will give you an introduction to fundamental research skills, help you to build a conceptual framework, and teach you how to organise and write an academic report (this is fundamental knowledge also for your graduation project). 2. Moreover, the course will introduce you to some of the key theoretical issues underpinning much of the current debates in urbanism, including those on the topics of socio-spatial justice and socio-technical transitions to sustainability. This course is different to the studio because here you will focus on traditional academic research skills and methods, which complement the less traditional and designerly forms of research you will use in the studio. This connection between traditional and non-traditional (design-based) forms of research is one of the characteristics of education and research in the Department of Urbanism of the TU Delft. In connection with the first component (organisation of the report and conceptual framework), the methodology course will help you: -EXPLAIN what a conceptual framework is;  BUILD a conceptual framework that will sustain your research and design in Q3; -IDENTIFY a community of authors and practitioners who write about the core ideas of your theoretical framework; -DESIGN, ORGANIZE and WRITE an academic report, in which you will describe what are the main questions you will seek to answer in Q3 and the best methods to answer them; - EXPLAIN the values connected to and the ethical issues involved in the activity of planning and designing for people and explain what public goods are created with your design and strategy. In connection with the second component (socio-spatial justice and sustainability), the methodology course will help you: -EXPLAIN main issues of socio-spatial justice and sustainability in relation to issues of socio-technical transitions in urban and regional design and development; -BUILD upon those concepts to formulate your research questions, your conceptual framework, and your objectives; -CRITICALLY ASSESS issues of urban and regional planning and design using ideas connected to those concepts. In other words, formulating a regional design that includes those concepts is a task you will carry out in the studio. The methodology course can help you clarify those concepts and make them operational. Formulating your own problem statement, research questions, and methodology is are some of the goals of the Urbanism Master track. You should be able to design your research soundly. The conceptual framework is the foundation on which the whole research and design are based. Ultimately, the course operates as an introduction to several issues you will have to deal with in your academic and professional life, such as: -Issues of validity and relevance of knowledge; -Underpinning of claims in spatial planning and design; -Integration of research and practice; -Integration of text and image (communication); -Formulation and communication of original knowledge.	
Study Goals	In connection with the first component (organisation of the report and conceptual framework), the methodology course will help you: EXPLAIN what a conceptual framework is; BUILD a conceptual framework that will sustain your research and design in Q3; IDENTIFY a community of authors and practitioners who write about the core ideas of your theoretical framework; DESIGN, ORGANIZE and WRITE an academic report in which you will describe what are the main questions you will seek to answer in Q3 and the best methods to answer them; EXPLAIN the values connected to and the ethical issues involved in the activity of planning and designing for people, and explain what public goods are created with your design and strategy. In connection with the second component (socio-spatial justice and sustainability), the methodology course will help you: EXPLAIN main issues of socio-spatial justice and sustainability in relation to issues of socio-technical transitions in urban and regional design and development; BUILD upon those concepts to formulate your research questions, your conceptual framework, and your objectives; CRITICALLY ASSESS issues of urban and regional planning and design using ideas connected to those concepts.	
Education Method	Workshops and lectures. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Combination of written and oral examination (integrated in the AR2U086 design examination). A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	

Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	(External) students can only enroll for this course when also enrolling for the AR2U086 course, since the AR2U086 and the AR2U088 course are intertwined.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 3.
Concept Schedule	The studio sessions of AR2U086 are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and Friday afternoons. The AR2U088 course schedule is tuned with the AR2U086 course, thus not scheduled on Tuesdays and Fridays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

(Intra)disciplinary elective (15 EC or a combination of 10 + 5 EC)

AR0139	MEGA	15

Course Coordinator	Prof.dr. M. Overend
Course Coordinator	Dr. A. Luna Navarro
Course Coordinator	Dr. S. Brancart
Course Coordinator	Dr. M. Turrin
Instructor	Dr.ir. M.C. Lugten
Responsible for assignments	Dr. S. Brancart
Education Period	4
Start Education	4
Exam Period	none
Course Language	English
Course Contents	MEGA is a collaborative design studio where students work in multi-disciplinary teams to make an integrated design of a large multi-functional building. Students work in teams of 5 to 8 students in which they are each responsible for one discipline. The disciplines cover different (technical) aspects of building design like architectural design, structural design, climate design, façade design, computational design, and urban design. The course targets master students in Building Technology, Architecture and Urbanism, welcomes students from Civil Engineering and is open to non-TU Delft students (conforming with TU Delft regulations). By working in multi-disciplinary teams, MEGA closely matches the design process of large international projects in the competition phase. This allows students to specialize in one discipline and teaches them to collaborate and communicate with students that represent other disciplines. The disciplines are divided amongst the team members and each member is responsible for the contribution and integration of these aspects in the collective design. The design collaboration is supported by an integrated design process that supports on computational modelling (including parametric design, performance analysis, multi-disciplinary computational optimization or design exploration). The course is supported by external international design and engineering offices. As examples from recent years, support was given by Arup, UNStudio, ABT Neutelings Riedijk Architecten, MVRDV, DGMR, Techniplan, Packhunt, Thornton Tomasetti, Foster & Partners, Stichting Hoogbouw. Sustainability runs throughout the course and the different disciplines, and becomes apparent in the interdisciplinary collaboration. More specifically, this course regards the construction of large buildings in a perspective of sufficiency: how can we increase the density in urban regions with a limited amount of energy and materials? In doing so, it touches upon aspects of climate adaptation and resilience, circularity, renovation.
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: After successful completion of the course, the student will be able to: - work in an interdisciplinary design process; - understand and apply discipline-related knowledge in projects for large or tall buildings. - develop design strategies to achieve high building performance; - integrate numeric analysis and simulations data to address design choices.
Education Method	In this course, the education methods are: - Lectures - Collaborative workshops with other students - Exposure to external architectural practice and external experts - Tutoring - Presentations The course is structured in three phases. The first phase consists of lectures, workshops and a site visit. Many external guests (international practitioners) will join for the lectures and workshops. Through the workshops, students learn specific skills that are useful for the group work or for their specific discipline. In the second phase, students develop a sketch design, exploring several design options. They are supported in this by a team of tutors, for the different disciplines. In a third phase, they elaborate the design for a specific option.
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.
Assessment	Assessment is twofold: - Group assessment for integral group design based on presentations - Individual assessment for discipline report The students mark is a combination of the group assessment and individual assessment. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. A retake can be done in the next year.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Period of Education	Quarter 4
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 90.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in Architecture and Urbanism. If you are in a different program: please consult coordinators before enrolling and ask for approval. MSc track Architecture: students are expected to follow a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained concerning architectural/urban history, theory, art, and technology and relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces, and societys needs, including environmental aspects.	
Course Contents	Architectural and Urban Design in Times of Transitions. Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programs cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities, and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on the one hand in the urban intervention in the built environment and its effect on architecture, and on the other hand in the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series that provides an overview of vested theories and cutting-edge research on people movement, urban vitality, and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, and Lynch, and research work by Cullen, Smithsons, and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public spaces is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course The course material will come alive through active discussions and the direct application of theories in analyzing real urban settings.	
Study Goals	At the end of this course, the student: knows key literature and recent research on people, movement and public space understands main theories on people, movement and public space applies these theories in analysing real urban settings evaluates critically on these theories creates presentations analysing the subject on an academic level. And, the student: understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership, and meaning creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical, and functional requirements, including materialisation. Special attention is dedicated to the commitment to several Sustainable Development Goals, in particular SDG 11 (Sustainable Cities and Communities), SDG 9 (Resilient Infrastructures), SDG 3 (Good Health and Well-Being), and SDG 13 (Climate Action). In addition, you are assisted in developing cross-cutting key competencies for achieving these SDGs, such as anticipatory competency, collaboration competency, critical thinking competency, and self-awareness competency.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers In class, you are encouraged to question the course material, the case, the lecturer, and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight into the problematique highlighted in the course. The case will be updated annually. It serves as a test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborate design. These will be presented on the last day of class. *) the discussion groups ideally consist of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures:	

	Class participation and homework assignments together with the final presentation (including 5 pages of individual contribution to a collaborative report, 1 group poster (A1), and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Special Information	This course includes AR0168 - People, Movement, and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform the course coordinators what this year's case studies are.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesday and Friday. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design | Design of Public Space Architectural Crossovers
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 45.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Todays new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student in the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programs. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK enrolment	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday, Friday	
Minimum number of participants	15	
Maximum number of participants	25 students NB. For MSc track Urbanism students only 3 spots are available.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW0501	Applied Spatial Analysis for Sustainable Urban Development	5

Course Coordinator C. Forgaci

Course Coordinator Dr. D. Cannatella

Contact Hours / Week x/x/x/x 5 hours per week starting from week 1 and ending in week 9.

Education Period 4

Start Education 4

Exam Period 4

Course Language English

Course Contents

Spatial analysis is a crucial part of projects carried out in urbanism and landscape architecture and a core application domain for the field of geomatics. After having encountered basic spatial analysis with short targeted workshops in the first three quarters of their MSc tracks, Urbanism and Landscape Architecture students following this intradisciplinary course work on an integrated, end-to-end analytical assignment, combining, advancing and extending their data-related skill- and tool-sets, while focusing on advanced workflows. The course offers the opportunity to MSc Urbanism Landscape Architecture, and Geomatics students to advance their technical skills required for scalable, automated and reproducible spatial analyses, practicing working in an intra-disciplinary setting with a real-world assignment. All students have the chance to learn skills from the other tracks and in the process strengthen their analytical skills for their upcoming thesis.

The course will start with an introduction to multi-criteria decision analysis, typology construction and their corresponding workflows suitable for a sustainable spatial development challenge. The higher-level principles of scalability, automation and reproducibility, as well as their implementation through computational notebooks, are also introduced early in the course. The course emphasizes data visualization as a way to explore data iteratively, generate new knowledge and communicate project results. Students learn about different types of thematic analysis, such as spatial analysis, network analysis, and landscape analysis which they can integrate in their analytical workflows. To apply the workflows and analytical methods introduced throughout the quarter, students work in intra-disciplinary groups on a given location.

The workflows presented in the course involve open-source tools using graphical user interfaces, such as QGIS, as well as programming in R and/or Python for spatial analysis. In the era of open data, the course emphasizes the importance of leveraging accessible and shared datasets to enhance the quality and depth of spatial analyses. In addition, the course underscores the significance of transparent and collaborative research practices by incorporating a dedicated segment on publishing datasets, workflows, and results. MSc Urbanism, Landscape Architecture and Geomatics students will be given the opportunity to contribute to the broader academic community through the dissemination of their work.

Study Goals

After following this course, students will be able to:

1. Construct analytical workflows in the context of spatial design and planning.

2. Employ spatial data visualization both for exploratory data analysis and for the presentation of their final project results.

3. Conduct integrated spatial analysis to understand and describe the complexity of urban environments and to inform spatial design and planning decisions targeting their transformations.

4. Adopt open science practices, including FAIR data publication, reproducible workflows, and usage of open-source software.

5. Report on analytical findings as part of a multidisciplinary team.

Education Method

The course has a blended learning set-up that makes use of online learning material developed or curated as part of the Spatial Data & GIS education platform of the Department of Urbanism. During contact hours, the focus will be on lectures introducing key concepts and hands-on workshops addressing the methodological and technical challenges of the workflows and methods introduced in the course. Self-study time will involve both individual and group work, with an emphasis on the latter. In addition to the online software tutorials, peer troubleshooting will be facilitated.

Literature and Study Materials

Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.

Assessment

Students keep track of their progress and present their final products in a group report set up as a computational notebook in a version control system. This allows them to reflect on their process throughout the quarter and the instructors to assess individual contributions in detail. Peer feedback as an essential component of the course is employed in the midterm and the thematic workshops.

A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.

Students should achieve a minimum grade of 6.0 to pass this course.
Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.
In case of a final grade below 6,0 the student needs to reenroll for the course.

Enrolment / Application

Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see:
https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams
Additional enrolment information for non-BK students:

Course enrolment via Brightspace only is not considered a valid registration
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The course can be taken only after confirmation of placement in BIS by BK-enrolment

Remarks The maximum marking period is 15 working days.

Period of Education Quarter 4.

Concept Schedule

The sessions are scheduled on one day part per week; on Tuesday or Friday mornings or afternoons.
The actual schedule will be available via Bright Space > My Timetable

Minimum number of participants For this course the minimum number of participants is 15.

Maximum number of For this course the maximum number of participants is 30.

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW0502	**Plantscapes**	**5**
Course Coordinator	J.R.T. van der Velde	
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Responsible for assignments	Dr.ir. N.M.J.D. Tillie	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	During the first weeks, we build the FOUNDATION of knowledge/skills regarding planting in the built environment. After this, we work with DIMENSIONS OF DESIGN, build upon the foundation and focus on planting strategies for gardens. Following, we strengthening PLANTING ENGINEERING strategies to improve ecosystem services. Finally, we integrate the knowledge gained in the past weeks into the design of (parts of) a park, while focussing on DRAWING REPRESENTATION.	
Study Goals	By the end of this course, the student should be able to: - recognize typologies that suit specific planting strategies. - recognize planting potential. - design with plants for climate adaptation, biodiversity and aesthetics. - develop his/her own library with knowledge about a variety of plants, trees and shrubs.	
Education Method	Studio work Workshops Instruction sessions Excursions Fieldwork	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	2D drawings and supporting visuals Physical/digital 3D-model Oral presentations Written reflection A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on two day parts per week; on Tuesday and Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable	
Leerstoel	Landscape Architecture	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW0503	Urbidata: Responsible Design for Fair Data Cities	5
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 7.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this course we explore the main principles of data ethics, and the relevance of applied ethics in the domain of the data-driven city. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence. Data, processed by AI and increasingly powerful computers, that promise mankind the opportunity to influence all aspects of life, especially in the urban setting. The concept of the 'smart city' assumes that big data leads to better (after all smart) solutions for planners, designers, industry and, at the end, citizens. Urbidata, the data-driven city of 2050, will be optimally planned, with smart mobility that allows for efficient and safe use of space and time, and citizens living in smart homes and working in smart offices that cater optimally their needs, based on their personal profiles. Does this technology driven perspective provide the answer how we want the life in the future city of Urbidata to look like? Should all developments of data-driven solutions (e.g. by architects, urban planners and geo-specialists) and their use be permissible in the light of the consequences? The choices we make today determine our world of tomorrow. Taking a critical, responsible design perspective it's clear that our data-driven future cannot be just a question of engineering and economic models. How about values as autonomy, privacy, transparency and consent? At first sight this interpretation of the city of tomorrow appears to be a question of policy. Policy that might lead to a normative (legal) framework setting standards. However in this course we focus on the preceding fundamental discussion. What are the ethical concepts and principles that guide us to determine what solutions are right or wrong? And next: how to apply data ethics in a given situation? This in order to make the future city of Urbidata a fair data city.	
Study Goals	After this course the student is able: - to explain the relevance of applied ethics in relation to the design and governance of the data driven city in general; - to identify concrete ethical issues within the field of data processing in a concrete situation; - to identify and apply the relevant ethical principles for this concrete situation; - to create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours). Self-study, Individual assignments (60 hours). Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The student will design an artefact and submit a written report, providing explanation of the artefact and containing a reflection. The artefact itself can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The student needs to give one presentation on an assigned topic during the course in order to pass the course. The course is concluded with an oral exam. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. A rubric will be used for grading. The rubric will be available on the course specific Brightspace page. Students should achieve a minimum grade of 5,75 to pass this course. Grades between 5,0 5,75 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the first week of Summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 5,75 the student needs to re-enroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on two day parts per week; on Tuesday and Friday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARFW1001	Integrated Metabolic Design for Sustainable Neighborhoods, Cities and Regions	10

Course Coordinator	Dipl.-Ing. U.D. Hackauf
Course Coordinator	Dr. A. Wandl
Responsible for assignments	Dipl.-Ing. U.D. Hackauf
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	In the course, students address sustainability challenges of society metabolism based on a systemic understanding, analysis and design: through different spatial scales (from urban block and street to neighbourhood, city and region) through different disciplines of engineering, design and governance through interaction with different institutions and stakeholders A theoretical foundation of society's metabolism and the methods used to improve it is introduced, discussed and applied through a methodological, multiscale, iterative cycle of analysis, design and assessment. Presentations are used to share findings with stakeholders and gather feedback for further design iterations.
Study Goals	After the course, the student can: Apply the concept of societal metabolism to analyse integrated sustainability challenges in a specific context develop a multiscale design proposal based on their analysis. Evaluate their proposal in the context of Dutch policymaking. Communicate and discuss proposals with stakeholders in the concerned context. Assess the effectiveness of design variants of environmental technologies through scales.
Education Method	systemic design, lectures, group work, stakeholder workshops and self-study
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	The final assessment has two parts: (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) a written reflection reacting to the feedback received on the poster presentation. Each individual week finishes with a formative assessment, either a review by the course instructors or a peer review between the students. This way, the students get weekly formative assessments. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	4
Concept Schedule	The sessions are scheduled on two different day parts per week, most likely Monday, Wednesday and Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 36.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1002	Heritage Landscape Design	10
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Responsible for assignments	Dr.ir. G.A. Verschuure-Stuip	
Contact Hours / Week x/x/x/x	Wk 1-6: 10 hrs per week Wk7: 28 hrs per week Wk 8-9: 8 hrs per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	The Heritage Landscape Design elective offers spatial landscape and/or urban design training in a challenge-based (heritage) transformation by participating in a design workshop (charette) in interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies to create a visionary spatial plan for a transformation site based on local needs.	
Course Contents	Identity, continuity and transformation are essential notions of todays spatial design practice. In this course, you will learn how to analyze a given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site in contact with the public. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the interactive workshop, you study literature on theory and methods of heritage transformation, landscape design, value assessment and participation with local communities in weekly assignments. Youll team up with students from different disciplines. You will debate theories on place-making. During the workshop you will work with experimental analysis visualization methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. In this course, the Landscape Architecture section wants to strengthen the interaction between landscape transformations and the role of (local) stakeholders in a multidisciplinary setting.	
Study Goals	By the end of this course, you should be able to: - systematically identify, visualize and value the main characteristics of the (cultural) landscape design project as a continuous process in time; - debate and apply methods and tools on landscape transformation, place-making and heritage management; - use methods and tools to create a visionary transformation plan and present it to a larger audience in different visualization methods; - understand the processes of creating plans and the role of spatial designers in citizen engagement, facilitating dialogue in participation processes; - analyse and communicate effectively on complex design (transformation) ideas within a multidisciplinary setting with and for local stakeholders.	
Education Method	Lectures and assignments Literature study Three-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; Tijdschrift voor Economische en Sociale Geograﬁe 2014, Vol. 105, No. 5, pp. 622629. Manual: Erfgoed en participatie (in prep.) Other recommended literature will be mentioned on the specific Brightspace page at least one week prior to the start of the course.	
Assessment	A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on two different day parts per week, most likely Monday, Wednesday and Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 30.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1500	Metropolitan Regions in Times of Environmental and Economic Transitions	15

Course Coordinator	R. Ordonhas Viseu Cardoso
Course Coordinator	Dr. D.A. Sepulveda Carmona
Responsible for assignments	Dr. D.A. Sepulveda Carmona
Contact Hours / Week x/x/x/x	12 hours per week starting from week 1 and ending in week 9.
Education Period	4
Start Education	4
Exam Period	4
Course Language	English
Course Contents	The main aim of the course is to examine how metropolitan regions respond to, and navigate the impact of, the imperatives of sustainable and integrated development. The course is embedded in the field of spatial and environmental planning and design in TU Delft and builds upon urban and regional planning theories that are aligned with the UN sustainable development goals. These seek to effectively address environmental and social challenges facing cities and regions. One of the courses primary goals is to advance a critical understanding of metropolitan processes within (post-)globalization, as well as their effects. It seeks to articulate these through both theory and design to move towards new urban and regional development paths that are better able to respond to the ever-increasing challenges faced by cities and city-regions. The course programme involves learning about current and upcoming challenges of sustainable development through a process of research and design. It focuses on urban and regional development models globally and develops an integrated approach, providing key skills that students can use as planners, designers, and landscape architects in different geographical contexts. It also enables them to work on different levels of spatial and temporal scales. The students are trained in research methods relating to spatial planning, urban studies, environmental and urban design, with an emphasis on the diversity of planning cultures and the need to incorporate socio-environmental concerns into urban development. The course has a transdisciplinary ethos, and uses a series of different developing regions as case studies. Students have the chance to meet stakeholders from diverse backgrounds, including academics and governments, who act as teachers and trainers. The course is also part of the International Forum on Urbanism (IfoU) and works in partnership with other universities from that network to deliver work that builds on theoretical knowledge in order to advance strategies for spatial and environmental planning. The results are location-sensitive strategies that outline real opportunities for moving towards more a sustainable future for the regions selected for study. Such regions include Greater Buenos Aires, Argentina, the Greater Bay Area in China (including Hong Kong, Guangzhou, Macao) and the Greater Paris region. Previous students work has been published and presented at several biennales (including Venice 2014, Buenos Aires 2016, and Shenzhen Hong Kong 2019).
Study Goals	At the end of this course the student is able to: Critically understand and articulate metropolitan processes in the framework of globalization effects on urban and regional development. Design metropolitan and local development strategies, considering infrastructural development, social integration, socio-economic wellbeing, and environmental adaptation. Facilitate the articulation of urban and regional design within complex and multidisciplinary perspectives. Understand and develop planning strategies and actions across scales and borders using a variety of methods. Advance stakeholder analysis and translate diverse interests within local, regional and cross-regional scales. Define a coherent communication perspective at various scales and levels of analysis to facilitate communication across a broader set of actors.
Education Method	Approach: The course is a collaboration between the research groups Planning Complex Cities, Design of the Urban Fabric and Territories in Transition. This allows a combined multidisciplinary approach that facilitates the definition of assessment and intervention criteria under a complex analytical framework. The approach starts by seminar readings on regionalization and metropolisation processes, perspectives, and effects, allowing students to choose critical approaches supported by theory and advance an assessment and possible key challenges of the selected metropolitan region of study. Then, a series of sub-themes are advanced as concrete tools for strategic design focused on the nexus of at least three topics - e.g., water management, soil, urban expansion, demographics trends, economic redistribution, and environmental risk management, among others. This stage is part of a studio approach, which is then exposed through workshops and interviews to concrete stakeholders from the selected regions, who facilitate the validation of the approaches and contextualize them within local planning frameworks. This is then strategized through value setting, morphological analysis, and environmental and landscape perspectives to give to the selected nexus a concrete focus. Mapping is used to explain and support the diverse possibilities, in the form of evolutionary strategies. The selected criteria of analysis are revised under the specific conditions and trends in the selected area and scenario are devised to cope with possible uncertainties. Overall, this facilitates the proper landing of the ideas on very concrete and properly scanned regional conditions. Finally, forecast projections are built up, using adaptive pathways (policy and design compositions) to define the frameworks for the integral perspectives (diverse systems, from socio-environmental perspectives- to decision making) Format: The students will form groups to tackle, evaluate and propose new visions for diverse scenarios of regional and metropolitan development, which vary in terms of maturity and consideration of broader challenges. A series of strategic actions will be defined and tested to consolidate the proposals. The groups will consult with local partners to advance their local knowledge and revise the diverse stakeholder perspectives.
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.
Assessment	Oral presentation and report. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris.

	In case of a final grade below 6,0 the student needs to reenroll for the course.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Minimum number of participants	For this course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 35.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW1502	Peri-Urban Transformations: Analysis, Development, Design	15
Course Coordinator	T.N. Broekmans	
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	15 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this new elective course the interaction between spatial design, area and project development tools and social, economic, political, and environmental issues is at stake. Students from the MBE, Landscape Architecture and Urbanism tracks and the MSc CME Design & Integration Domain (from the coupled course Transformative Design), collaborate on the transformation of a peripheral urban area in the Netherlands. The pressure on these urban fringes is both urban and rural and characterised by urban sprawl, fragmented agricultural or industrial activity, remnants of natural and landscape values. Working together in groups you will develop a master plan and elaborate individually a more detailed, imaginable transformation proposal. Students confront spatial qualities, opportunities, limitations and challenges of the chosen area with societal needs, transition goals, ownership issues and stakeholder claims. You design and define proposals on different scales and complexities including area, portfolio and project level. You will apply development strategies, feasibility criteria and research by design activities. This course is organised as a multidisciplinary experience combining approaches, knowledge and skills from the viewpoints of management, urban design and landscape development and is therefore representative for area development practice in the real world. This collaboration provides insight in the complementarity of viewpoints, data, tools, vocabulary, methods and fun.	
Study Goals	(1) students can analyse the characteristics, conditions, stakeholders, opportunities and potentialities of peri-urban transformation areas (using a theoretical framework) (2) students can develop a tentative spatial perspective (e.g. spatial, social, economic, political, environmental) in a peri-urban context including primary notions of phasing: from initiative to implementation (masterplan version 0.5) (3) students can elaborate a crucial, domain-specific aspect or component of the masterplan like a portfolio, structure, block typology or building (4) students can synthesise varied input into an integrated, substantiated master plan for a peri-urban transformation area (masterplan version 1.0) (5) students can reflect on the implications of the domain-specific theory on the application of the masterplan and its elaborations	
Education Method	In this course four main educational methods are used: (1) Lectures (2) Studio work in groups (2) Studio work individually (4) Plenary presentations and discussions	
Assessment	This course has two assessments (1) group: atlas and integrated masterplan (2) individual: elaboration proposal and reflection report A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 4.	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 75.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 Urbanism

AR3U105	Graduation Orientation	3
Course Coordinator	L. Qu	
Course Coordinator	Dr. L.M. Calabrese	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	15 hours per week in week 1, 2, 3, 4, 6 and 7, and 5 hour per week in week 5 and 8, 1 hour per week in week 9 and 15 hours per week in week 10 (AR3U105 + AR3U110 together).	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	Within this Graduation Orientation course, students will connect their graduation project research to the main research questions of the (cross-cutting) themes of the Urbanism research programme.	
Course Contents	Within the Graduation Orientation course, students are asked to develop a first, preliminary proposal for their graduation project and to propose their studio of preference and their 1st and 2nd mentor of preference. Students will be connected to the Urbanism graduation studios based on their individual proposal. Students will be asked to critically reflect on the variety of urbanism research themes and to position their intended graduation project within.	
Study Goals	At the end of this course, the student is able to: develop a first proposal for his/her graduation project. develop an overview of the different research themes in the department of Urbanism, and is able to state his/her own position within. provide clear arguments of the connection between his/her own graduation topic, the Urbanism research program, and the preferred graduation studio(s). communicate his/her first proposal for his/her graduation project in an academic way. demonstrate skills of researching and academic writing.	
Education Method	Seminars and interactive peer-feedback (and expert-reflection) sessions. Combination of mainly individual but also partially group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment. The assignment, developing a first proposal for his/her graduation project, will be assessed in two ways. The course coordinator team will assess the assignment according to the format (downloadable from Brightspace), based on the following criteria: Critical reflection skills on the positioning within the Urbanism research portfolio Academic writing and referencing Scientific and societal relevance Reflection on ethical issues The proposed studio coordinator or main mentor will assess the assignment based on the following criteria: Clarity of motivation, problem (field) statement, aim of study, project approach In depth analysis and underpinning of the project/thesis proposal Appropriateness (of argumentation) for the proposed main mentor and studio As this assignment is a start to build towards P1/P2, the grading will be done by a Fail, Repair, or Pass. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade 'Pass' to pass this course. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student before P2. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1 and quarter 3.	
Concept Schedule	The sessions are scheduled on Mondays and Thursdays / Friday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3U115	Graduation Lab Urbanism	15
Course Coordinator	L. Qu	
Course Coordinator	Dr. L.M. Calabrese	
Instructor	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. D.A. Sepulveda Carmona	
Instructor	V.E. Balz	
Instructor	Dr. A. Wandl	
Instructor	Ir. M. Lub	
Instructor	T. Kuzniecow Bacchin	
Instructor	Dr. M.M. Dabrowski	
Instructor	Dr. D. Maiullari	
Instructor	T.N. Broekmans	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 & 2 of the Master track Urbanism.	
Summary	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. This course is the first part of the graduation laboratory; students (individually) develop their graduation project.	
Course Contents	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. These semesters give the student therefore a unique opportunity to do an in-depth research and design project in the field of urbanism. This course is the first part of the graduation laboratory; students (individually) develop their graduation project. After getting to know the main Urbanism research topics and research questions, the student works in content-driven studios with specialized researchers. The studios provide activities for the students, such as lectures, peer review sessions, master classes dedicated to improve students skills and lectures, symposiums and workshops organised by students themselves according to their own interests.	
Study Goals	At the end of this course, the student: is able to describe and map the problem field of his graduation work on the basis of a motive, fascination, or question (Problem field). is able to define a relevant field of graduation objectives, concerning research questions and design tasks (Field of graduation objectives). is able to define an approach, with specific methods, techniques and design instruments for the graduation work (design and research), based on the results of the Master 3 AR3U013 course, which suits the objectives, the design task and the research questions. (Approach). is able to present a consistent and adequately constructed theoretical framework for the graduation topic, based on the results of the Master 3 course AR3U023 (Theoretical framework). is able to define and describe the project location and design task, together with an urban analysis, in line with the formulated problem field (Design and research location). is able to define the in-between and end products appropriate for the aim of the graduation project (In-between and end products). is able to put forward arguments on how the graduation work will provide a substantial contribution to society and science (Relevance). is able to present a first concept or hypothesis, in which a first solution or direction for the design task or the main question is embedded (Concept);. is able to provide the agreed time frame with the formulated in-between and end products (Planning). See the assessment criteria of the 'Graduation Criteria Urbanism P1/P2' of the master track of Urbanism.	
Education Method	In this quarter, and according to their topic of interest, students will be assigned to work in studios where they will work closely with researchers specialized in their specific topic. Students will choose their first mentor from this studio, and a second mentor from a different section.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course and by the studio coordinator and mentors.	
Assessment	The assessment is imbedded in the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on the assessments of the graduation project. In this course the project is evaluated two times: at the first Evaluation: Compulsory Progress Review (P1) at the second Evaluation: Formal Assessment (P2) A rubric (EMMA) will be used for grading. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on retake possibilities.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	On set conditions, Urbanism students have the possibility to carry out their graduation project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved	

	by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found in the year guide.
Remarks	The maximum marking period is 15 working days. The final product of this course is the still growing thesis report, an integral product that the student has to deliver at the end of Master 3, 1 week prior to the P2 presentation. See the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences and the assessment criteria of the 'Graduation Criteria Urbanism P1/P2' of the Master track of Urbanism.
Period of Education	Semester 1 and semester 2. The MSc3 Urbanism courses start both in September and February. The majority of students starts the MSc3 in September. In February just one studio will be on offer, the number of intensives will be reduced and the number of available mentors will be limited.
Concept Schedule	The sessions differ per week. The important dates and deadlines are represented in the academic graduation calendar and in the year guide available via Bright Space. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Choose 1 course

AR3U110	Graduation Exploration	12
Course Coordinator	L. Qu	
Course Coordinator	Dr. L.M. Calabrese	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	15 hours per week in week 1, 2, 3, 4, 6 and 7, and 5 hour per week in week 5 and 8, 1 hour per week in week 9 and 15 hours per week in week 10 (AR3U105 + AR3U110 together).	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course consists of 2 intensives/workshops. In the Graduation Orientation course students define their graduation topic, find a studio and 1st mentor. The studio choice defines which studio intensive the student will join. In this studio intensive the essential topics for the specific studio are addressed. Together with their 1st mentor and studio coordinator students select which other intensive is most fit for their graduation project. These intensives focus on a specific skill, technique, approach or method. Students used the gained knowledge of the intensives to sharpen their graduation project.	
Study Goals	At the end of this course, the student: is able to analyse the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. is able to evaluate the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. is able to reflect on the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. is able to use the content of the intensives to create and polish his/her graduation project. shows social awareness and professional responsibility.	
Education Method	two intensives/workshops of 2 weeks each. Combination of individual and group work	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus for some intensives/workshops design examination. A rubric will be used for grading. The rubric will be available in the year guide on the course specific Bright Space page. Each intensive results in an assignment. The combination of and reflection on these assignments results in a final assignment which supports the more precise definition of the graduation project. The assignments together result in 1 final grade. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student before P2. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1 and quarter 3.	
Concept Schedule	The sessions might be scheduled on all day parts of the week. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

IFM4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	To enter JIP, a student should be a 2nd-year master student; have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. be full time available (40 hours per week) in Q5; meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV Students will be selected by the JIP core staff based on motivation, study progress and recommendation.	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman,VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (Ilo;K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.	

	The minimum grade for the summative assessment is =/> 5.8. Process grade is 20% The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 4 Urbanism

AR4U010	Graduation Lab Urbanism	30
Course Coordinator	L. Qu	
Course Coordinator	Dr. L.M. Calabrese	
Instructor	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. D.A. Sepulveda Carmona	
Instructor	V.E. Balz	
Instructor	Dr. A. Wandl	
Instructor	Ir. M. Lub	
Instructor	T. Kuzniecow Bacchin	
Instructor	Dr. M.M. Dabrowski	
Instructor	Dr. D. Maiullari	
Instructor	T.N. Broekmans	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	1 hours per week starting from week 1 and ending in week 9.	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	MSc1, 2 & 3 of the Master track Urbanism.	
Summary	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. This course is the second part of the graduation laboratory; students continue (individually) developing their graduation project that was approved at the end of the previous semester (GO for P2).	
Course Contents	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. These semesters give the student therefore a unique opportunity to do an in-depth research and design project in the field of urbanism. This course is the second part of the graduation laboratory; students continue (individually) developing their graduation project that was approved at the end of the previous semester (GO for P2).	
Study Goals	At the end of this course, the student: is able to describe and carry out research in the field of urbanism and process the research results as well as use drawings/maps/graphics as a means to research; is able to describe the problem field of the selected topic and translate it into a field of graduation objectives (and design task) with associated research questions and research approach; is able to describe a clear theoretical framework which is appropriate for the selected topic; is able to carry out research by design in a methodological way focused on the research questions; is able to process the research results in the final report adequately: i.e. formulated and/or imagined by means of analytical drawing(s); is able to use drawings / maps / graphics as a means to research; is able to define urban design methods, choices, aspects, effects and consequences by means of plan forms and design-instruments; is able to define and visualised the own working method(s) and the (design) choices within the design process with sound arguments; is able to define the spatial, functional, technical, and/or social aspects of the design adequately: clear, transparent and with a proper justification; is able to use plan forms and design-instruments which suit his/her design task; is able to describe, imagined and justified the effects and consequences of the design proposal(s) with respect to the aimed field; is able to draw conclusions and define recommendations; is able to evaluate the urban design and research aims in the conclusions; is able to indicate clearly and logically which research questions are answered and how that has been processed in the urban design; is able to define clear, concrete, specific recommendations based on the results of the urban research and/or design; is able to specify for which questions is still additional (design) research necessary; is able to show an analytical capacity to present a complex matter in a brief and concise way; is able to describe the projects relevance, reflect on the products and present these; is able to describe clearly the innovative (scientific and/or social) insights of the graduation project in text and images and, if necessary, concrete strategies and/or application possibilities for the field of the urbanism; is able to position the own graduation project with respect to the field of the urbanism, as well as other adjacent scientific fields; is able to discuss and present the thesis products (urban design & research) and the process (design & research) in the form of an epilogue or evaluation. In short, the student has to show that he/she is able to deliver a project of professional quality and of academic level in line with the end terms of the master track Urbanism.	
Education Method	In this semester students work individually, supervised by two mentors, within the frame of the studio and graduation lab of Urbanism. Meetings and lectures are less frequent than in the previous semester, the independency of the student is enlarged.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course and by the studio coordinator and mentors.	
Assessment	The assessment is imbedded in the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on the assessments of the graduation project. In this course the project is evaluated three more times: at the third Evaluation: Compulsory Progress Review (P3) at the fourth Evaluation: Formal Assessment (P4) at the final Evaluation: Public Final Presentation (P5)	

	A rubric (EMMA) will be used for grading. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on retake possibilities.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Once enrolled in AR3U115 (start of the graduation) students automatically continue their graduation in AR4U010
Special Information	On set conditions, Urbanism students have the possibility to carry out their graduation project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found in the year guide.
Remarks	The maximum marking period is 15 working days. The final product of this course is an integral concise product that the student has to deliver at the end of Master 4, 1 week prior to the P5 presentation. See the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences, the 'Exit Qualifications' of the master Architecture, Urbanism and Building Sciences, the evaluation form 'Graduation Criteria Urbanism P3' and the assessment criteria of the 'Graduation Criteria Urbanism P3/P4/P5' of the Master track of Urbanism.
Period of Education	Semester 1 and semester 2. The MSc3 Urbanism courses start both in September and February. The AR4U010 course starts automatically after successfully finishing the MSc3 with a GO at the P2 presentation.
Concept Schedule	The sessions differ per week. The important dates and deadlines are represented in the academic graduation calendar and in the year guide available via Bright Space. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

Cross Domain City of the Future

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 3 Cross Domain City of the Future

AR3U105	Graduation Orientation	3
Course Coordinator	L. Qu	
Course Coordinator	Dr. L.M. Calabrese	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	15 hours per week in week 1, 2, 3, 4, 6 and 7, and 5 hour per week in week 5 and 8, 1 hour per week in week 9 and 15 hours per week in week 10 (AR3U105 + AR3U110 together).	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	Within this Graduation Orientation course, students will connect their graduation project research to the main research questions of the (cross-cutting) themes of the Urbanism research programme.	
Course Contents	Within the Graduation Orientation course, students are asked to develop a first, preliminary proposal for their graduation project and to propose their studio of preference and their 1st and 2nd mentor of preference. Students will be connected to the Urbanism graduation studios based on their individual proposal. Students will be asked to critically reflect on the variety of urbanism research themes and to position their intended graduation project within.	
Study Goals	At the end of this course, the student is able to: develop a first proposal for his/her graduation project. develop an overview of the different research themes in the department of Urbanism, and is able to state his/her own position within. provide clear arguments of the connection between his/her own graduation topic, the Urbanism research program, and the preferred graduation studio(s). communicate his/her first proposal for his/her graduation project in an academic way. demonstrate skills of researching and academic writing.	
Education Method	Seminars and interactive peer-feedback (and expert-reflection) sessions. Combination of mainly individual but also partially group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment. The assignment, developing a first proposal for his/her graduation project, will be assessed in two ways. The course coordinator team will assess the assignment according to the format (downloadable from Brightspace), based on the following criteria: Critical reflection skills on the positioning within the Urbanism research portfolio Academic writing and referencing Scientific and societal relevance Reflection on ethical issues The proposed studio coordinator or main mentor will assess the assignment based on the following criteria: Clarity of motivation, problem (field) statement, aim of study, project approach In depth analysis and underpinning of the project/thesis proposal Appropriateness (of argumentation) for the proposed main mentor and studio As this assignment is a start to build towards P1/P2, the grading will be done by a Fail, Repair, or Pass. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade 'Pass' to pass this course. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student before P2. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1 and quarter 3.	
Concept Schedule	The sessions are scheduled on Mondays and Thursdays / Friday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3U110	Graduation Exploration	12
Course Coordinator	L. Qu	
Course Coordinator	Dr. L.M. Calabrese	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	15 hours per week in week 1, 2, 3, 4, 6 and 7, and 5 hour per week in week 5 and 8, 1 hour per week in week 9 and 15 hours per week in week 10 (AR3U105 + AR3U110 together).	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course consists of 2 intensives/workshops. In the Graduation Orientation course students define their graduation topic, find a studio and 1st mentor. The studio choice defines which studio intensive the student will join. In this studio intensive the essential topics for the specific studio are addressed. Together with their 1st mentor and studio coordinator students select which other intensive is most fit for their graduation project. These intensives focus on a specific skill, technique, approach or method. Students used the gained knowledge of the intensives to sharpen their graduation project.	
Study Goals	At the end of this course, the student: is able to analyse the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. is able to evaluate the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. is able to reflect on the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. is able to use the content of the intensives to create and polish his/her graduation project. shows social awareness and professional responsibility.	
Education Method	two intensives/workshops of 2 weeks each. Combination of individual and group work	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus for some intensives/workshops design examination. A rubric will be used for grading. The rubric will be available in the year guide on the course specific Bright Space page. Each intensive results in an assignment. The combination of and reflection on these assignments results in a final assignment which supports the more precise definition of the graduation project. The assignments together result in 1 final grade. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student before P2. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1 and quarter 3.	
Concept Schedule	The sessions might be scheduled on all day parts of the week. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3U115	Graduation Lab Urbanism	15
Course Coordinator	L. Qu	
Course Coordinator	Dr. L.M. Calabrese	
Instructor	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. D.A. Sepulveda Carmona	
Instructor	V.E. Balz	
Instructor	Dr. A. Wandl	
Instructor	Ir. M. Lub	
Instructor	T. Kuzniecow Bacchin	
Instructor	Dr. M.M. Dabrowski	
Instructor	Dr. D. Maiullari	
Instructor	T.N. Broekmans	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 & 2 of the Master track Urbanism.	
Summary	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. This course is the first part of the graduation laboratory; students (individually) develop their graduation project.	
Course Contents	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. These semesters give the student therefore a unique opportunity to do an in-depth research and design project in the field of urbanism. This course is the first part of the graduation laboratory; students (individually) develop their graduation project. After getting to know the main Urbanism research topics and research questions, the student works in content-driven studios with specialized researchers. The studios provide activities for the students, such as lectures, peer review sessions, master classes dedicated to improve students skills and lectures, symposiums and workshops organised by students themselves according to their own interests.	
Study Goals	At the end of this course, the student: is able to describe and map the problem field of his graduation work on the basis of a motive, fascination, or question (Problem field). is able to define a relevant field of graduation objectives, concerning research questions and design tasks (Field of graduation objectives). is able to define an approach, with specific methods, techniques and design instruments for the graduation work (design and research), based on the results of the Master 3 AR3U013 course, which suits the objectives, the design task and the research questions. (Approach). is able to present a consistent and adequately constructed theoretical framework for the graduation topic, based on the results of the Master 3 course AR3U023 (Theoretical framework). is able to define and describe the project location and design task, together with an urban analysis, in line with the formulated problem field (Design and research location). is able to define the in-between and end products appropriate for the aim of the graduation project (In-between and end products). is able to put forward arguments on how the graduation work will provide a substantial contribution to society and science (Relevance). is able to present a first concept or hypothesis, in which a first solution or direction for the design task or the main question is embedded (Concept);. is able to provide the agreed time frame with the formulated in-between and end products (Planning). See the assessment criteria of the 'Graduation Criteria Urbanism P1/P2' of the master track of Urbanism.	
Education Method	In this quarter, and according to their topic of interest, students will be assigned to work in studios where they will work closely with researchers specialized in their specific topic. Students will choose their first mentor from this studio, and a second mentor from a different section.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course and by the studio coordinator and mentors.	
Assessment	The assessment is imbedded in the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on the assessments of the graduation project. In this course the project is evaluated two times: at the first Evaluation: Compulsory Progress Review (P1) at the second Evaluation: Formal Assessment (P2) A rubric (EMMA) will be used for grading. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on retake possibilities.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Special Information	On set conditions, Urbanism students have the possibility to carry out their graduation project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved	

	by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found in the year guide.
Remarks	The maximum marking period is 15 working days. The final product of this course is the still growing thesis report, an integral product that the student has to deliver at the end of Master 3, 1 week prior to the P2 presentation. See the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences and the assessment criteria of the 'Graduation Criteria Urbanism P1/P2' of the Master track of Urbanism.
Period of Education	Semester 1 and semester 2. The MSc3 Urbanism courses start both in September and February. The majority of students starts the MSc3 in September. In February just one studio will be on offer, the number of intensives will be reduced and the number of available mentors will be limited.
Concept Schedule	The sessions differ per week. The important dates and deadlines are represented in the academic graduation calendar and in the year guide available via Bright Space. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Architecture, Urbanism and Building Sciences**

MSc 4 Cross Domain City of the Future

AR4U010	Graduation Lab Urbanism	30
Course Coordinator	L. Qu	
Course Coordinator	Dr. L.M. Calabrese	
Instructor	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. D.A. Sepulveda Carmona	
Instructor	V.E. Balz	
Instructor	Dr. A. Wandl	
Instructor	Ir. M. Lub	
Instructor	T. Kuzniecow Bacchin	
Instructor	Dr. M.M. Dabrowski	
Instructor	Dr. D. Maiullari	
Instructor	T.N. Broekmans	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	1 hours per week starting from week 1 and ending in week 9.	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	MSc1, 2 & 3 of the Master track Urbanism.	
Summary	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. This course is the second part of the graduation laboratory; students continue (individually) developing their graduation project that was approved at the end of the previous semester (GO for P2).	
Course Contents	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. These semesters give the student therefore a unique opportunity to do an in-depth research and design project in the field of urbanism. This course is the second part of the graduation laboratory; students continue (individually) developing their graduation project that was approved at the end of the previous semester (GO for P2).	
Study Goals	At the end of this course, the student: is able to describe and carry out research in the field of urbanism and process the research results as well as use drawings/maps/graphics as a means to research; is able to describe the problem field of the selected topic and translate it into a field of graduation objectives (and design task) with associated research questions and research approach; is able to describe a clear theoretical framework which is appropriate for the selected topic; is able to carry out research by design in a methodological way focused on the research questions; is able to process the research results in the final report adequately: i.e. formulated and/or imagined by means of analytical drawing(s); is able to use drawings / maps / graphics as a means to research; is able to define urban design methods, choices, aspects, effects and consequences by means of plan forms and design-instruments; is able to define and visualised the own working method(s) and the (design) choices within the design process with sound arguments; is able to define the spatial, functional, technical, and/or social aspects of the design adequately: clear, transparent and with a proper justification; is able to use plan forms and design-instruments which suit his/her design task; is able to describe, imagined and justified the effects and consequences of the design proposal(s) with respect to the aimed field; is able to draw conclusions and define recommendations; is able to evaluate the urban design and research aims in the conclusions; is able to indicate clearly and logically which research questions are answered and how that has been processed in the urban design; is able to define clear, concrete, specific recommendations based on the results of the urban research and/or design; is able to specify for which questions is still additional (design) research necessary; is able to show an analytical capacity to present a complex matter in a brief and concise way; is able to describe the projects relevance, reflect on the products and present these; is able to describe clearly the innovative (scientific and/or social) insights of the graduation project in text and images and, if necessary, concrete strategies and/or application possibilities for the field of the urbanism; is able to position the own graduation project with respect to the field of the urbanism, as well as other adjacent scientific fields; is able to discuss and present the thesis products (urban design & research) and the process (design & research) in the form of an epilogue or evaluation. In short, the student has to show that he/she is able to deliver a project of professional quality and of academic level in line with the end terms of the master track Urbanism.	
Education Method	In this semester students work individually, supervised by two mentors, within the frame of the studio and graduation lab of Urbanism. Meetings and lectures are less frequent than in the previous semester, the independency of the student is enlarged.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course and by the studio coordinator and mentors.	
Assessment	The assessment is imbedded in the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on the assessments of the graduation project. In this course the project is evaluated three more times: at the third Evaluation: Compulsory Progress Review (P3) at the fourth Evaluation: Formal Assessment (P4) at the final Evaluation: Public Final Presentation (P5)	

	A rubric (EMMA) will be used for grading. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on retake possibilities.
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Once enrolled in AR3U115 (start of the graduation) students automatically continue their graduation in AR4U010
Special Information	On set conditions, Urbanism students have the possibility to carry out their graduation project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found in the year guide.
Remarks	The maximum marking period is 15 working days. The final product of this course is an integral concise product that the student has to deliver at the end of Master 4, 1 week prior to the P5 presentation. See the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences, the 'Exit Qualifications' of the master Architecture, Urbanism and Building Sciences, the evaluation form 'Graduation Criteria Urbanism P3' and the assessment criteria of the 'Graduation Criteria Urbanism P3/P4/P5' of the Master track of Urbanism.
Period of Education	Semester 1 and semester 2. The MSc3 Urbanism courses start both in September and February. The AR4U010 course starts automatically after successfully finishing the MSc3 with a GO at the P2 presentation.
Concept Schedule	The sessions differ per week. The important dates and deadlines are represented in the academic graduation calendar and in the year guide available via Bright Space. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

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